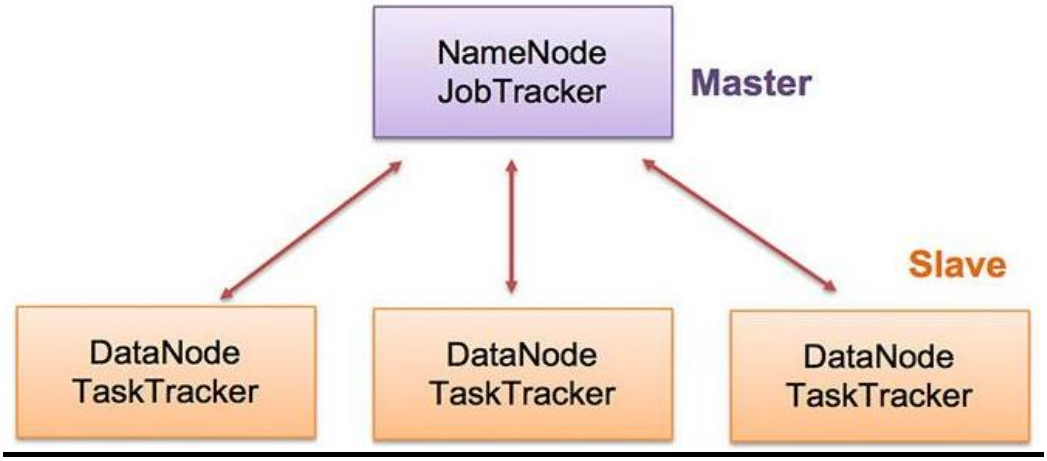
Tuần 2: Dựng mô hình Cluster gồm 1 master và nhiều slave bằng công nghệ hadoop, yarn, spark trên nền tảng ubuntu và đề demo bài toán wordcount.

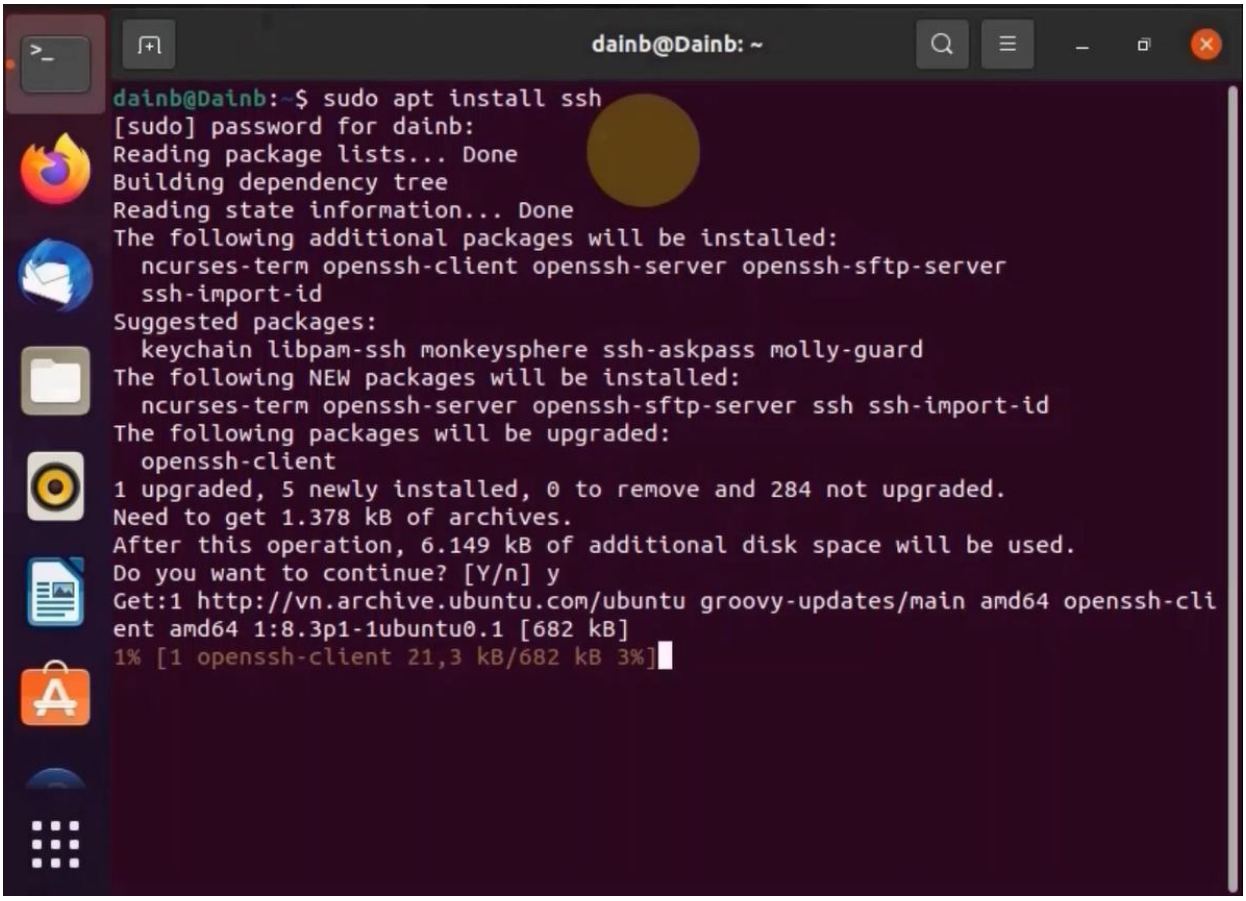
1. Mô hình triển khai hệ thống:



2. Cài đặt

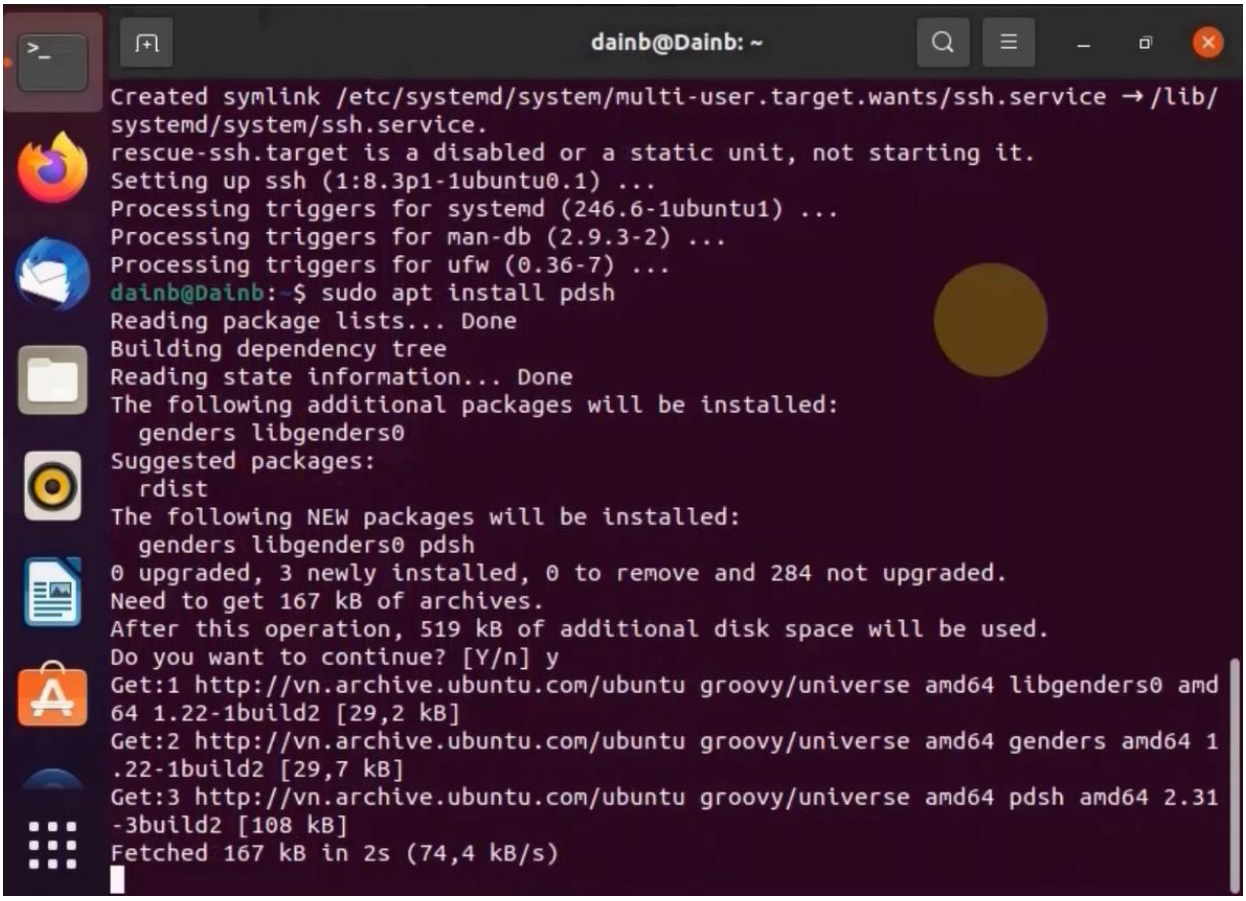
Bước 1: Cài đặt ssh

$ sudo apt install ssh



Bước 2: Cài đặt PDSH

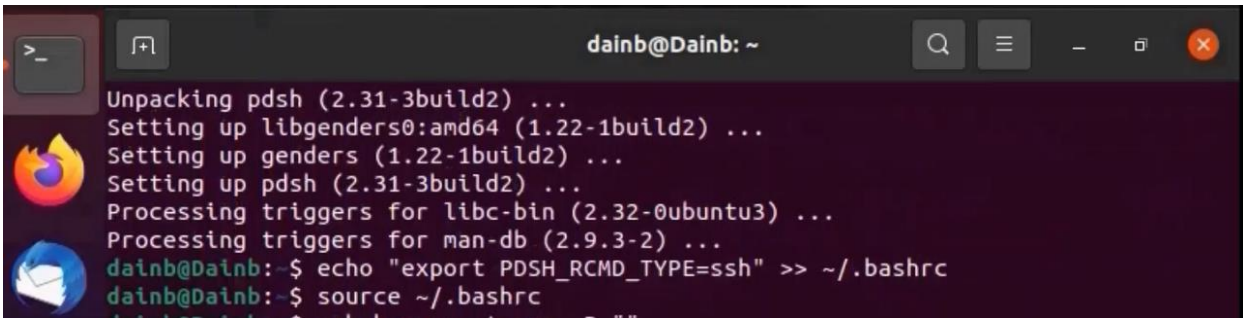
$ sudo apt install pdsh



Bước 3: Cấu hình file ~/.bashrc

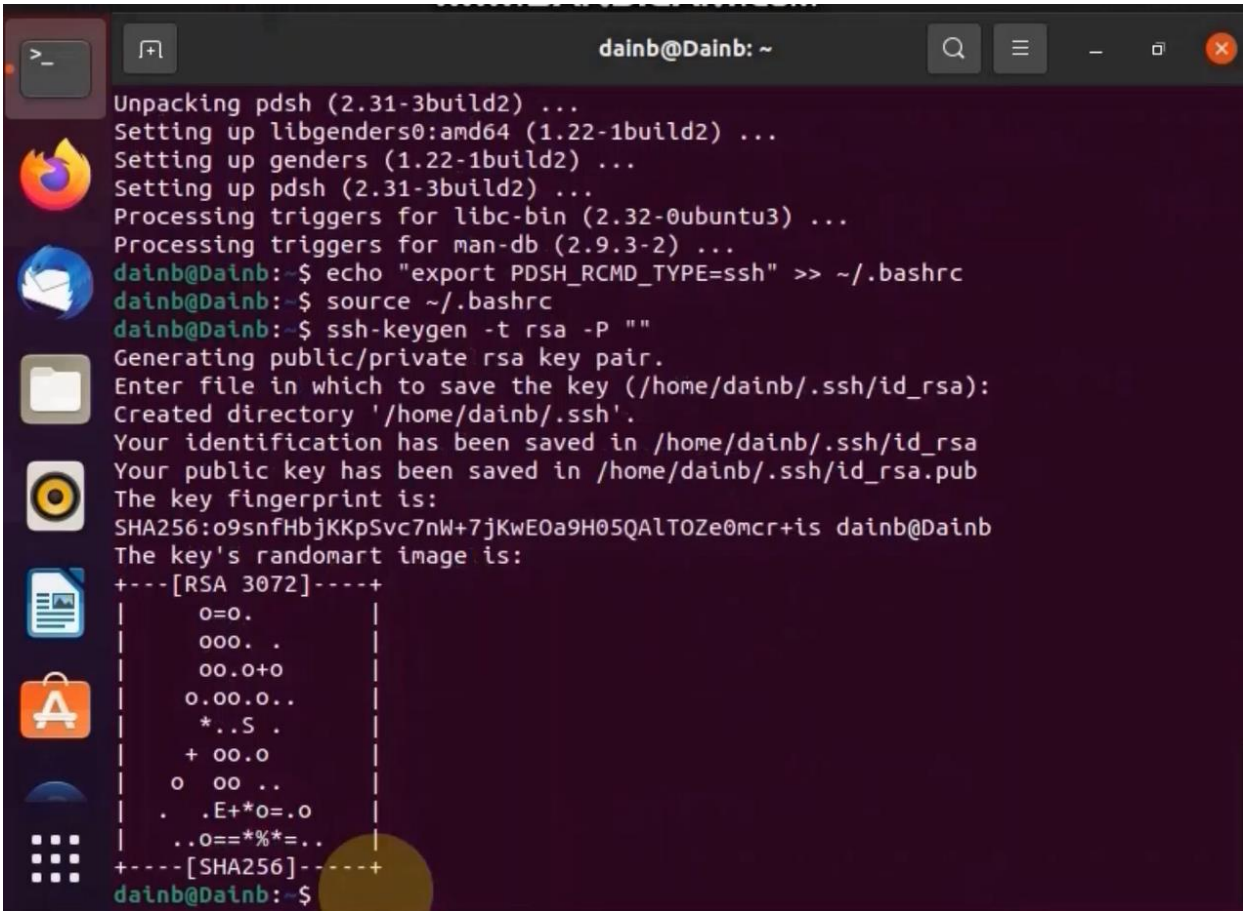
$ echo "export PDSH_RCMD_TYPE=ssh" >> ~/.bashrc

$ source ~/.bashrc



Bước 4: Tạo khóa cho ssh

$ ssh-keygen -t rsa -P ""



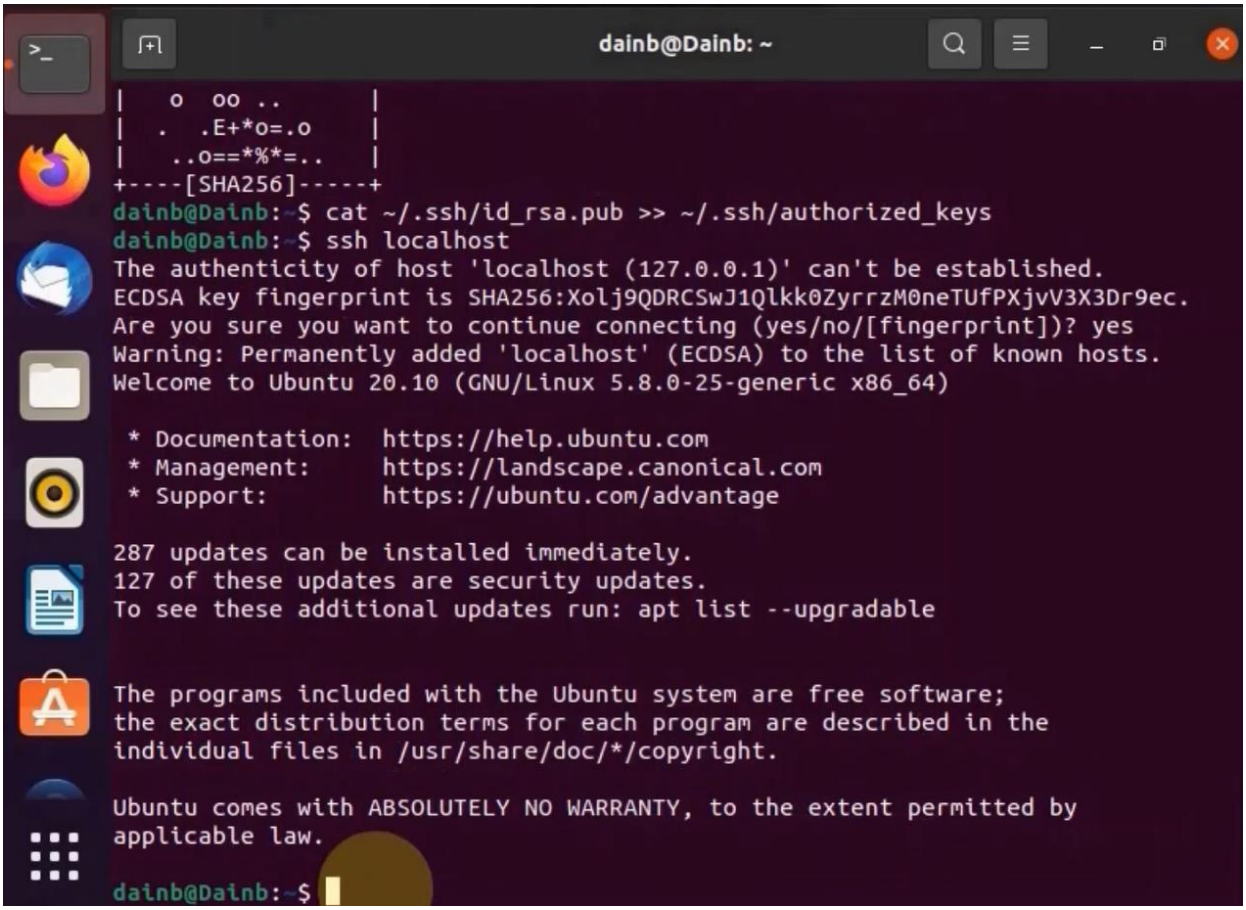
Bước 5: Cho phép ssh truy cập tới máy cục bộ

$ cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys



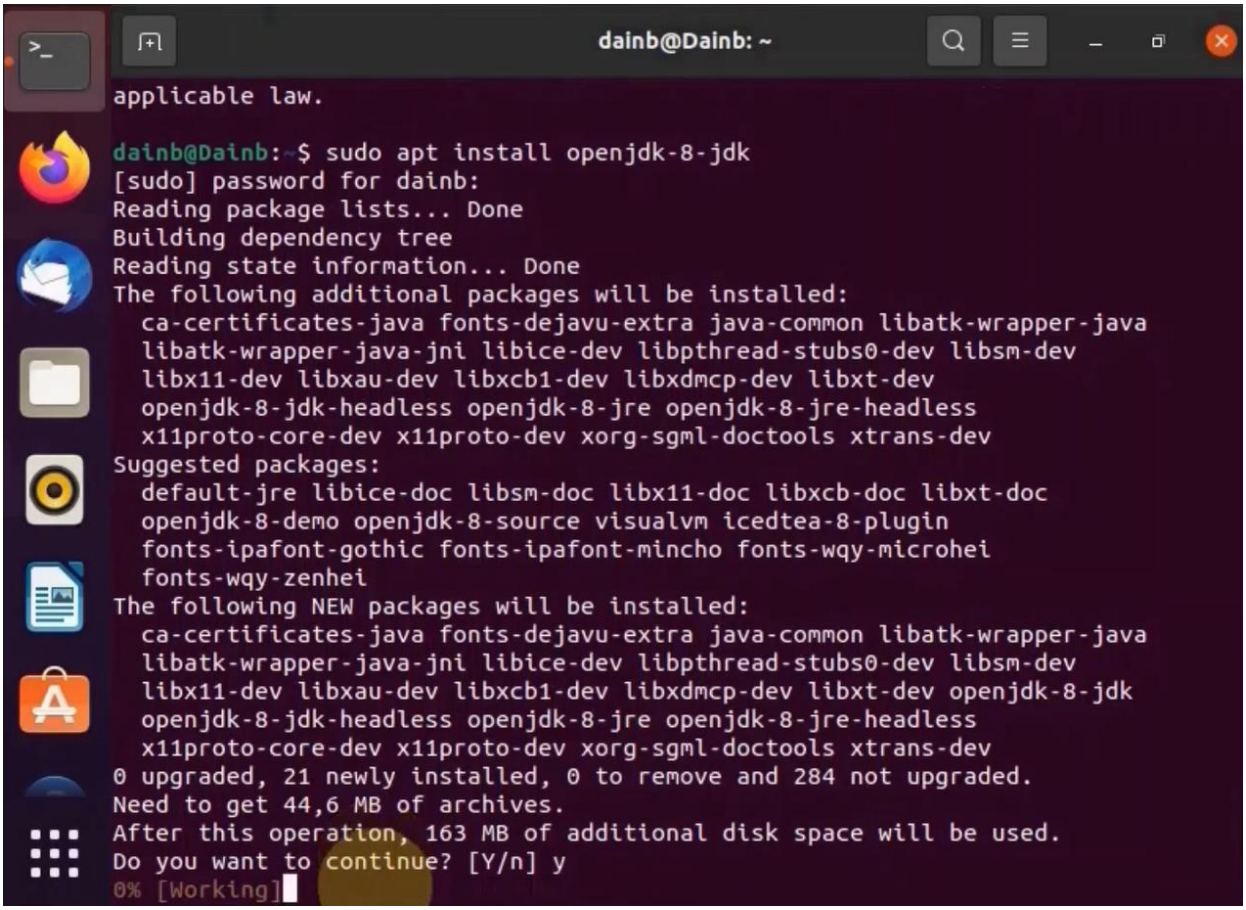
Bước 6: xác minh cấu hình ssh bằng kết nối với localhost

$ ssh localhost



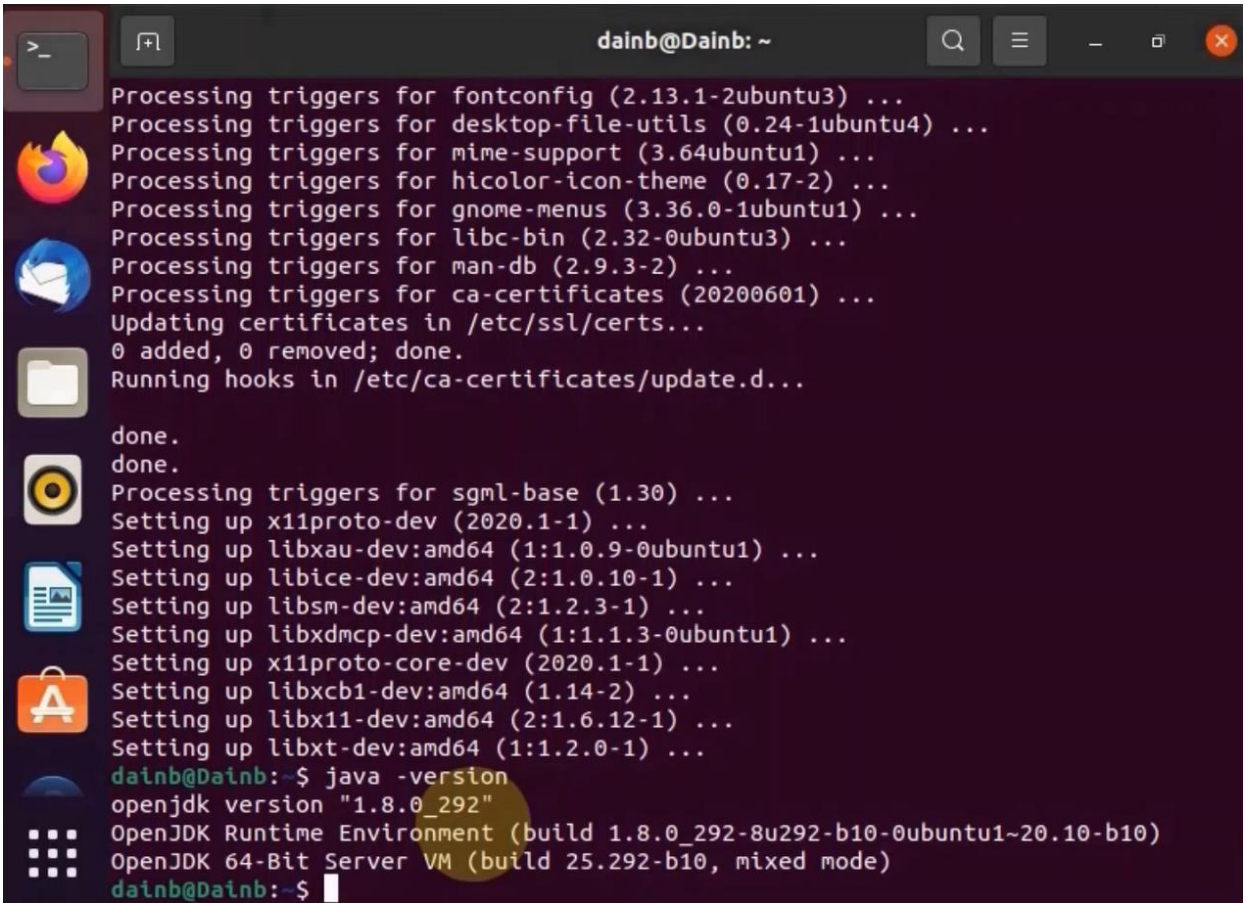
Bước 7: Cài đặt java version 8

$ sudo apt install openjdk-8-jdk



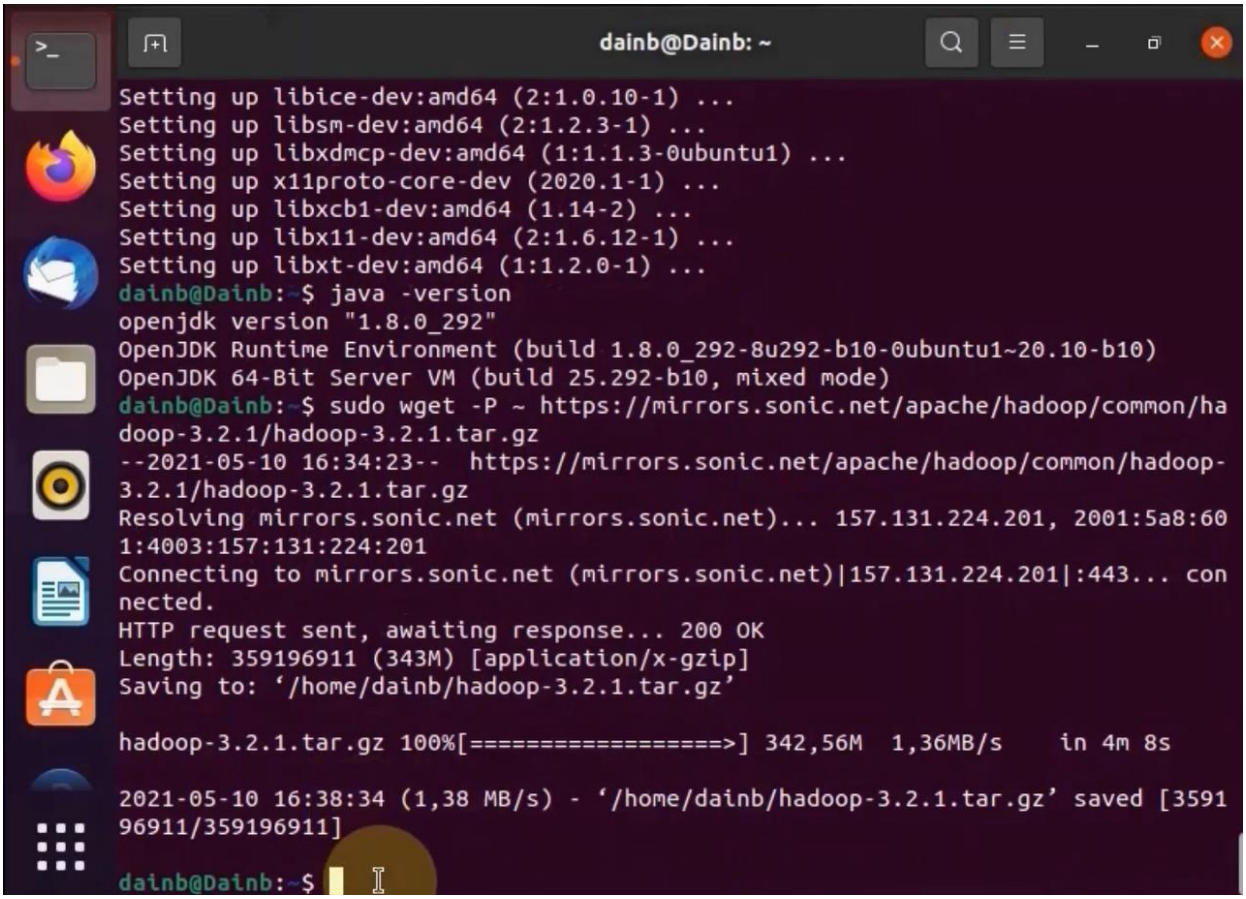
Bước 8: Kiểm tra cài đặt java

$ java -version



Bước 9: Tải hadoop

$ sudo wget -P ~ https://mirrors.sonic.net/apache/hadoop/common/hadoop-3.3.1/hadoop-3.3.1.tar.gz



Bước 10: Giải nén file hadoop-3.3.1.tar.gz

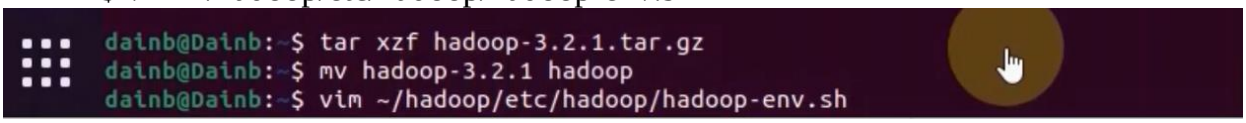
$ tar xzf hadoop-3.3.1.tar.gz

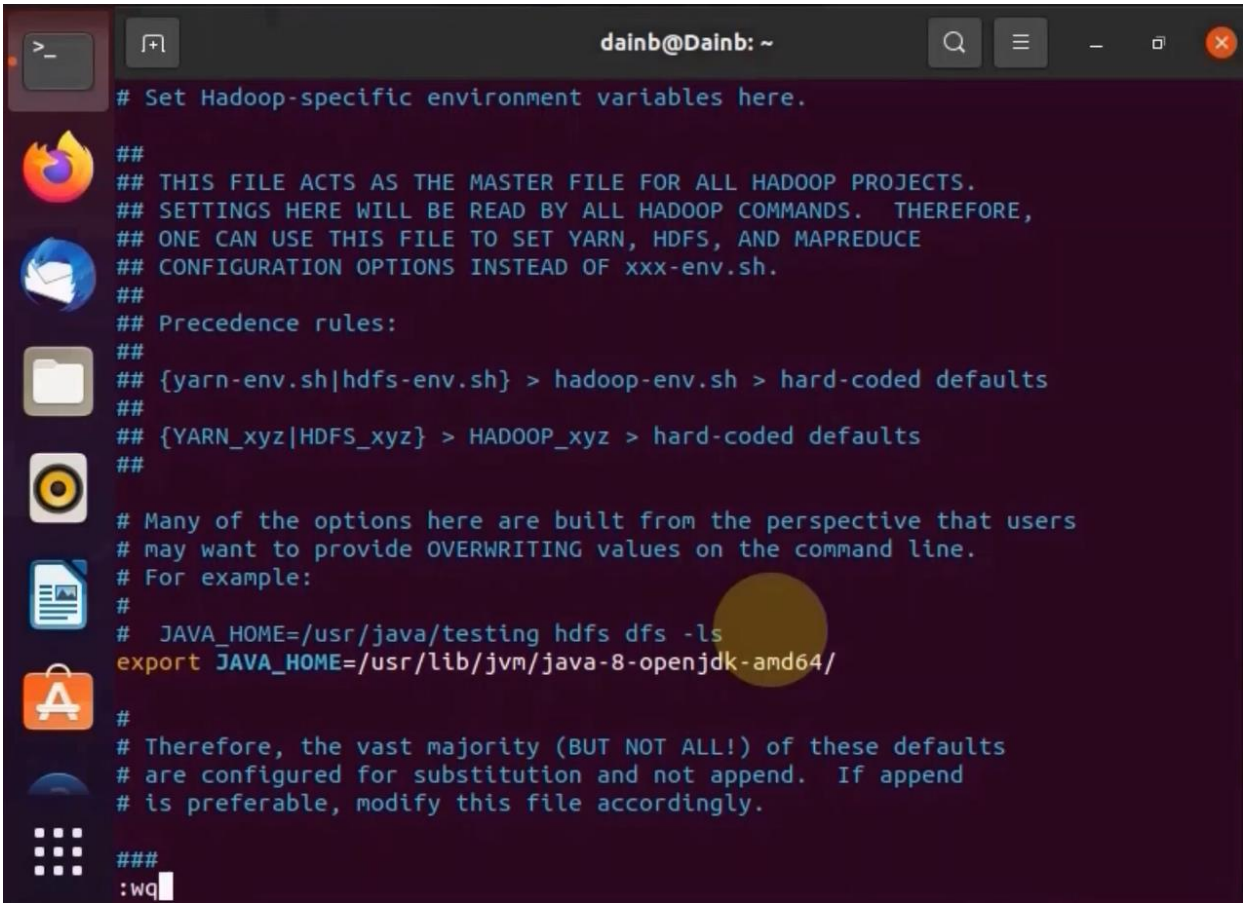
$ mv hadoop-3.3.1 hadoop



Bước 11: Mở file hadoop-env.sh để chỉnh sửa JAVA_HOME

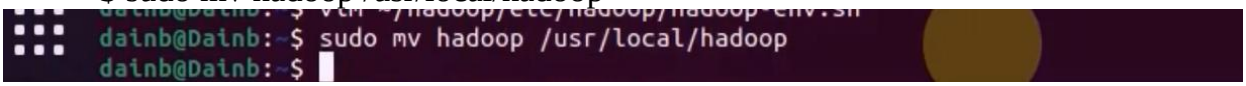
$ vim ~/hadoop/etc/hadoop/hadoop-env.sh





Bước 12: Di chuyển thư mục hadoop qua usr/local/hadoop

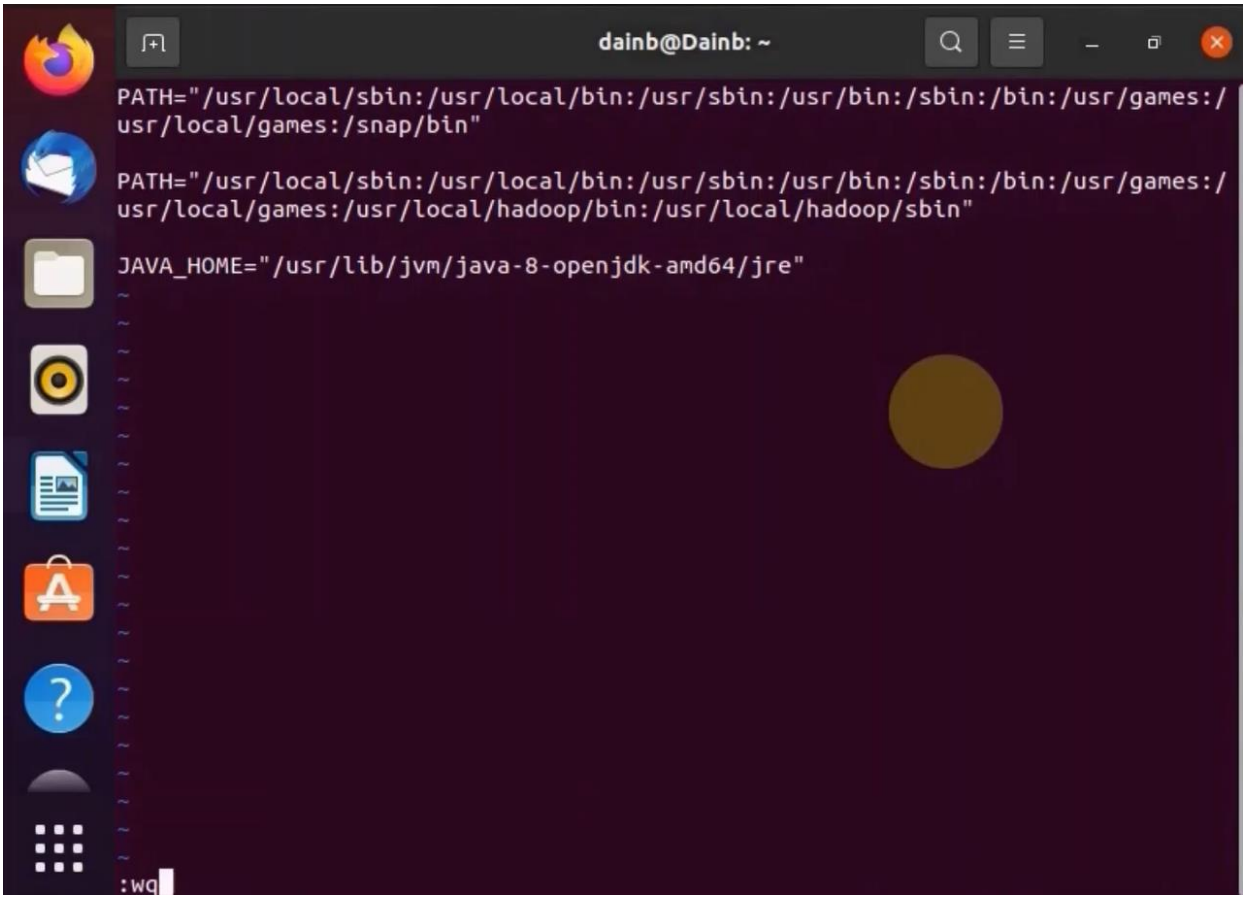
$ sudo mv hadoop /usr/local/hadoop



Bước 13: Mở file environment

$ sudo vim /etc/environment

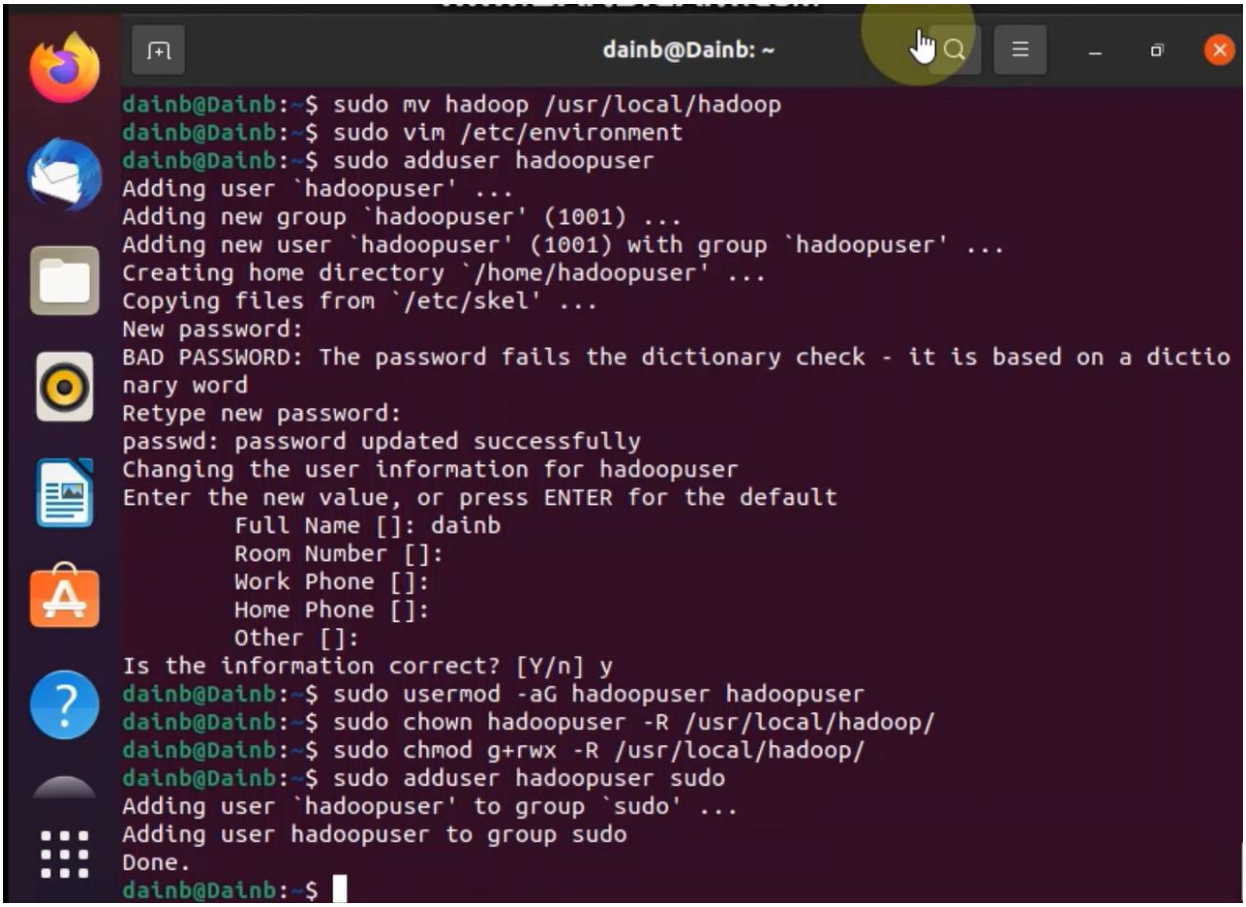




Bước 14: Thêm người dùng tên là hadoopuser

$ sudo adduser hadoopuser

$ sudo usermod -aG hadoopuser hadoopuser
$ sudo chown hadoopuser -R /usr/local/hadoop/
$ sudo chmod g+rwx -R /usr/local/hadoop/
$ sudo adduser hadoopuser sudo



Bước 15: Kiểm tra ip

$ ip addr

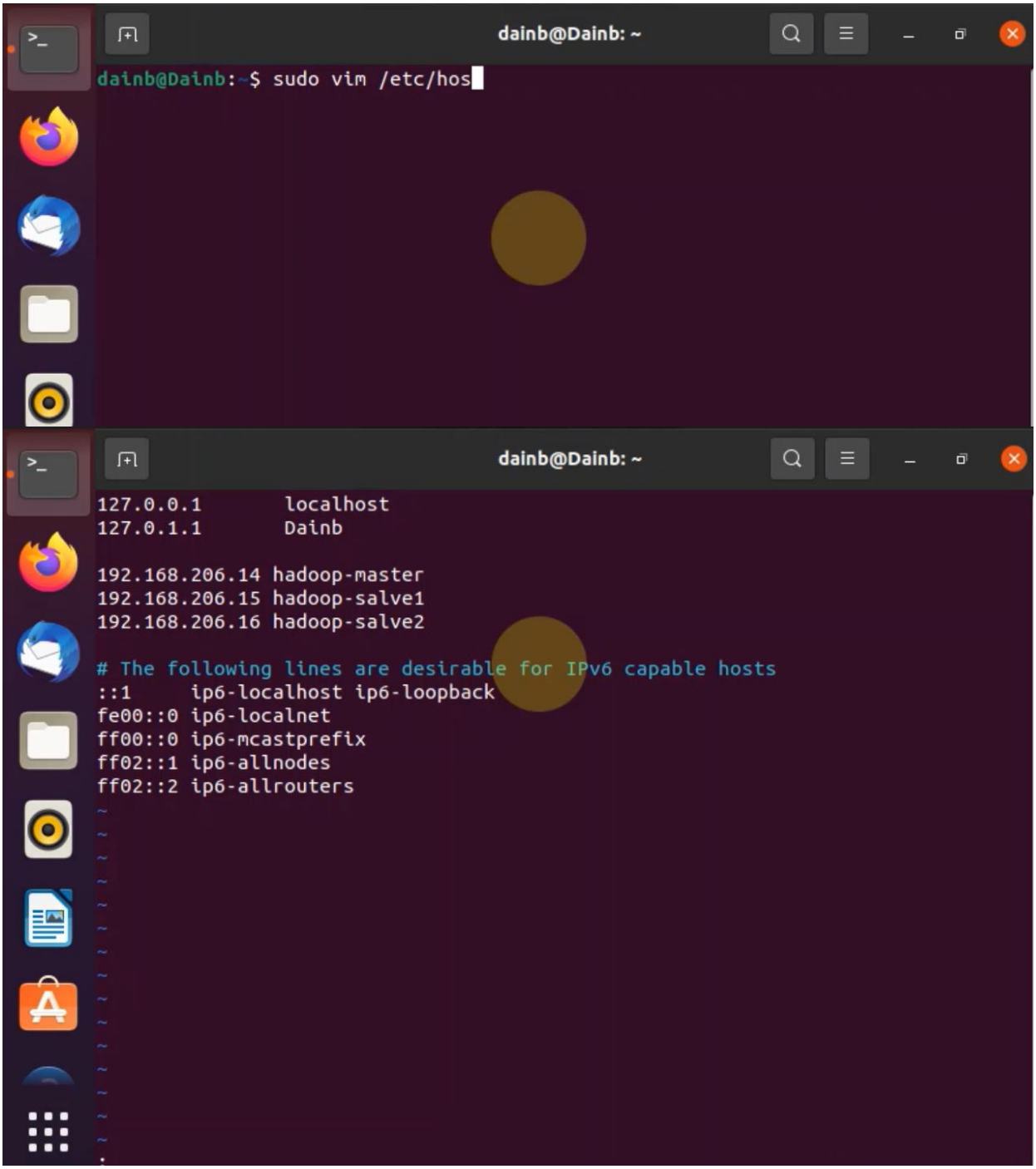
master 192.168.136.128

salve1 192.168.136.129

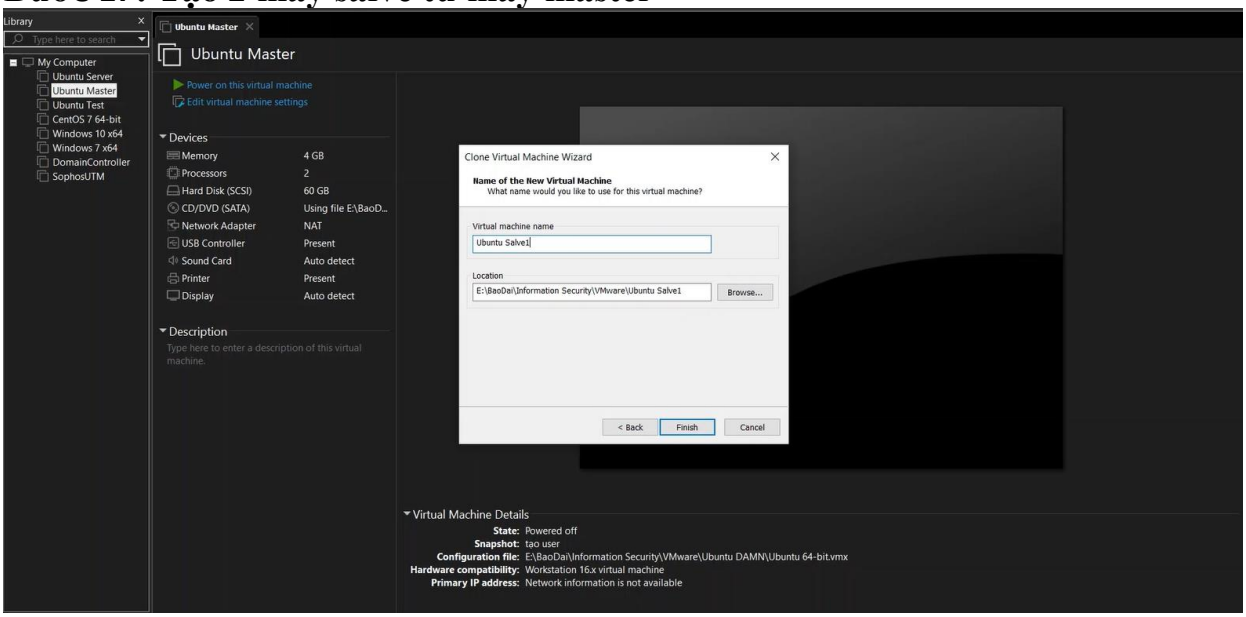
salve2 192.168.136.130

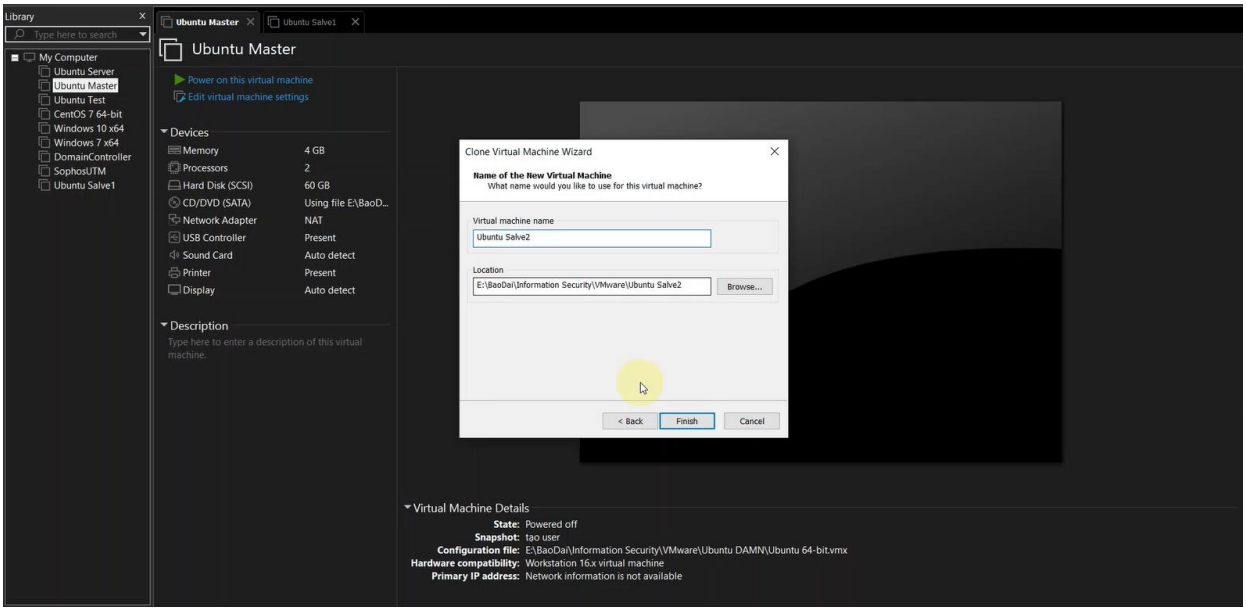
Bước 16: cấu hình file hosts

$ sudo vim /etc/hosts


dainb@Dainb: ~
dainb@Dainb:~$ sudo vim /etc/hos
dainb@Dainb: ~
127.0.0.1 localhost
127.0.1.1 Dainb
192.168.206.14 hadoop-master
192.168.206.15 hadoop-salve1
192.168.206.16 hadoop-salve2
The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters

Bước 17: Tạo 2 máy salve từ máy master





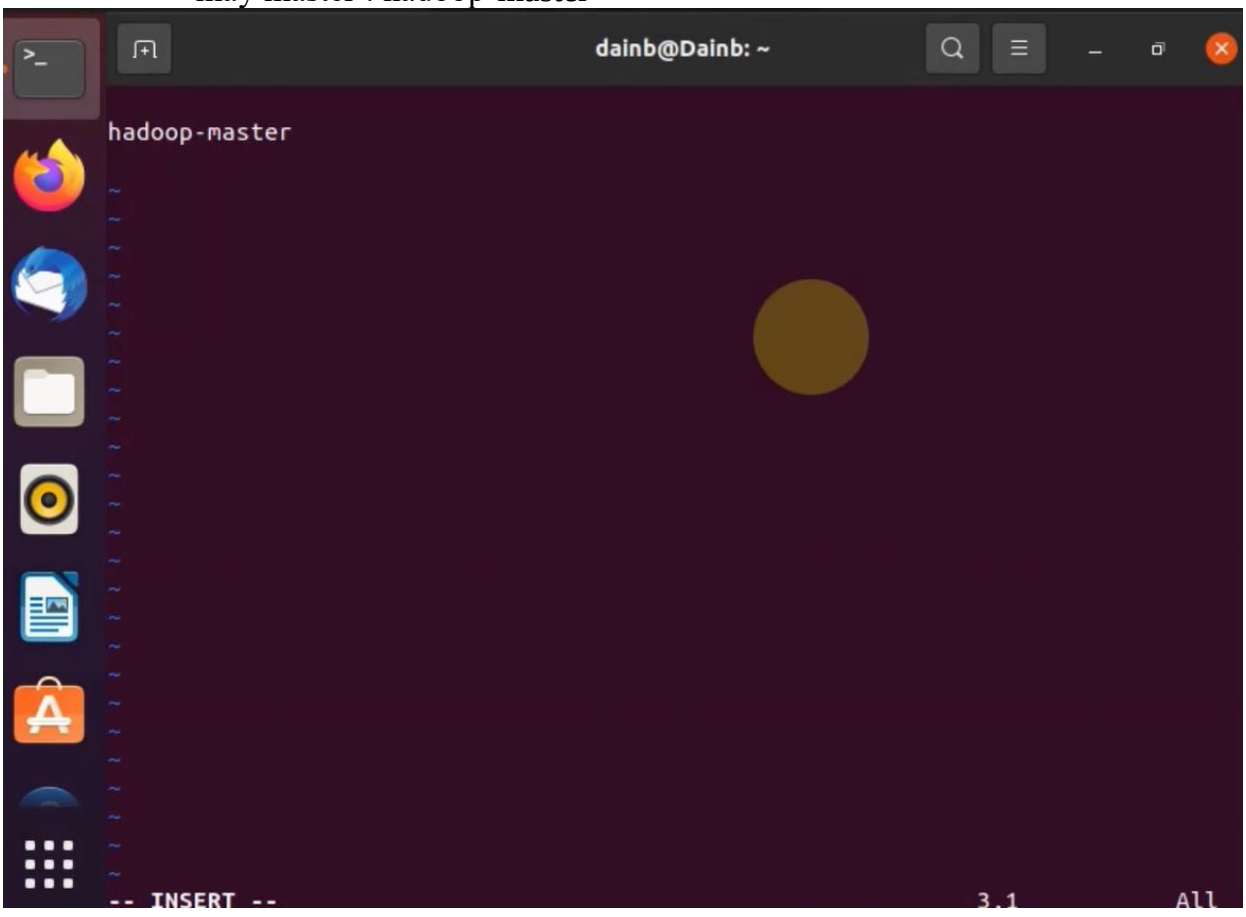
Bước 18: Cấu hình file hostname

$ sudo vim /etc/hostname

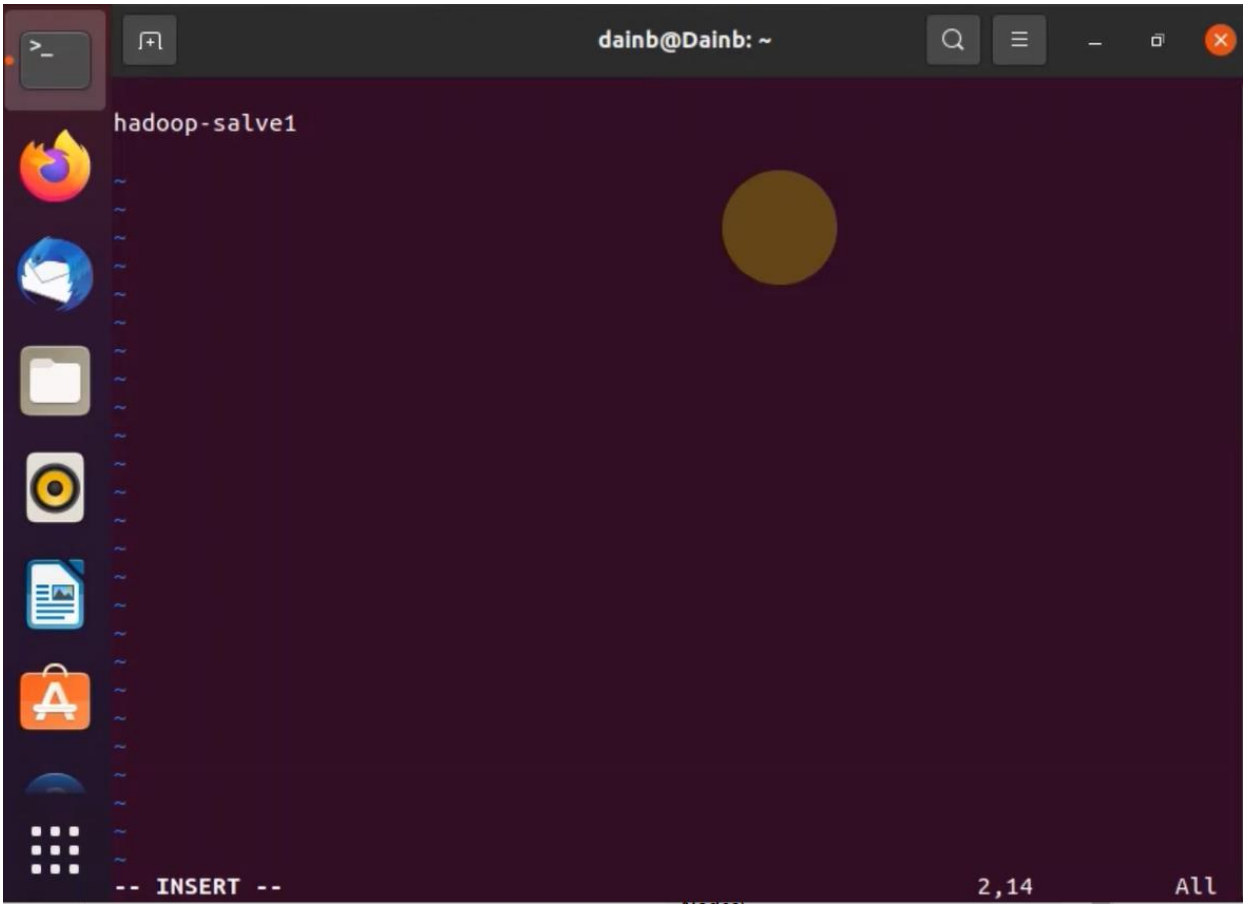


thêm vào

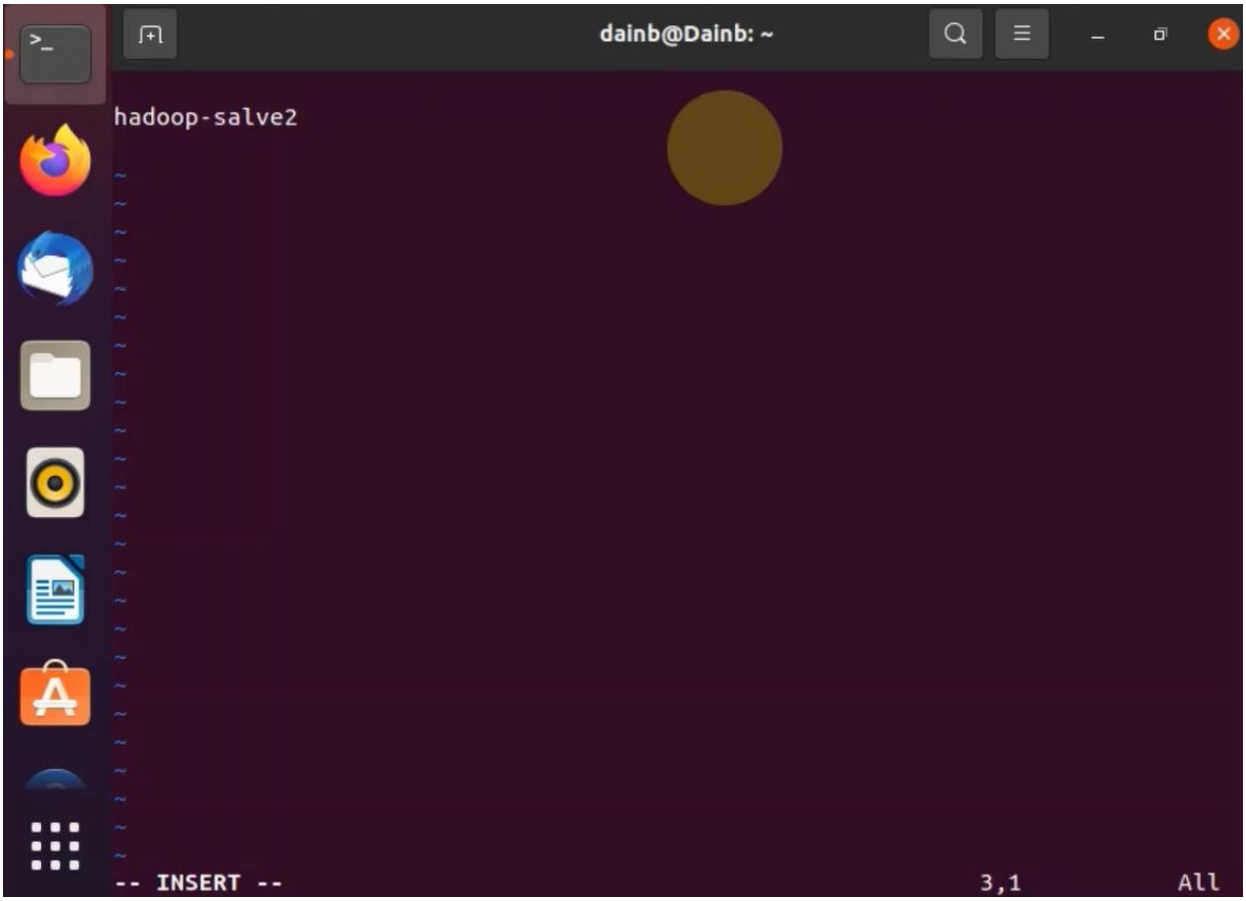
máy master : hadoop-master



máy salve1 : hadoop-salve1

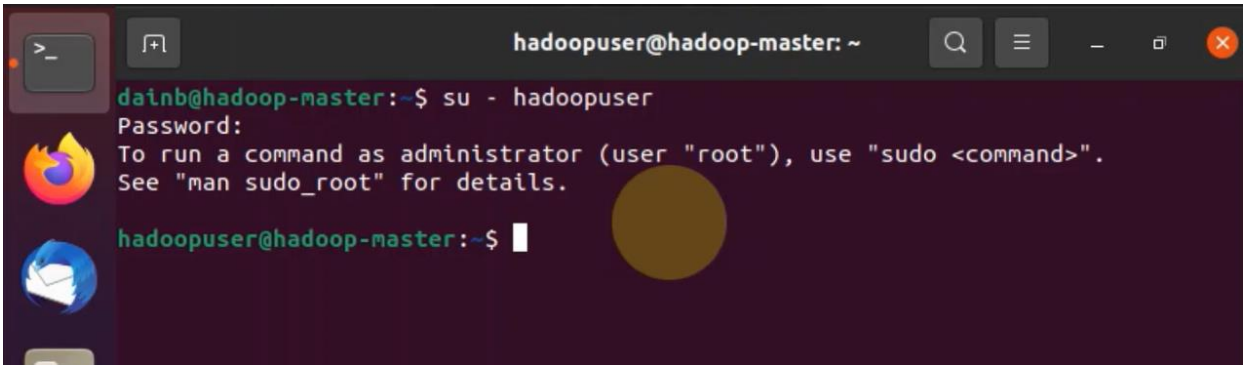


máy salve2 : hadoop-salve2



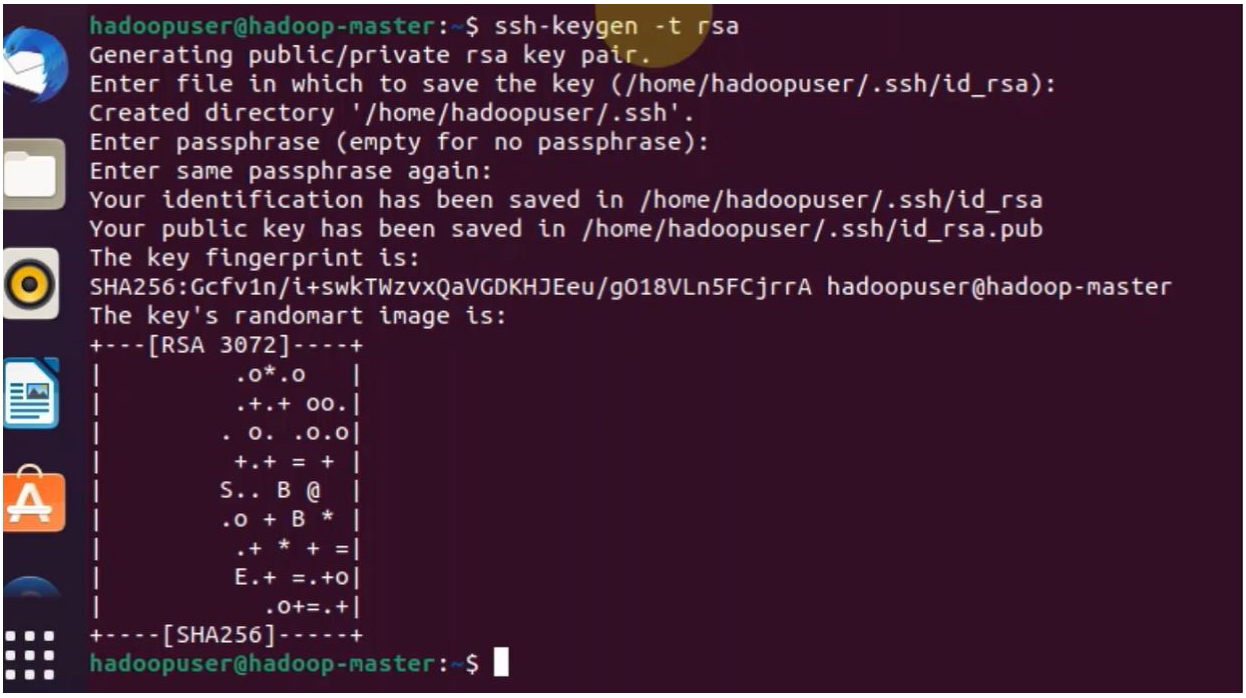
Bước 19: Cấu hình trên hadoop-master và hadoopuser

$ su - hadoopuser



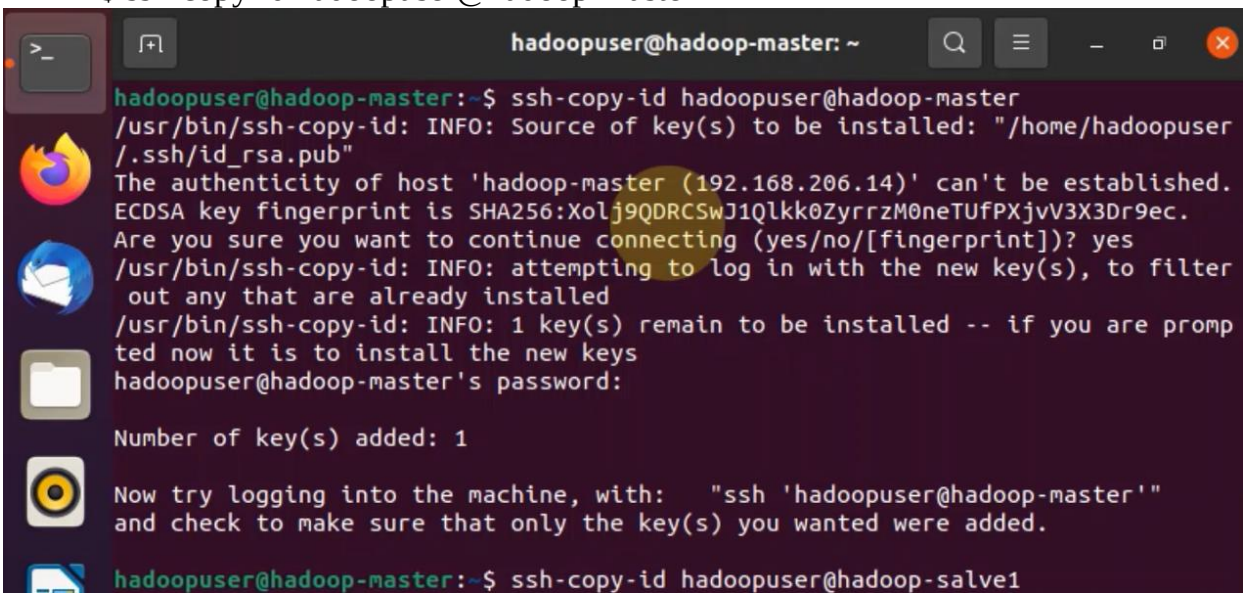
Bước 20: Tạo khóa ssh

$ ssh-keygen -t rsa

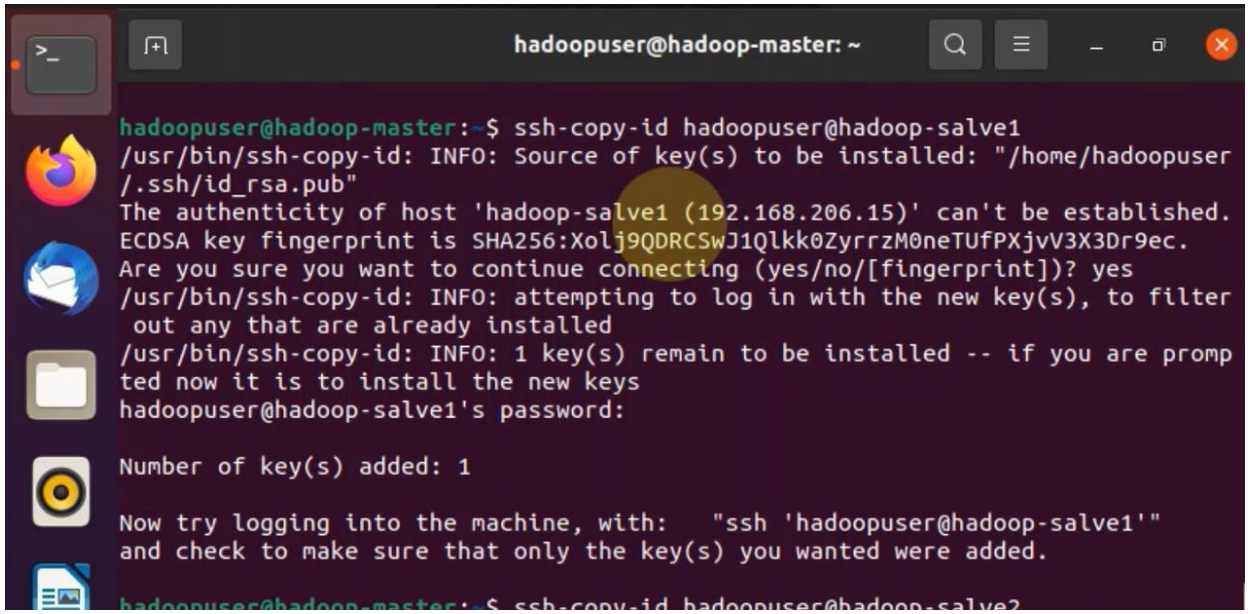


Bước 21: Sao chép khóa SSH cho các user

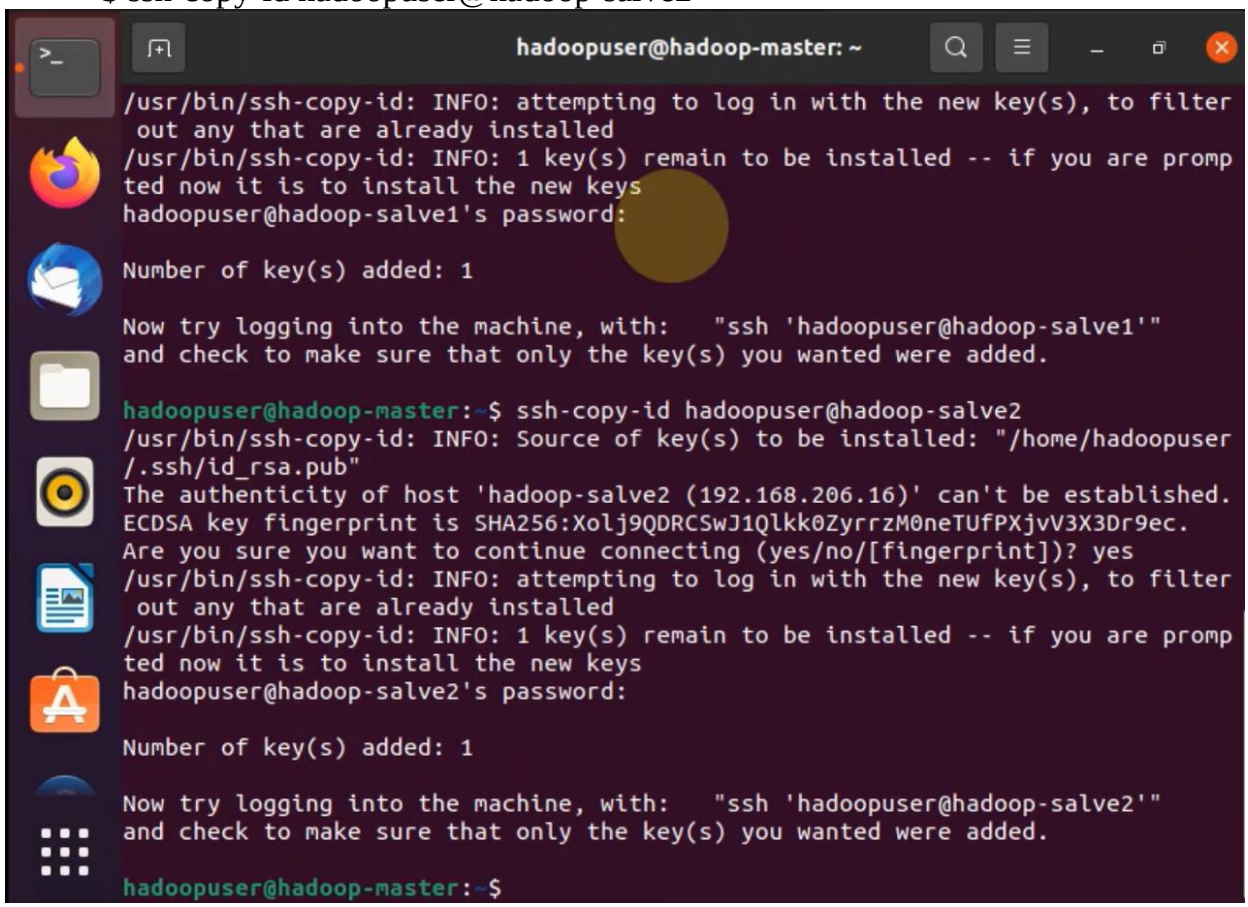
$ ssh-copy-id hadoopuser@hadoop-master



$ ssh-copy-id hadoopuser@hadoop-salve1

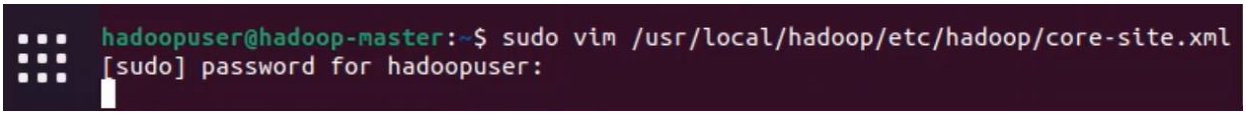


$ ssh-copy-id hadoopuser@hadoop-salve2



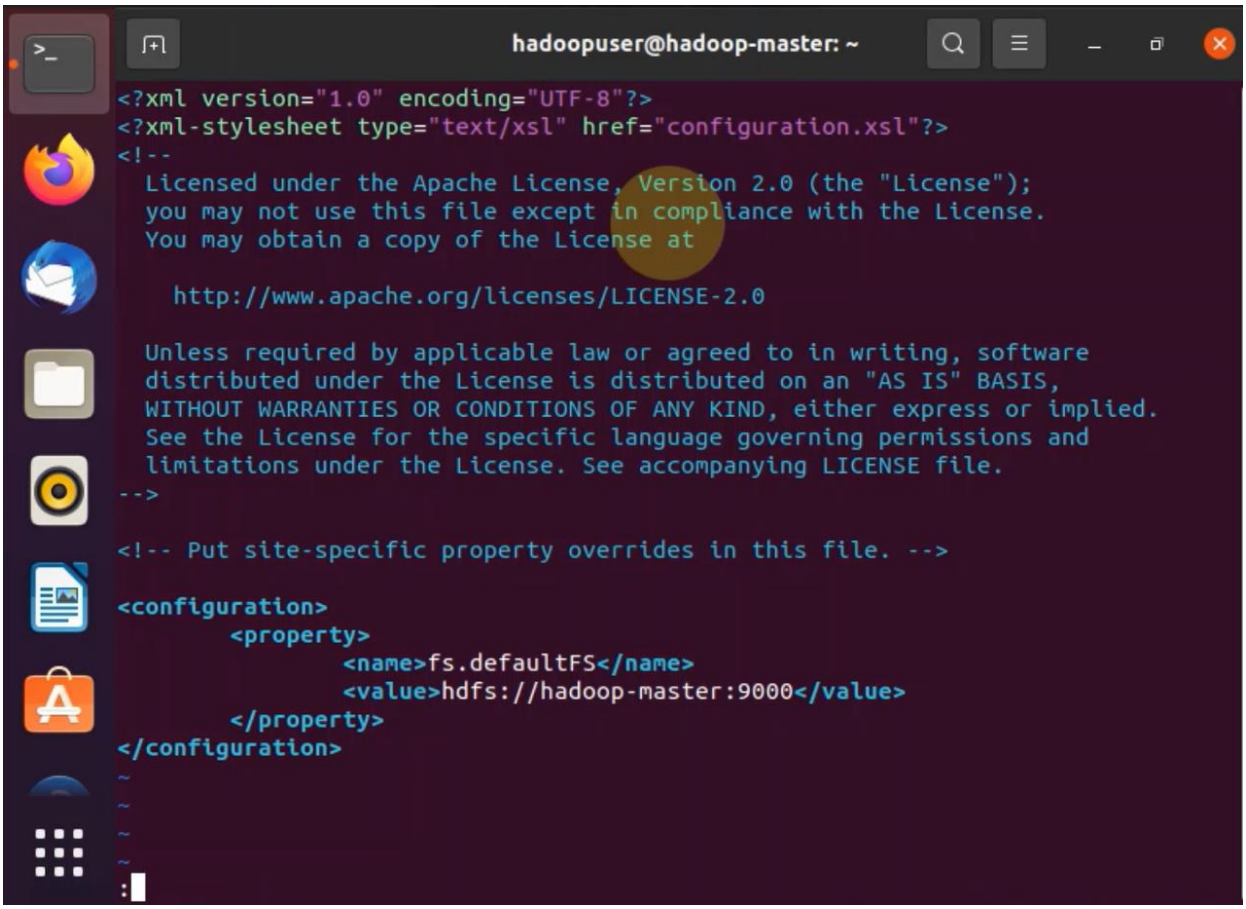
Bước 22: Cấu hình file trên hadoop-master

$ sudo vim /usr/local/hadoop/etc/hadoop/core-site.xml

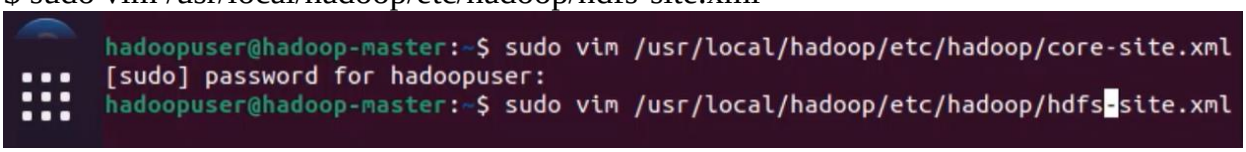


thêm cấu hình

```
<configuration>
        <property>
                <name>fs.defaultFS</name>
                <value>hdfs://hadoop-master:9000</value>
        </property>
</configuration>
```

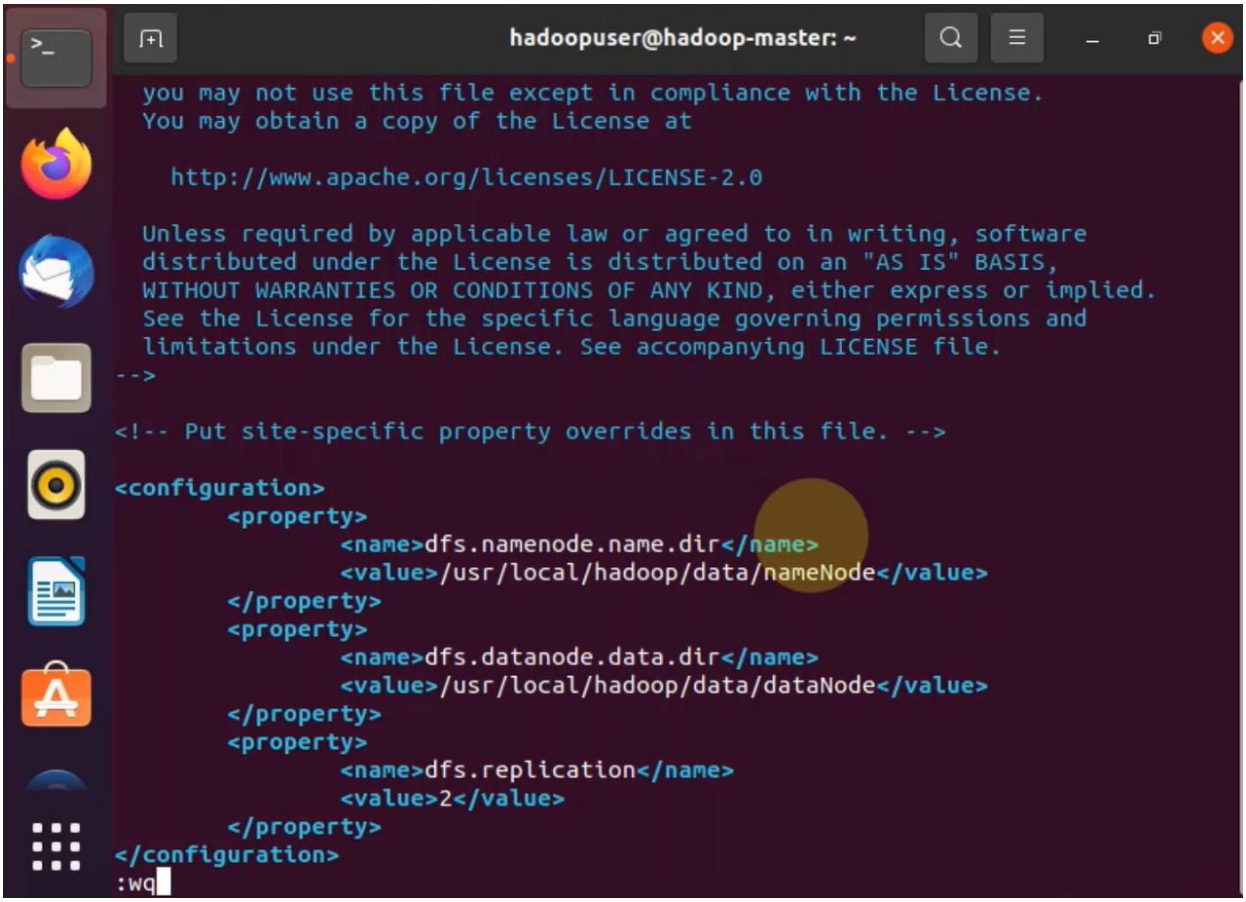


$ sudo vim /usr/local/hadoop/etc/hadoop/hdfs-site.xml

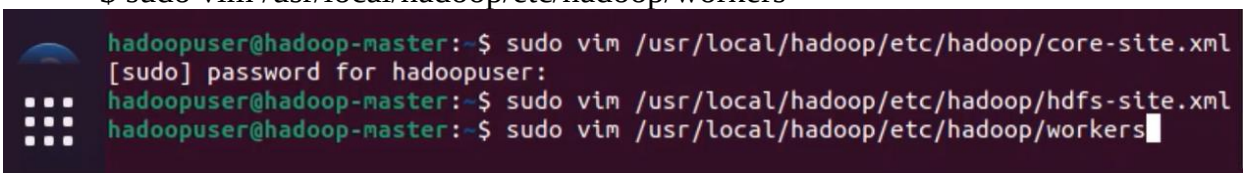


thêm cấu hình

```
<configuration>
        <property>
                <name>dfs.namenode.name.dir</name>
                <value>/usr/local/hadoop/data/nameNode</value>
        </property>
        <property>
                <name>dfs.datanode.data.dir</name>
                <value>/usr/local/hadoop/data/dataNode</value>
        </property>
        <property>
                <name>dfs.replication</name>
                <value>2</value>
        </property>
</configuration>
```



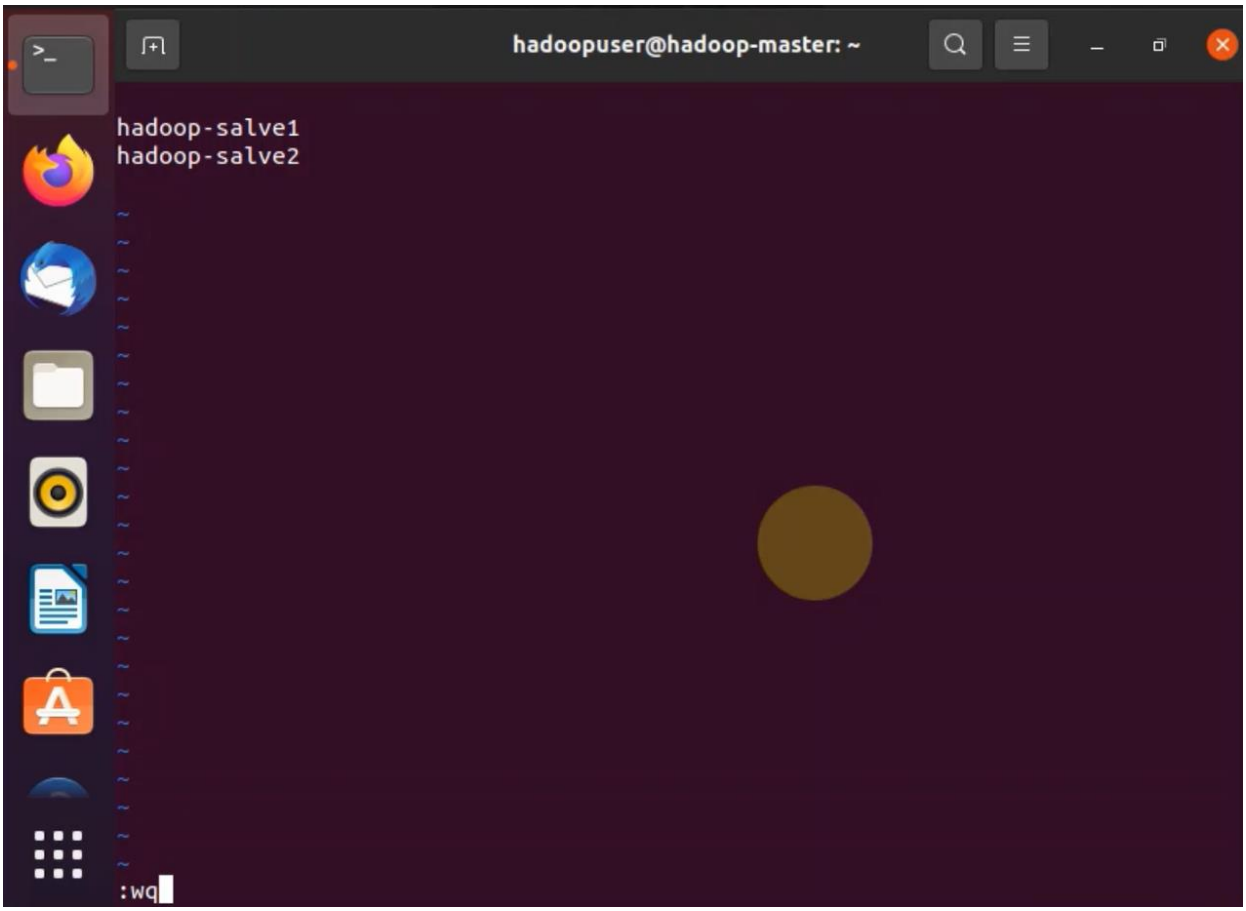
$ sudo vim /usr/local/hadoop/etc/hadoop/workers



thêm

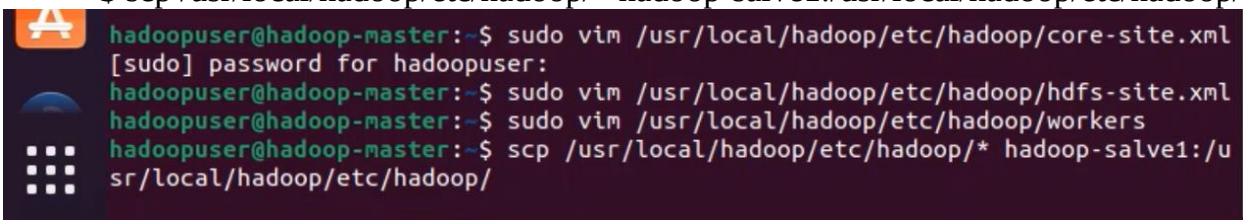
hadoop-salve1

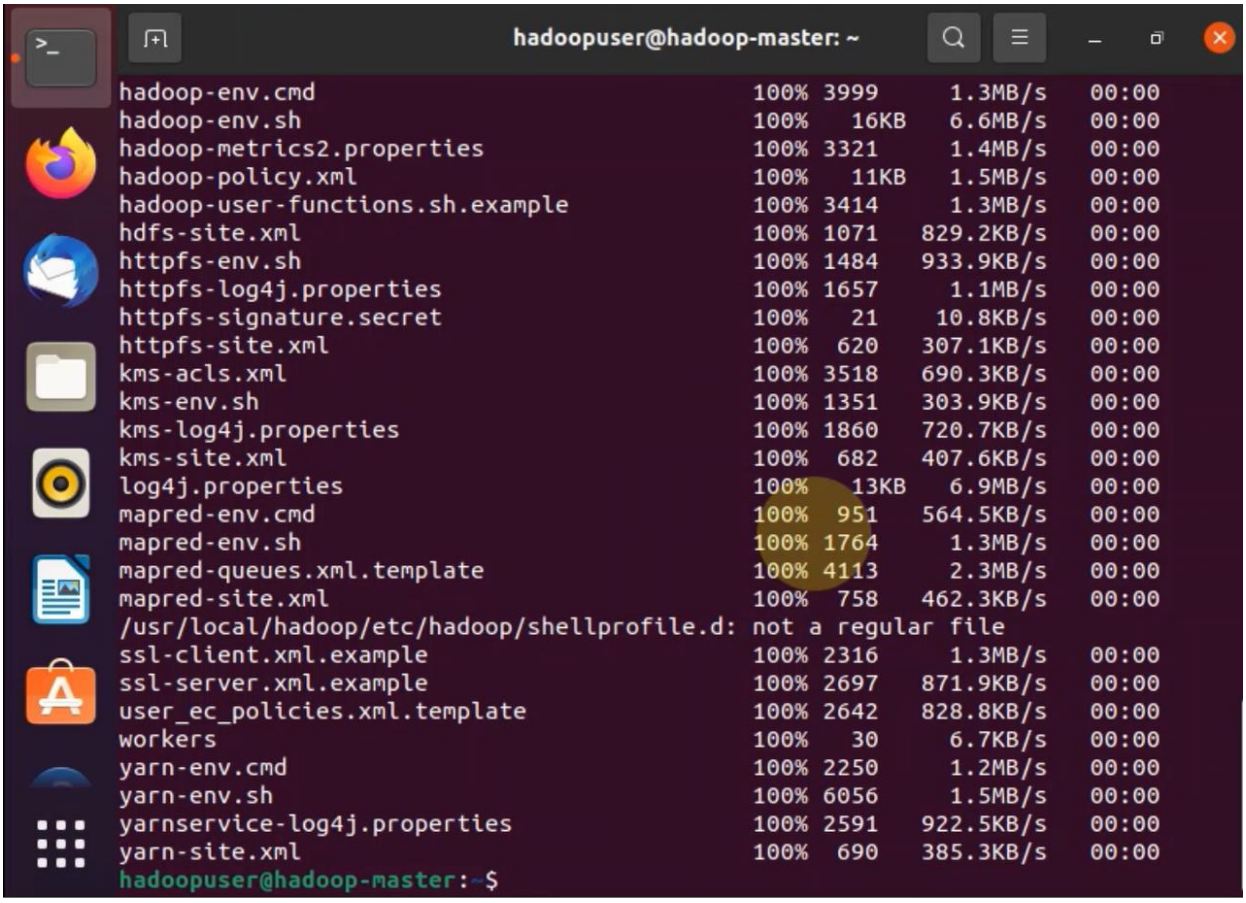
hadoop-salve2



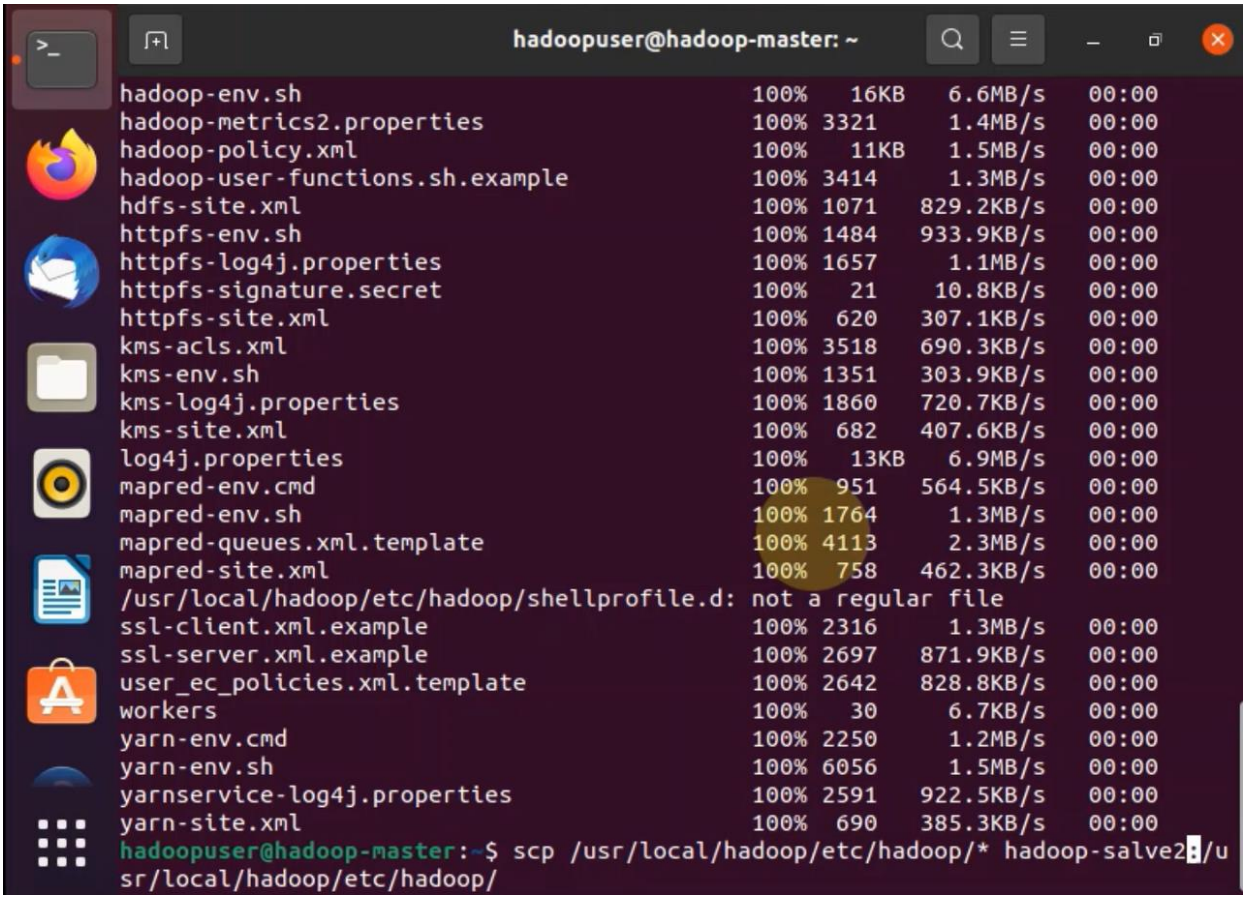
Bước 23: Sao chép cấu hình máy master cho các máy salve

$ scp /usr/local/hadoop/etc/hadoop/* hadoop-salve1:/usr/local/hadoop/etc/hadoop/





$ scp /usr/local/hadoop/etc/hadoop/* hadoop-salve2:/usr/local/hadoop/etc/hadoop/



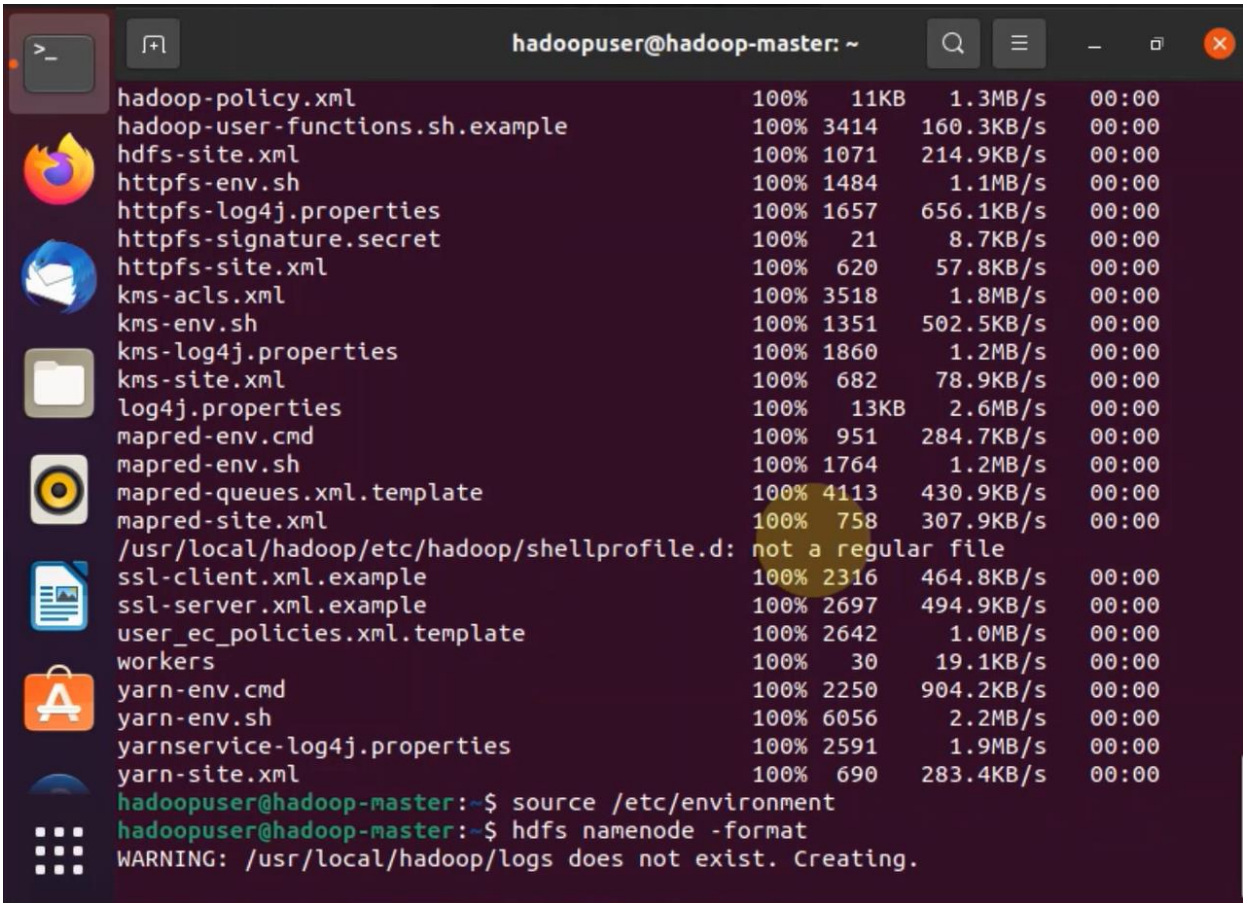
Bước 24: Định dạng lại file HDFS

$ source /etc/environment

$ hdfs namenode –format

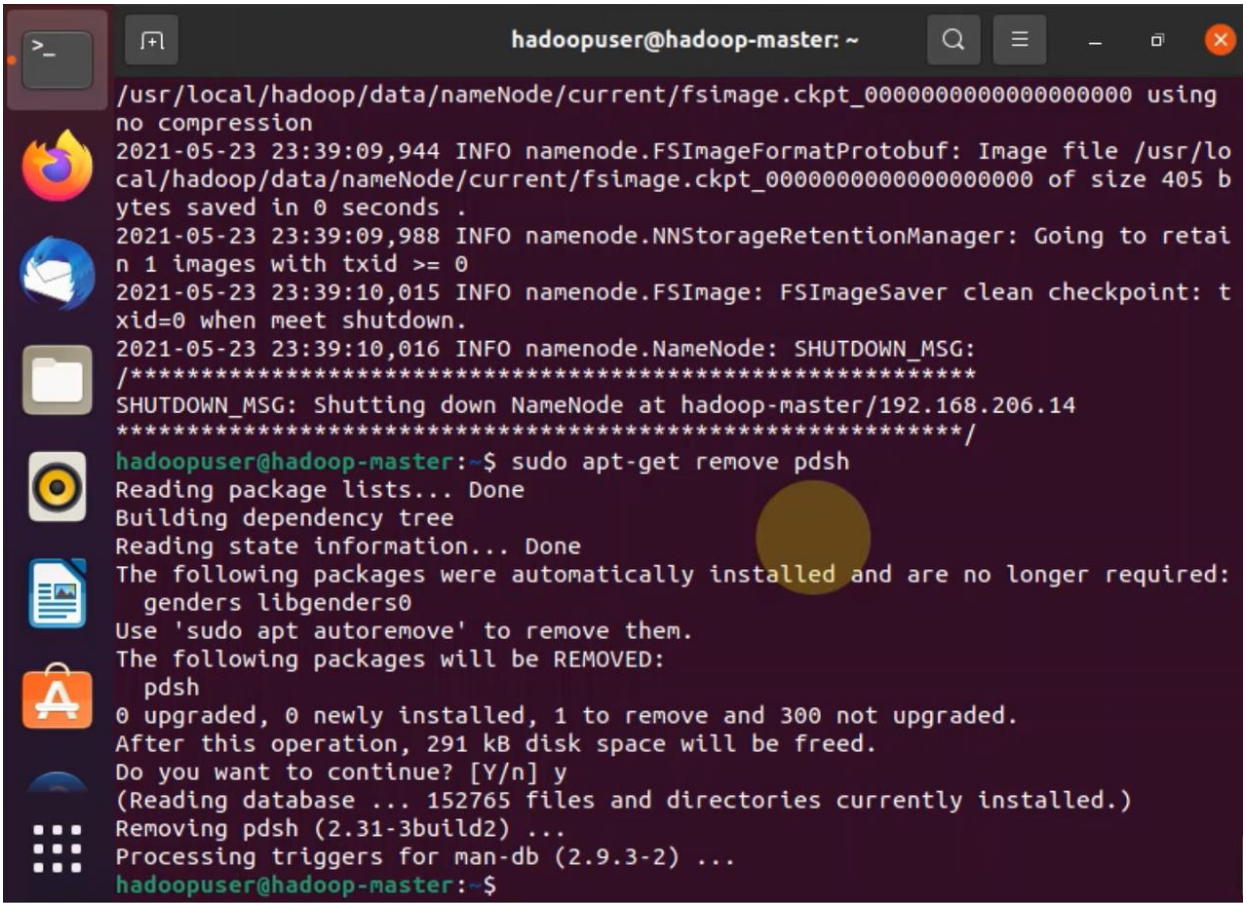
Nếu kg đủ quyền, cần chuyển quyền đọc và ghi vào thư mục nhật ký

```
sudo chmod -R 777 /usr/local/hadoop/logs
```



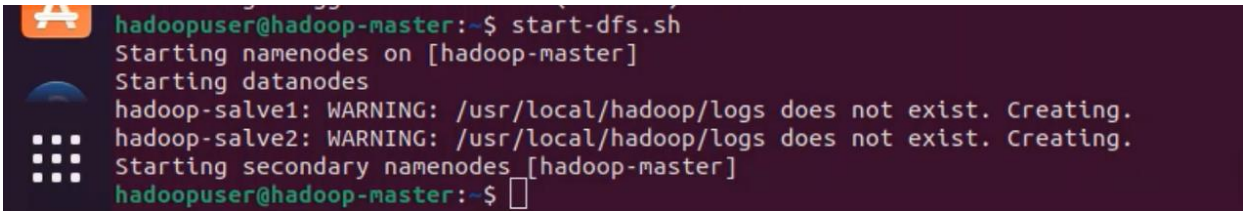
Bước 25: Gỡ cài đặt pdsh

$ sudo apt-get remove pdsh



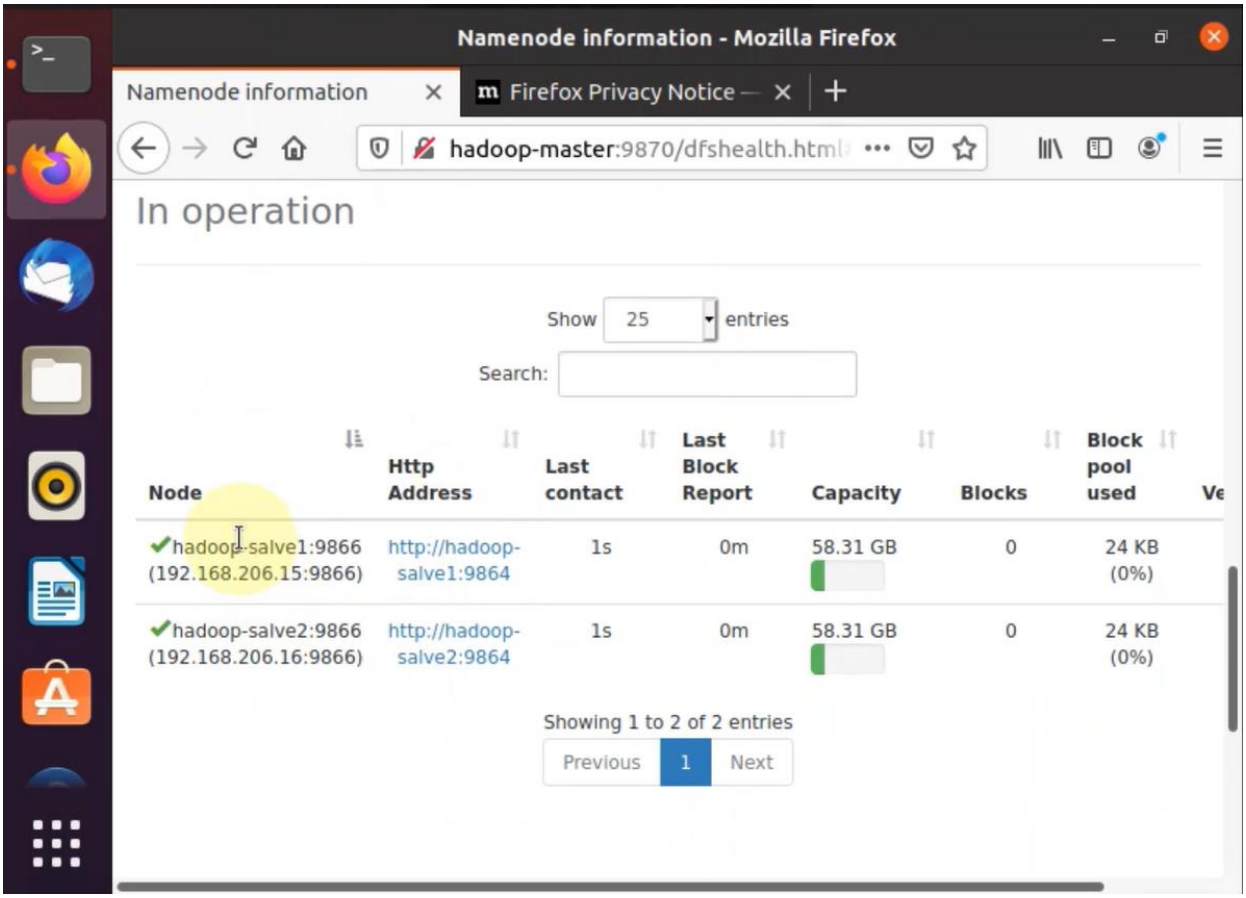
Bước 26: chạy file HDFS

$ start-dfs.sh



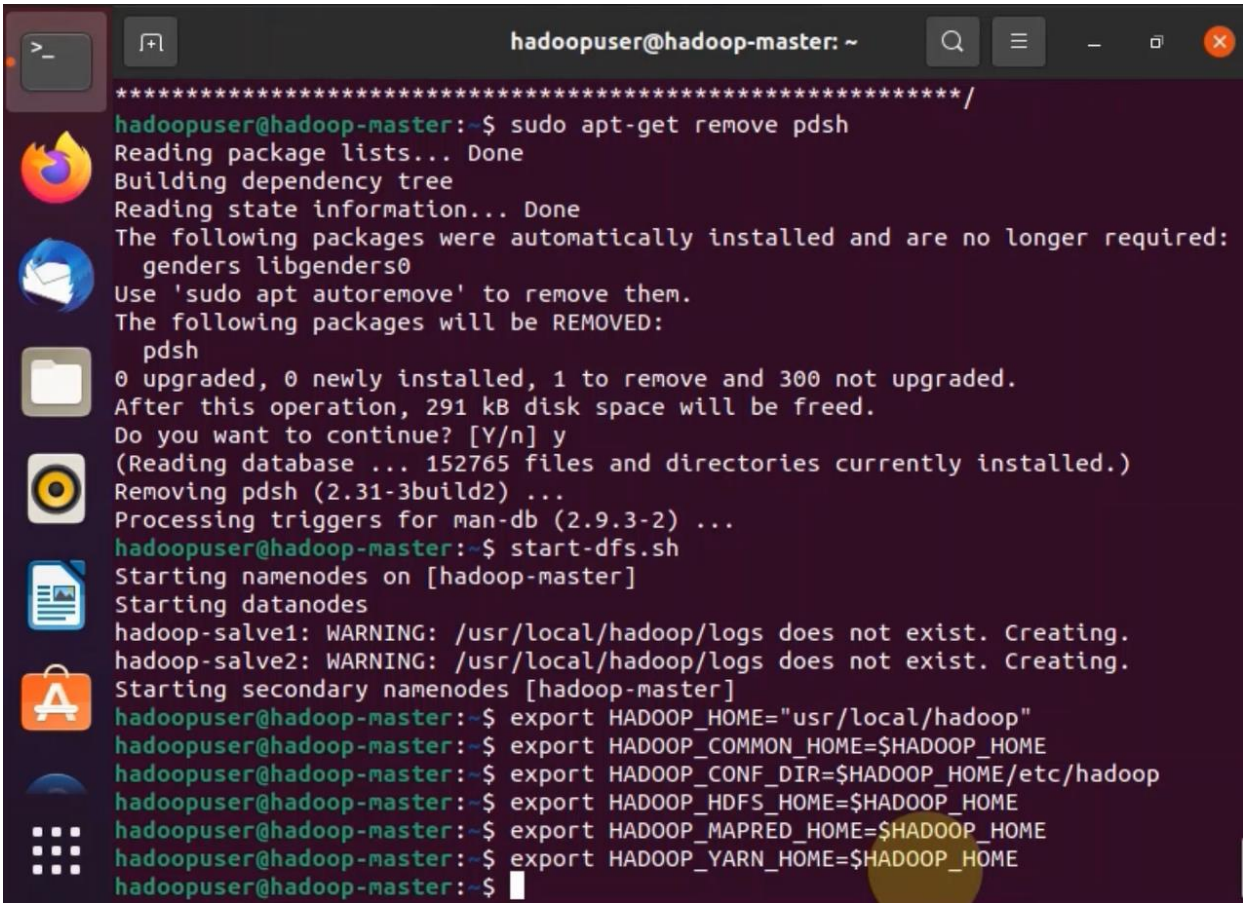
Bước 27: kiểm tra kết nối

Truy cập địa chỉ hadoop-master:9870



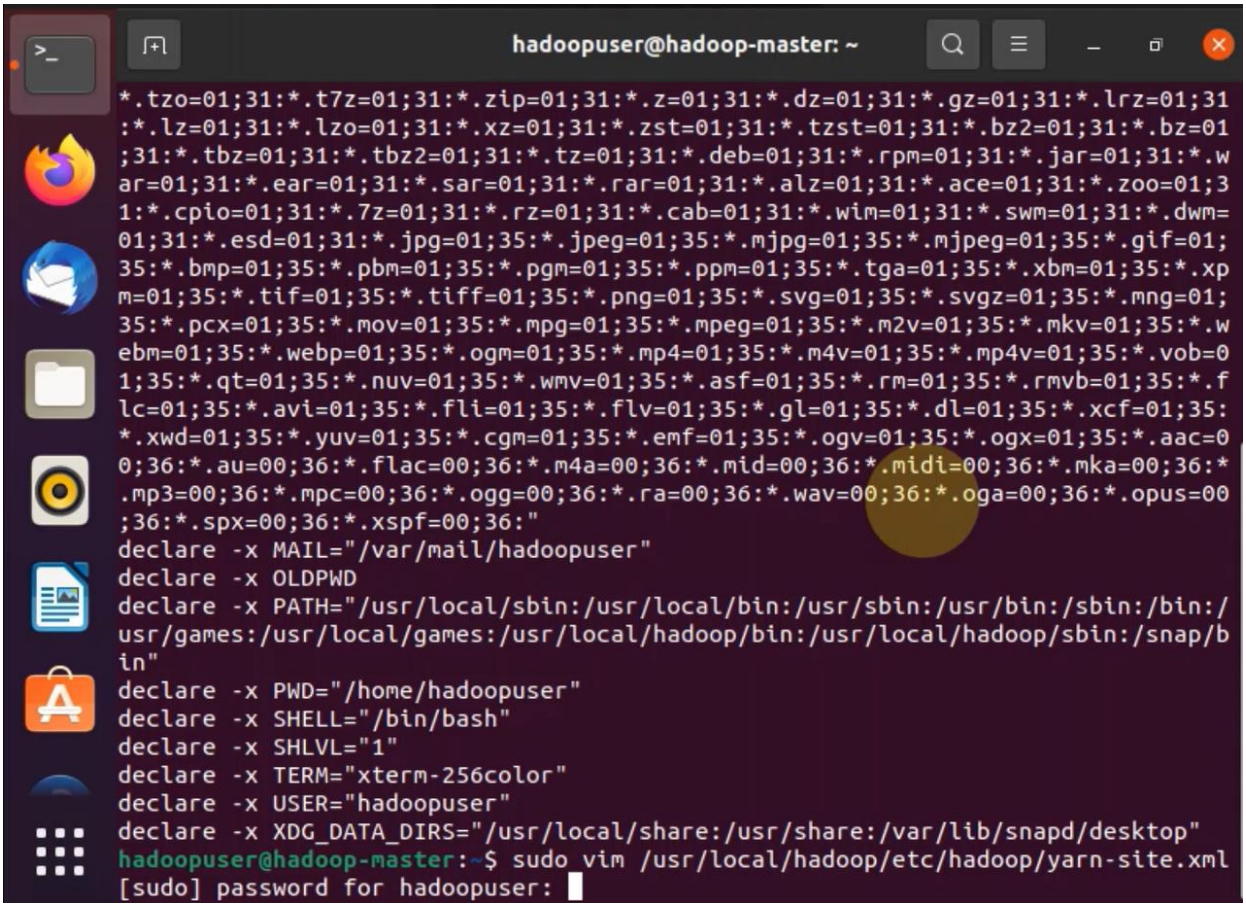
Bước 28: Cấu hình file YARN

```
$ export HADOOP_HOME="/usr/local/hadoop"
$ export HADOOP_COMMON_HOME=$HADOOP_HOME
$ export HADOOP_CONF_DIR=$HADOOP_HOME/etc/hadoop
$ export HADOOP_HDFS_HOME=$HADOOP_HOME
$ export HADOOP_MAPRED_HOME=$HADOOP_HOME
$ export HADOOP_YARN_HOME=$HADOOP_HOME
```



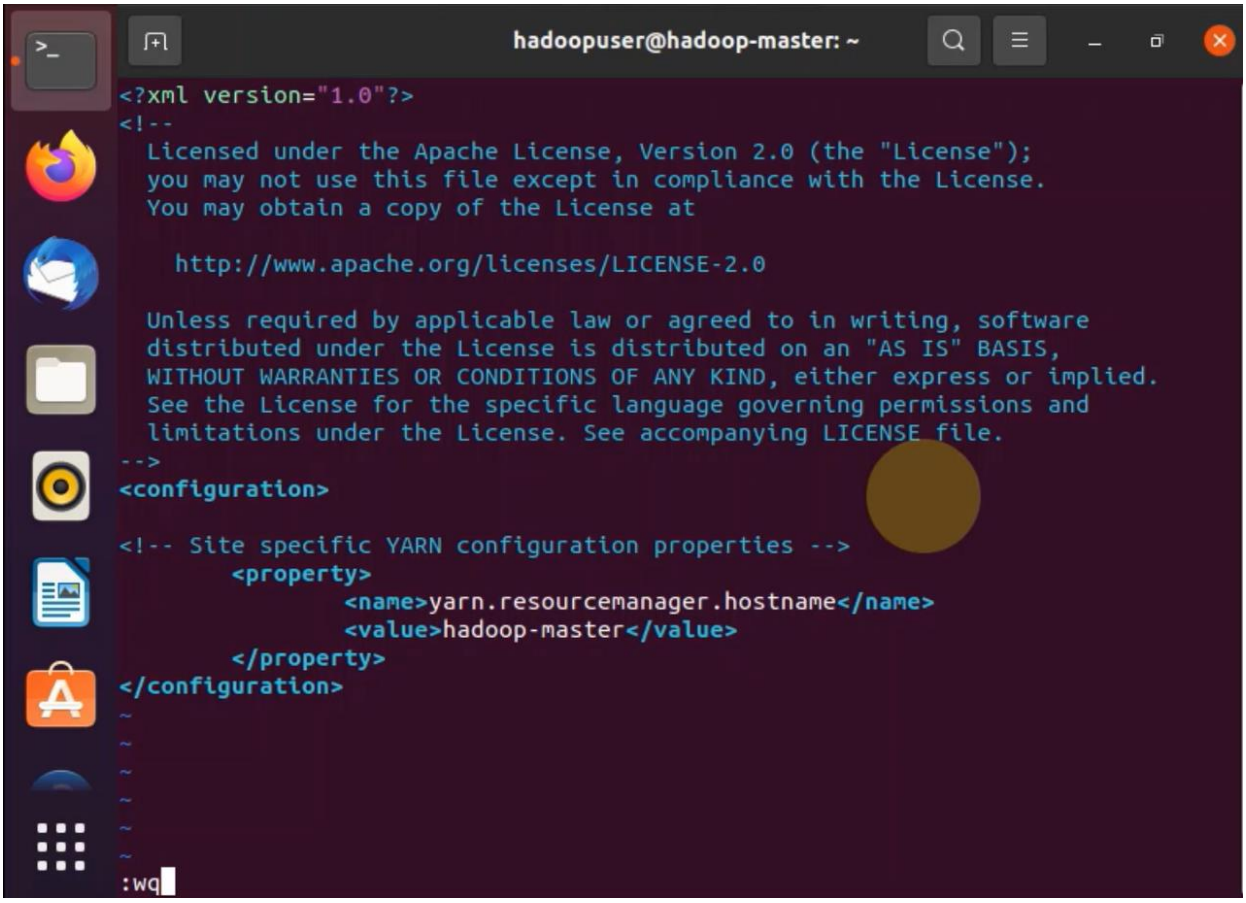
Bước 29: Mở file yarn-site.xml

$ sudo vim /usr/local/hadoop/etc/hadoop/yarn-site.xml

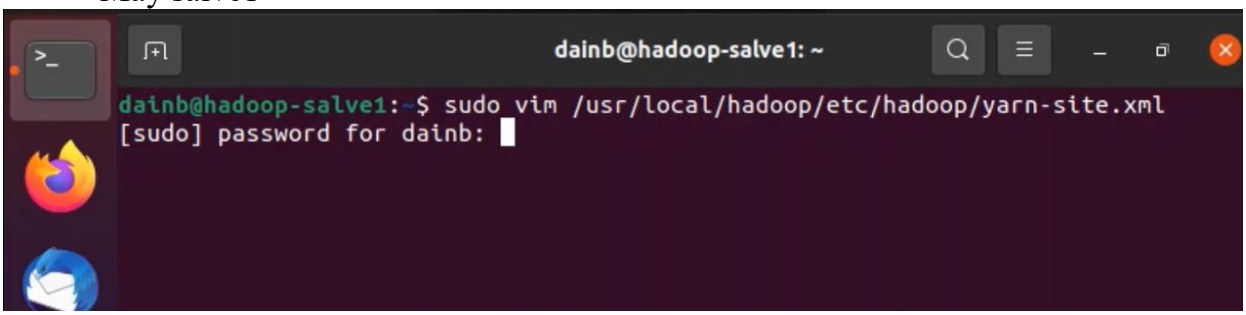


thêm

```
<property>
        <name>yarn.resourcemanager.hostname</name>
        <value>hadoop-master</value>
</property>
```

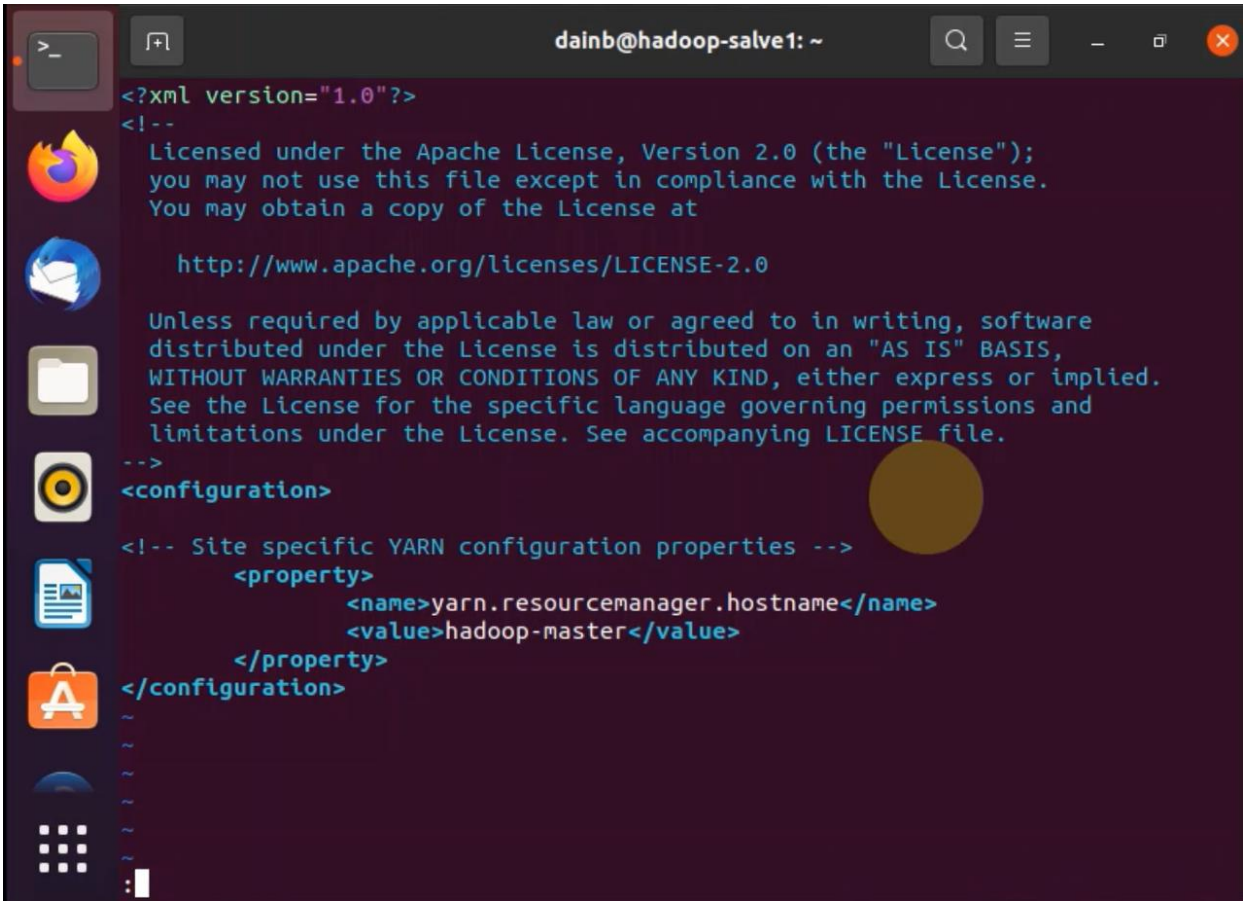


Máy salve1

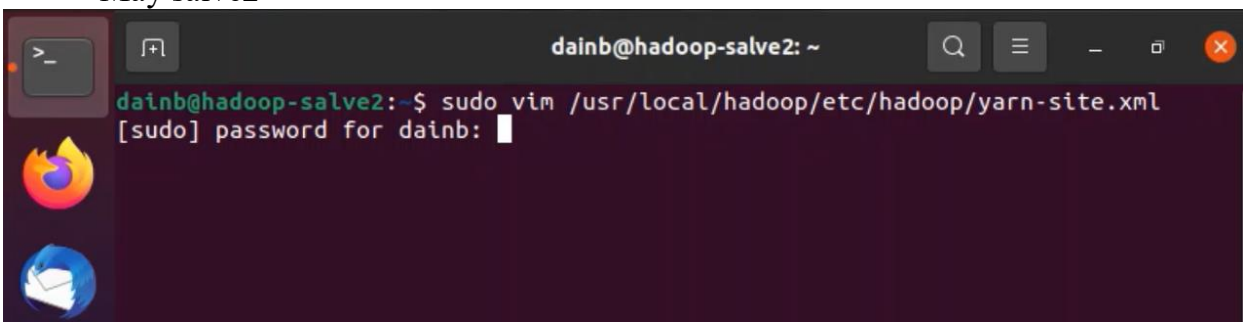


thêm

```
<property>
        <name>yarn.resourcemanager.hostname</name>
        <value>hadoop-master</value>
</property>
```

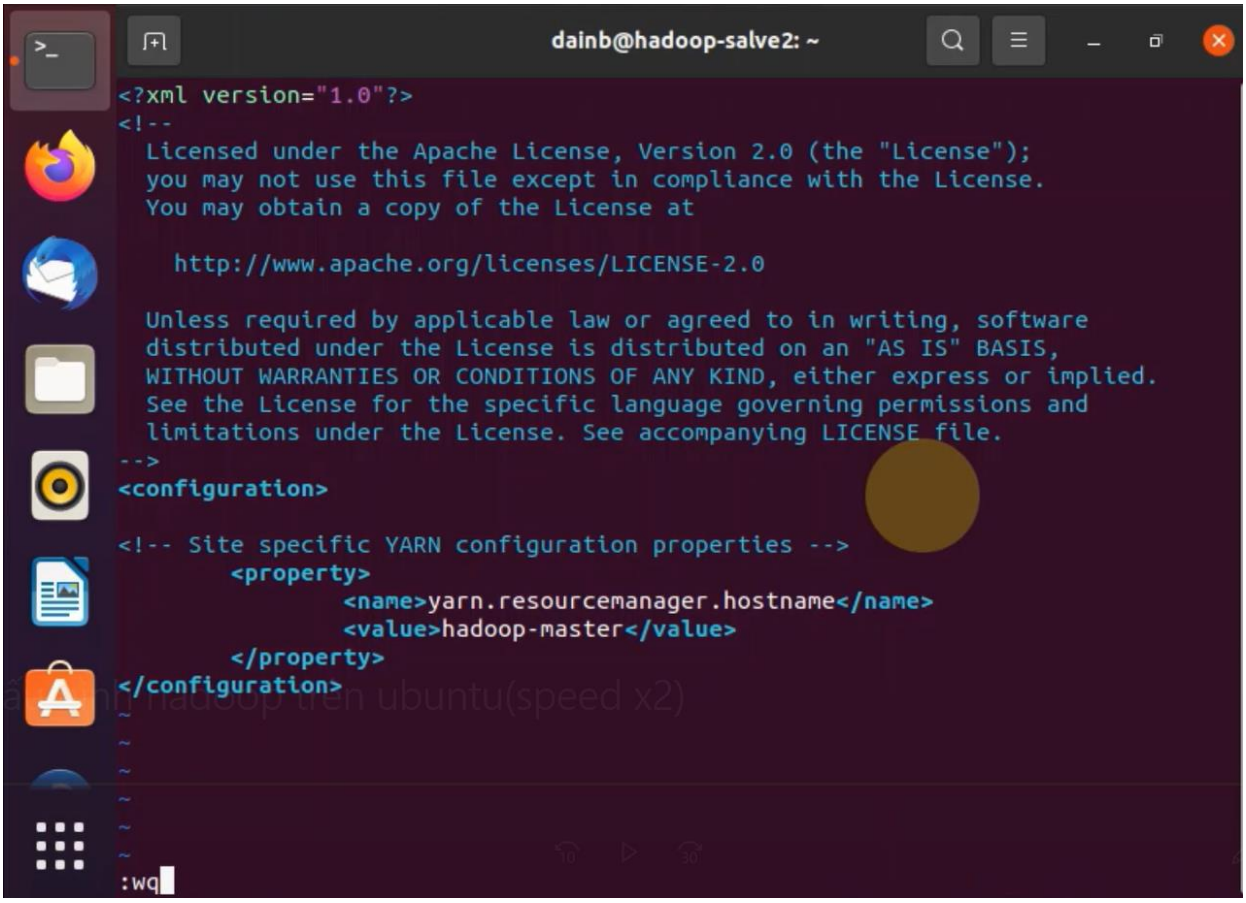


Máy salve2



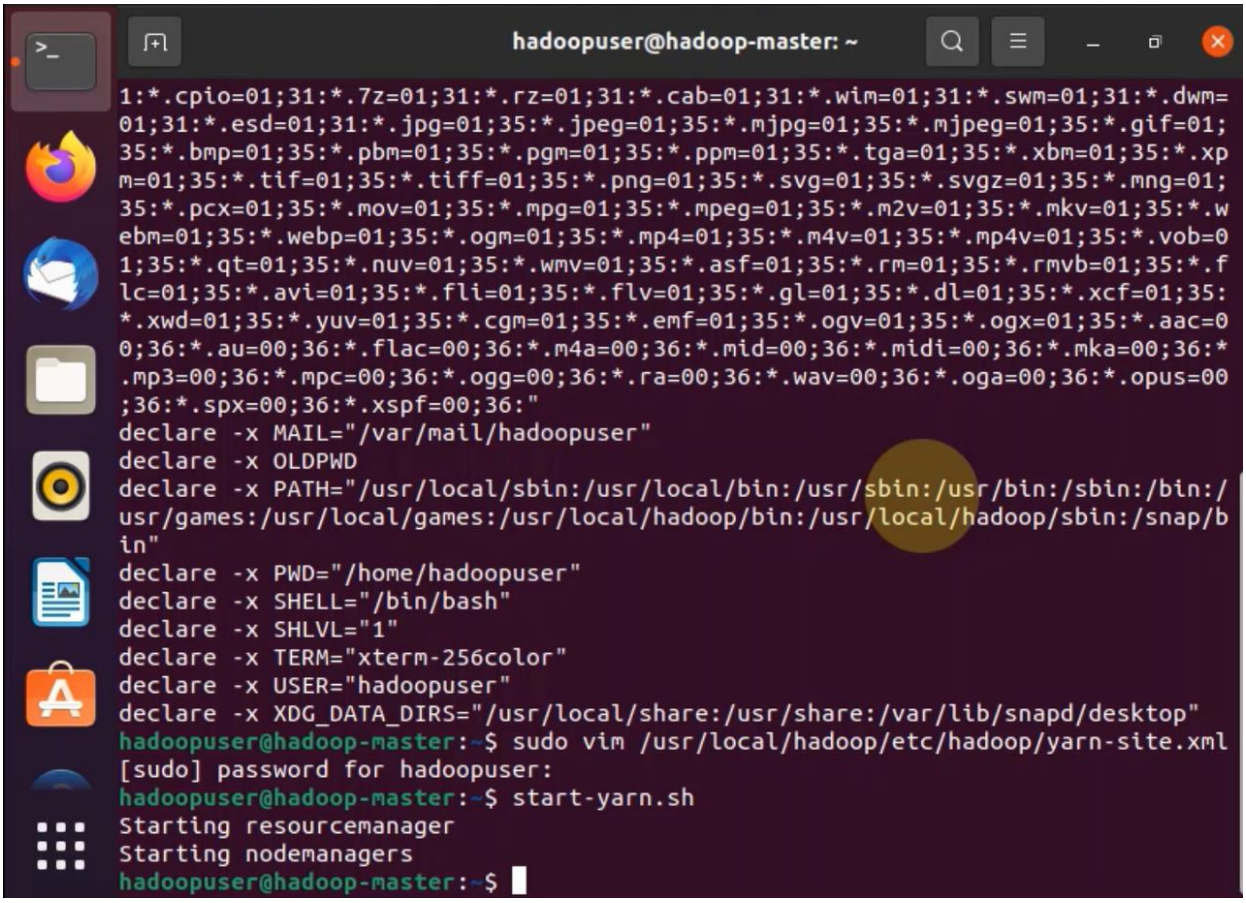
thêm

```
<property>
        <name>yarn.resourcemanager.hostname</name>
        <value>hadoop-master</value>
</property>
```



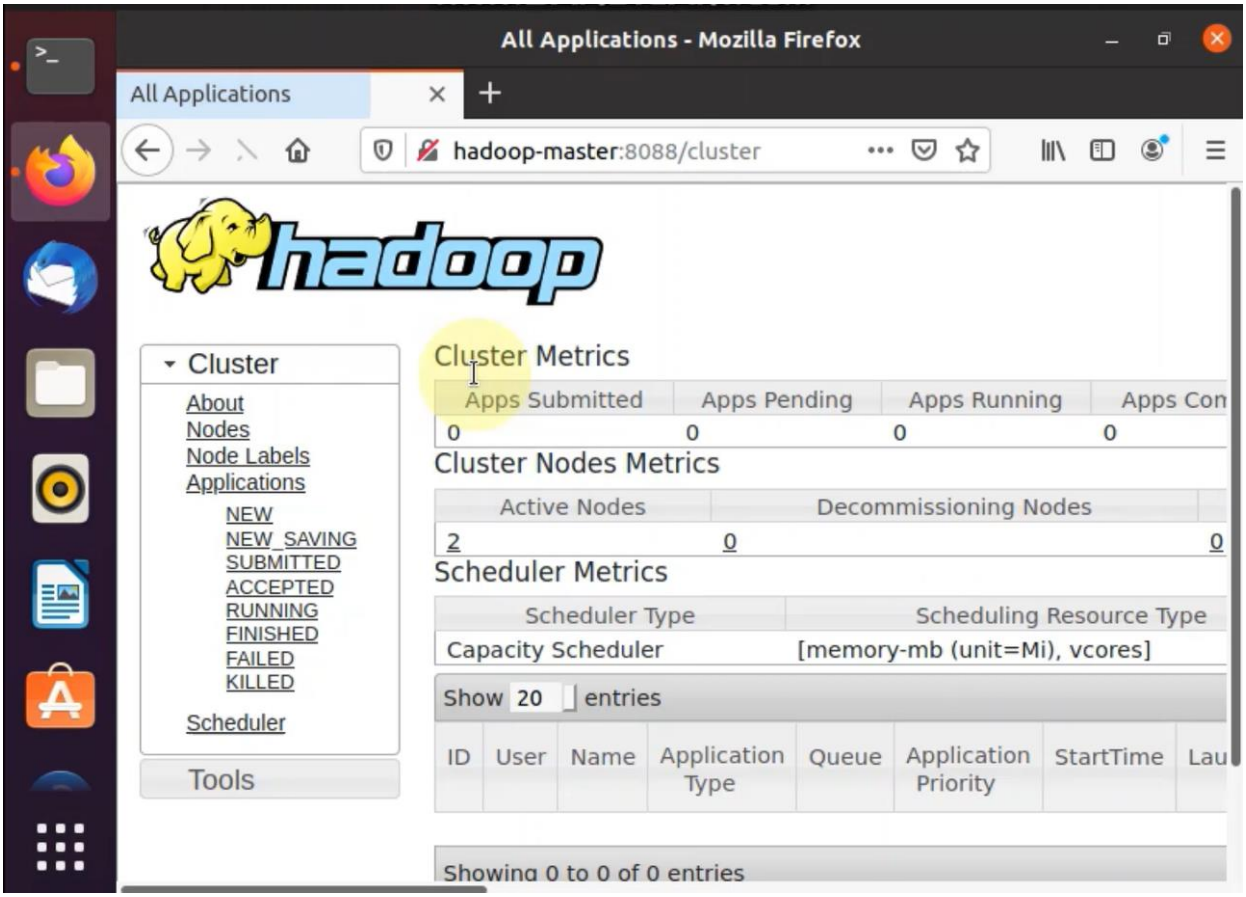
Bước 30: Chạy file yarn

$ start-yarn.sh



Bước 31: Kiểm tra

Mở trình duyệt -> http://hadoop-master:8088/cluster

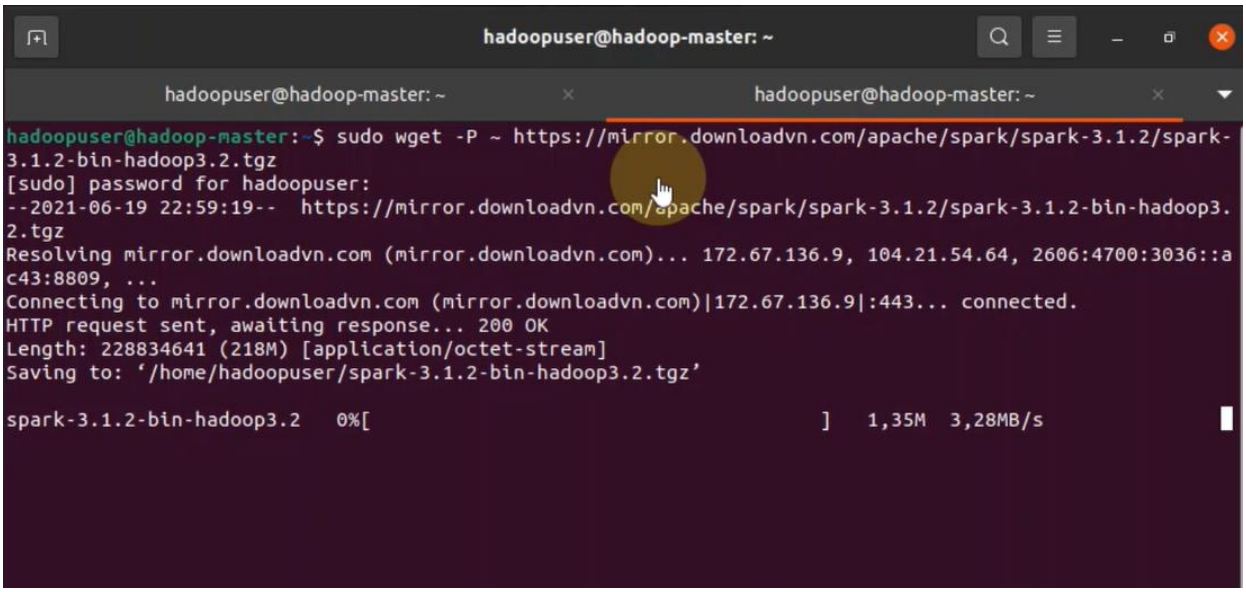

All Applications - Mozilla Firefox
All Applications
hadoop-master:8088/cluster
hadoop
Cluster
About
Nodes
Node Labels
Applications
NEW
NEW_SAVING
SUBMITTED
ACCEPTED
RUNNING
FINISHED
FAILED
KILLED
Scheduler
Tools
Cluster Metrics
Apps Submitted
Apps Pending
Apps Running
Apps Con
0
0
0
0
Cluster Nodes Metrics
Active Nodes
Decommissioning Nodes
2
0
0
Scheduler Metrics
Scheduler Type
Scheduling Resource Type
Capacity Scheduler
[memory-mb (unit=Mi), vcores]
Show 20 entries
ID
User
Name
Application Type
Queue
Application Priority
StartTime
Lau
Showing 0 to 0 of 0 entries

3. Cài đặt Spark

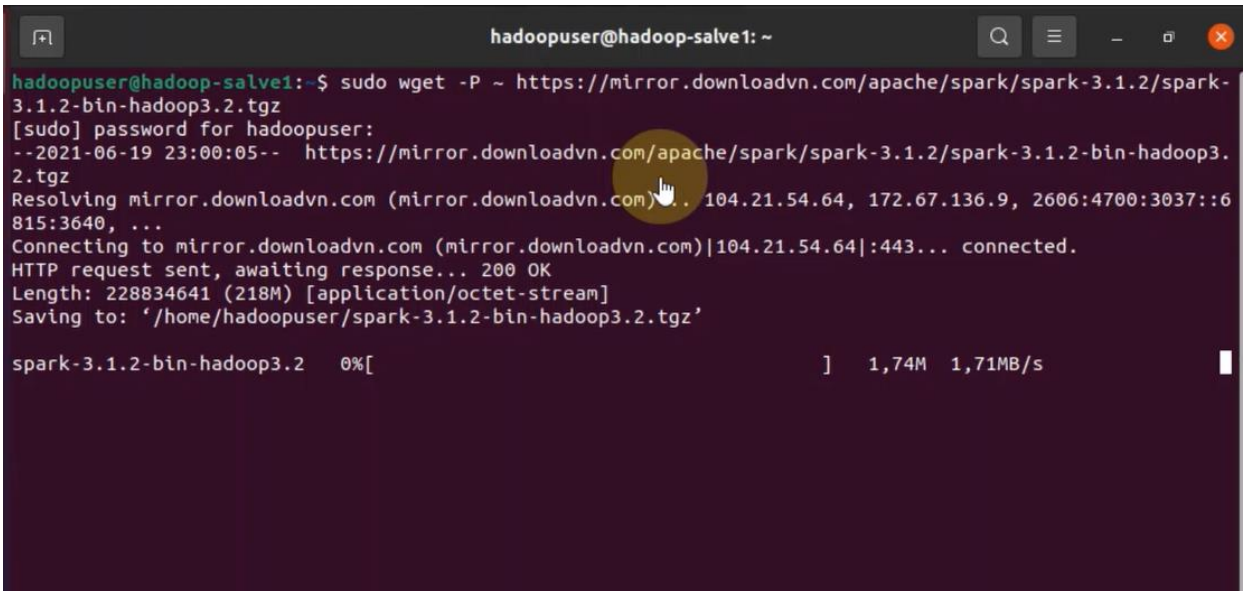
Bước 1: tải file nén spark

$ sudo wget -P ~ https://mirror.downloadvn.com/apache/spark/spark-3.1.2/spark-3.1.2-bin-hadoop3.2.tgz

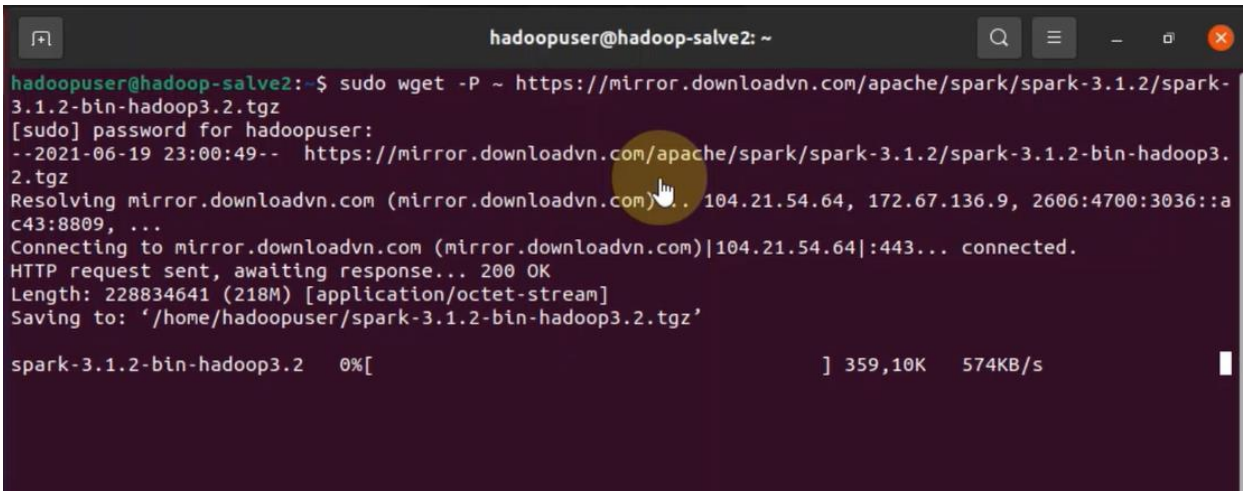
máy master



máy salve1



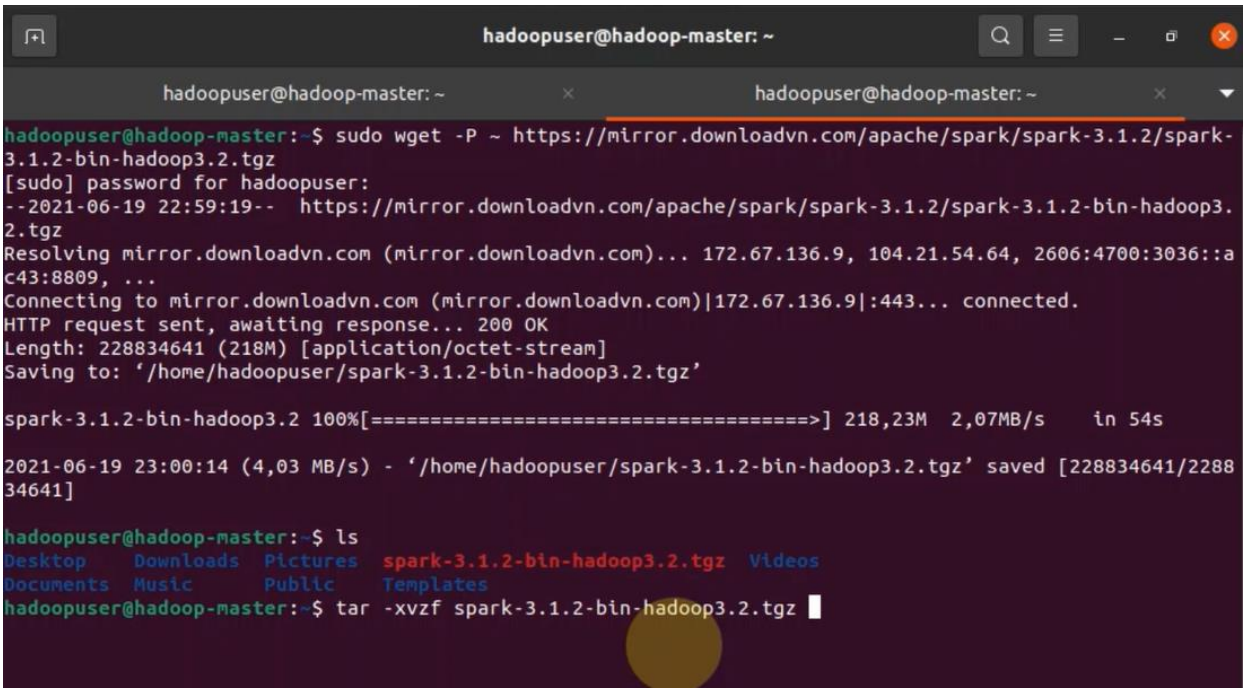
máy salve2



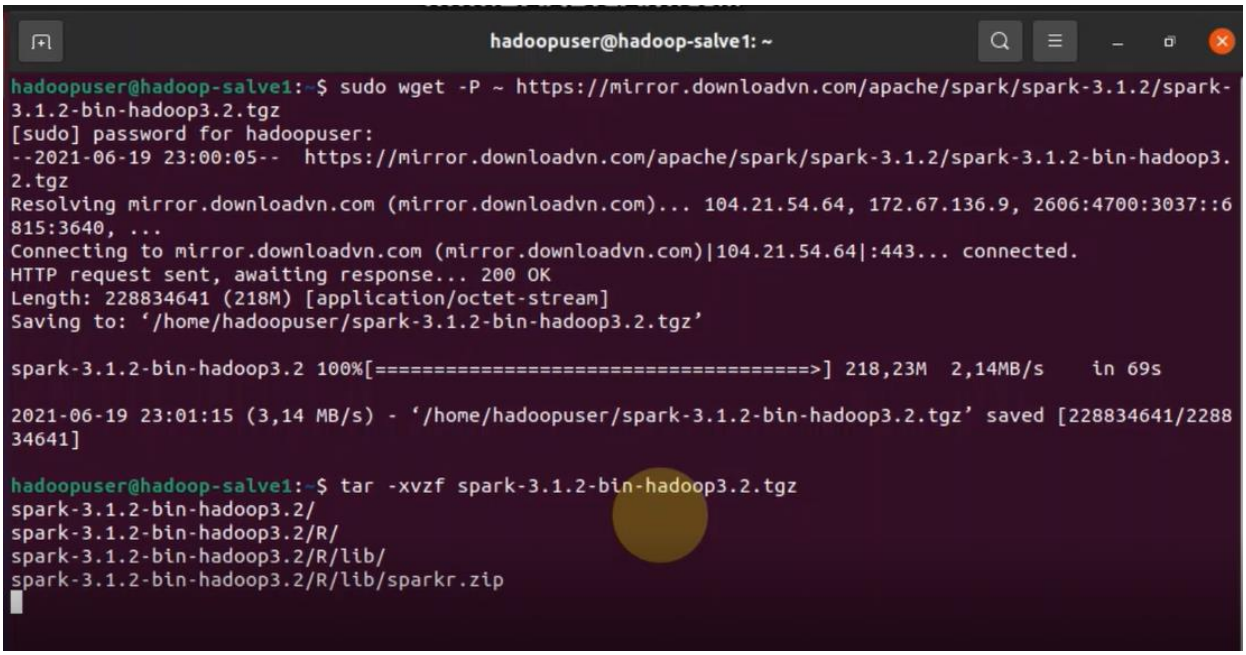
Bước 2: giải nén file spark-3.1.2-bin-hadoop3.2.tgz

$ tar -xvzf spark-3.1.2-bin-hadoop3.2.tgz

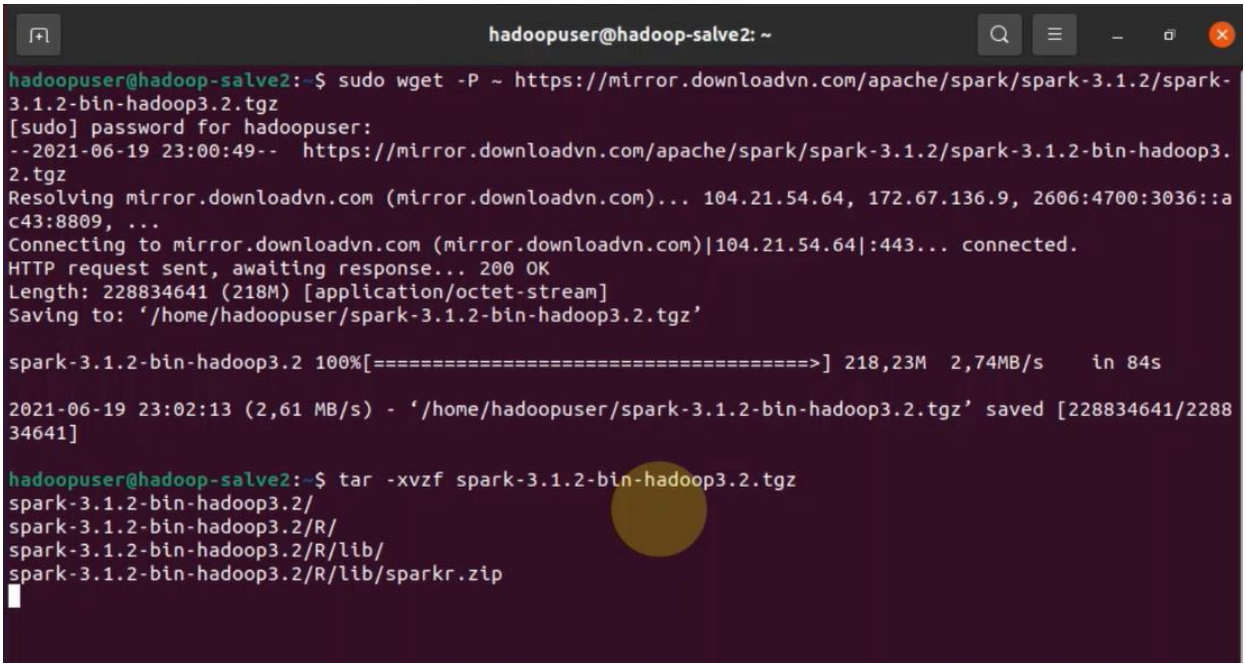
máy master



máy salve1

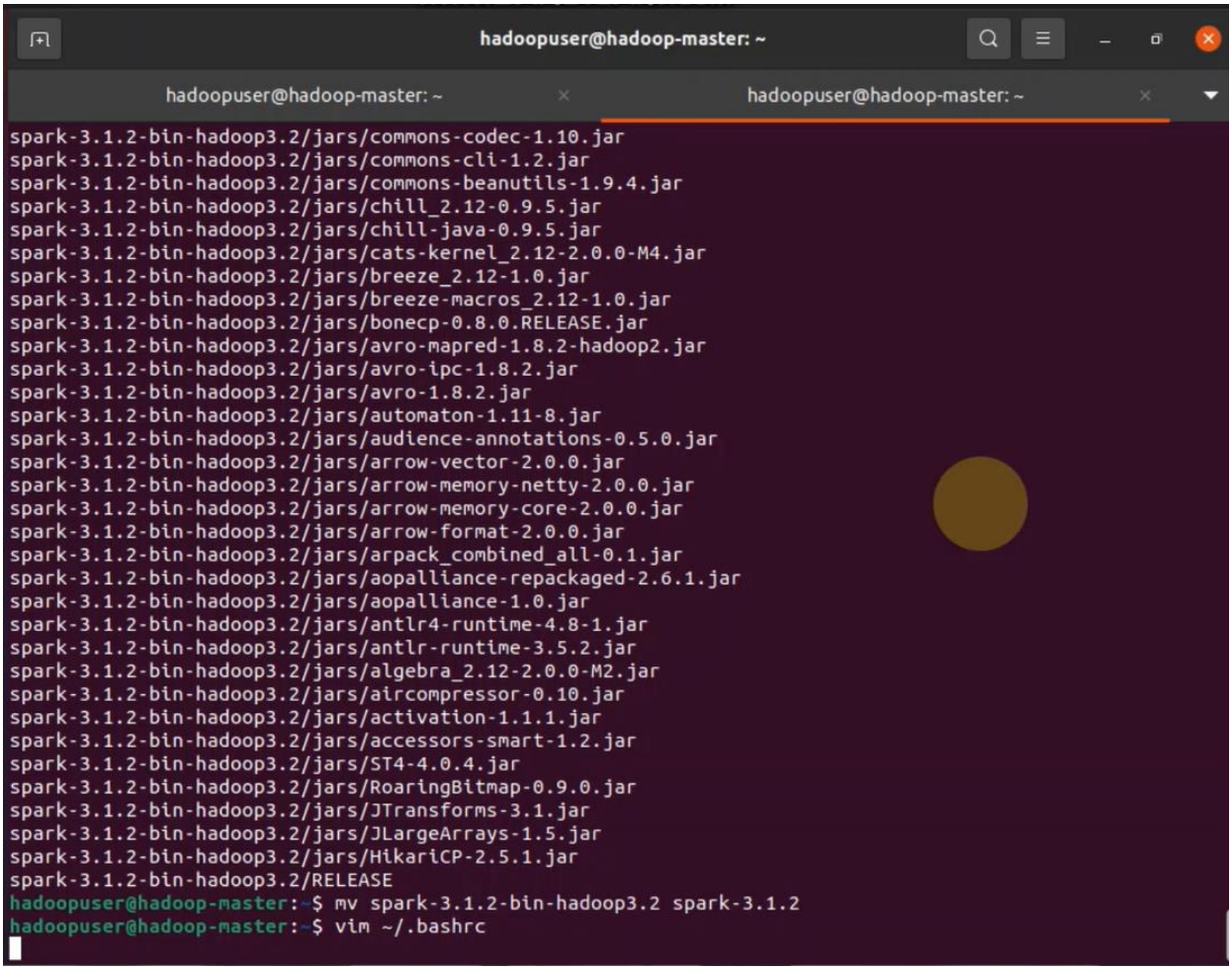


máy salve2



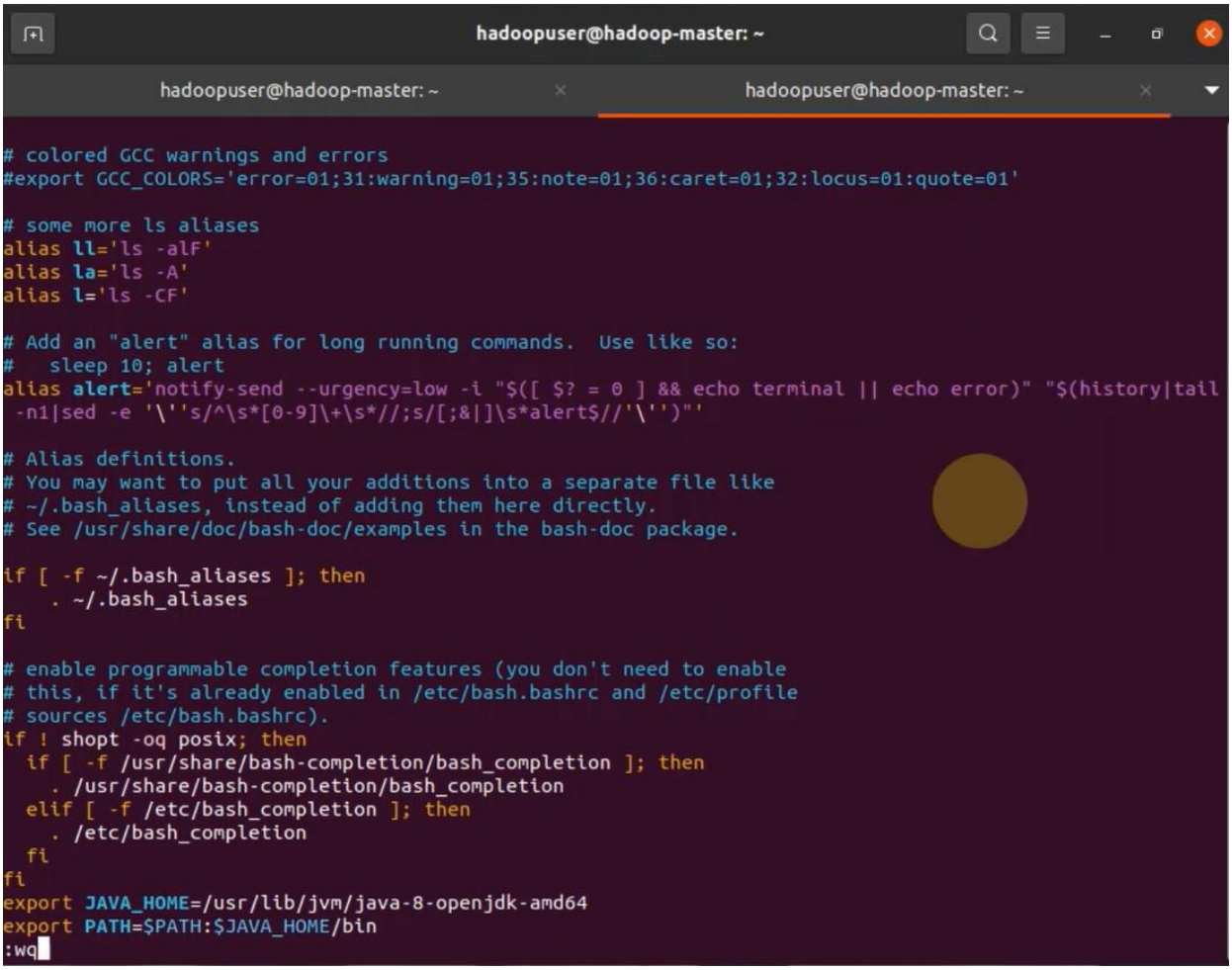
Bước 3: cấu hình file bashrc trên máy master

$ vim ~/.bashrc



thêm vào cuối file

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin
```

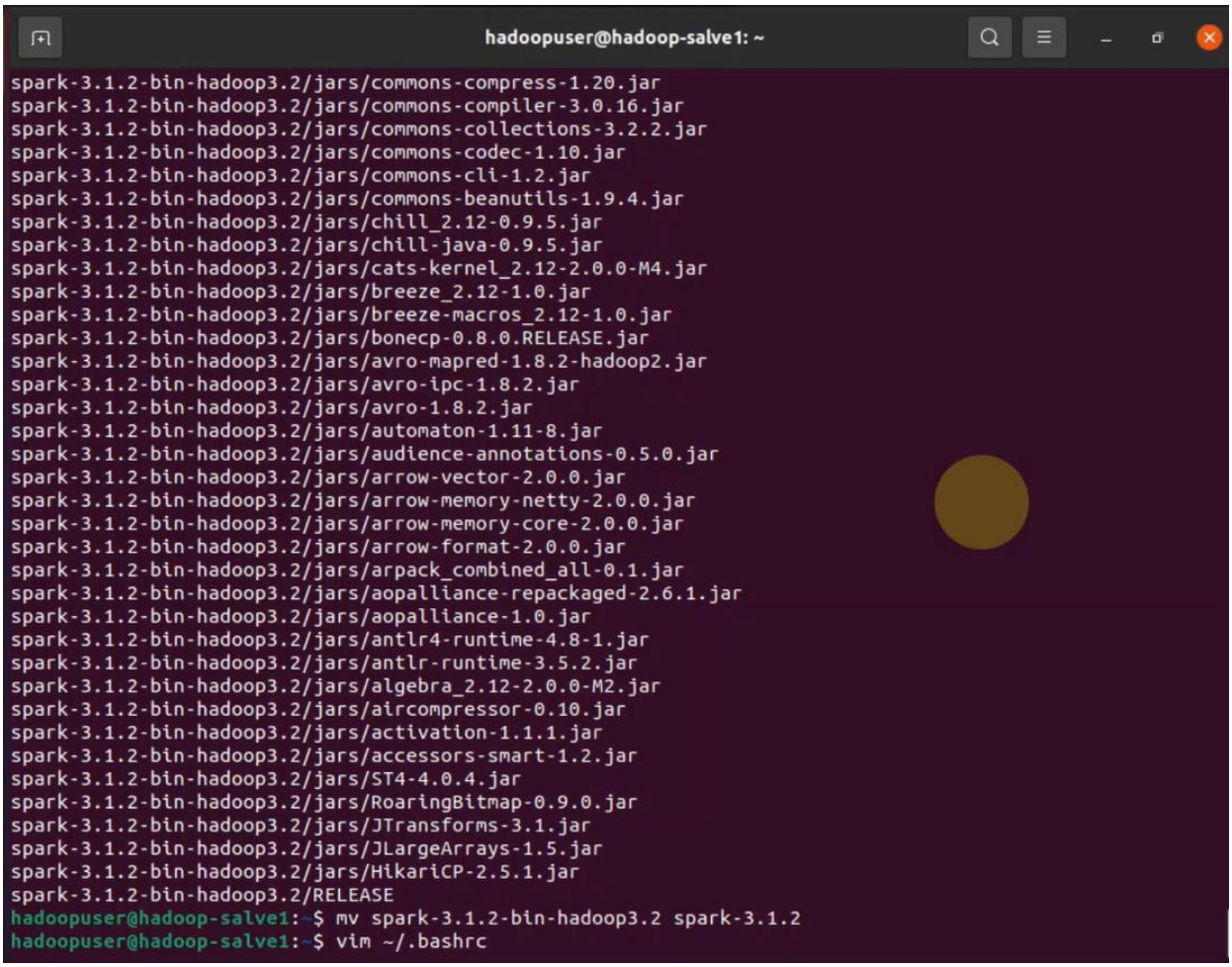


$ source ~/.bashrc



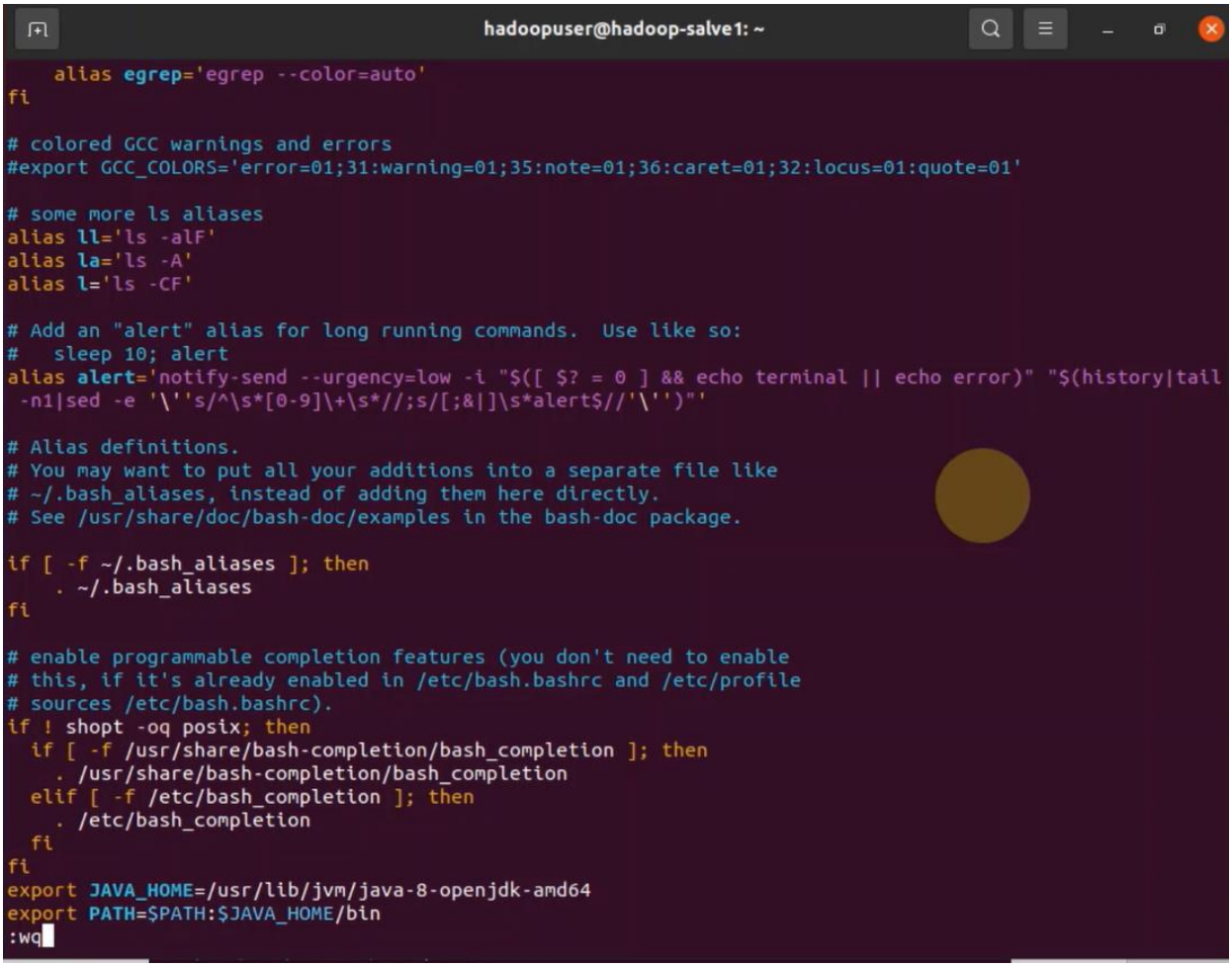
Bước 4: cấu hình file bashrc trên máy salve1

$ vim ~/.bashrc



thêm vào cuối file

```
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin
```

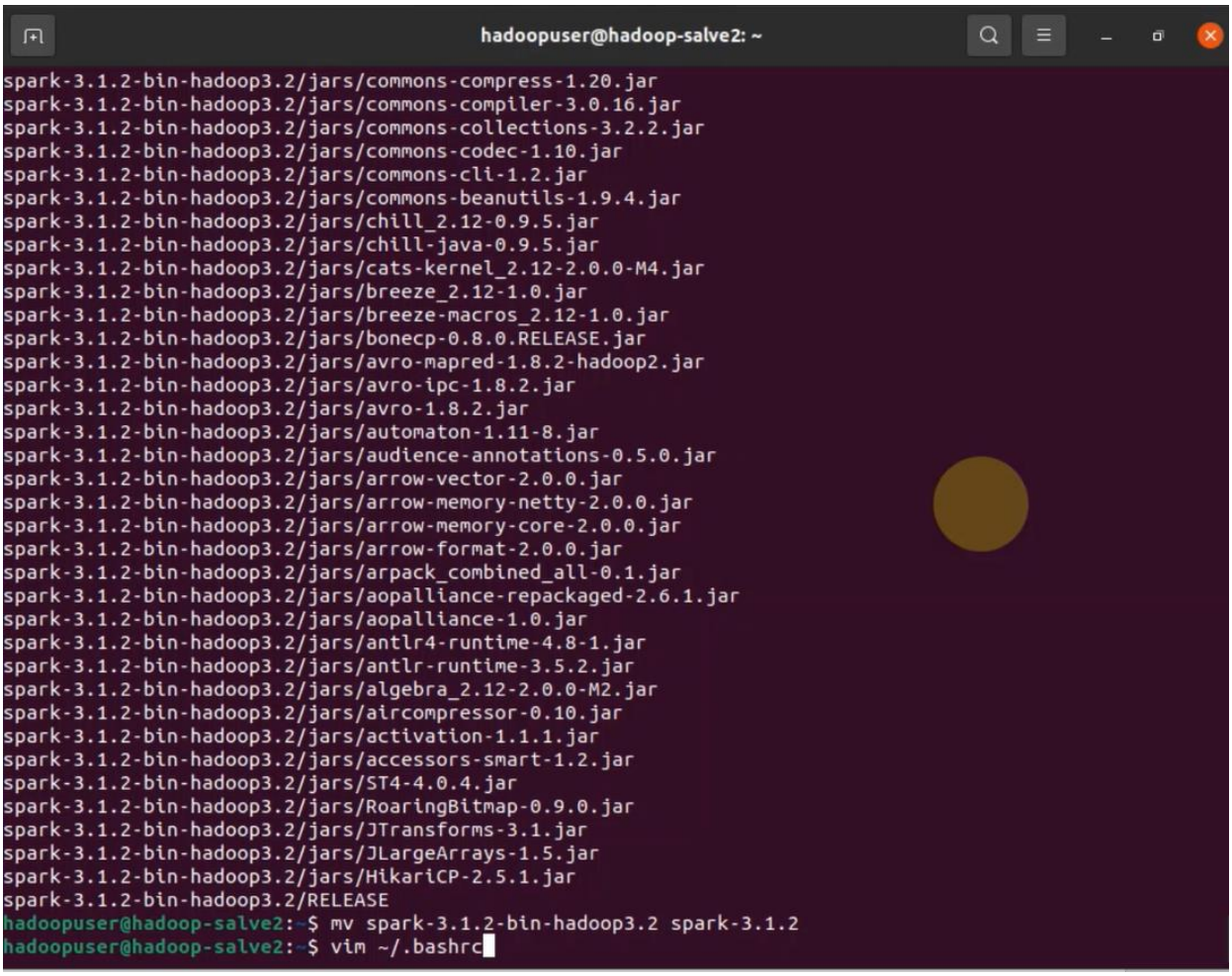


$ source ~/.bashrc



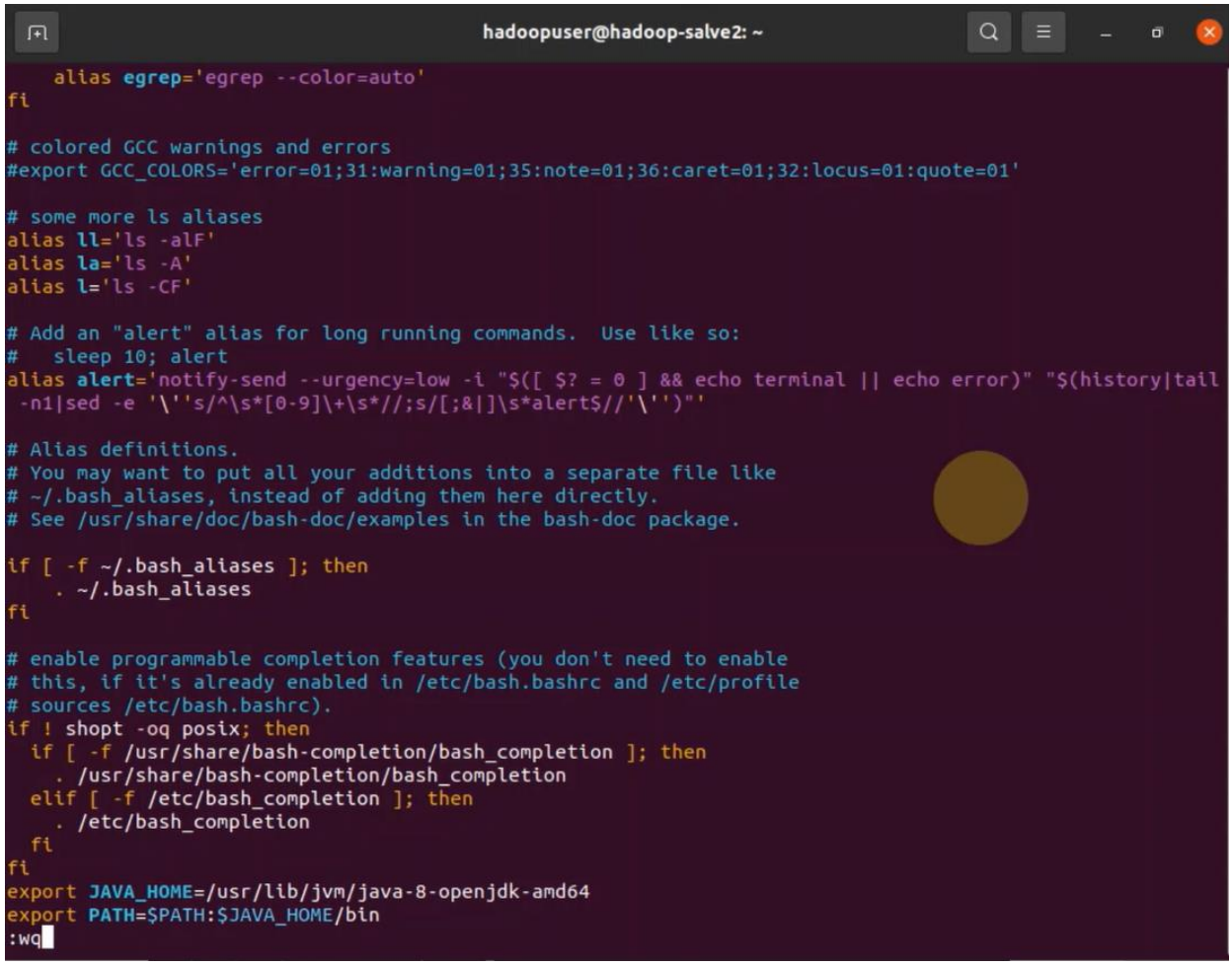
Bước 5: cấu hình file bashrc trên máy salve2

$ vim ~/.bashrc



thêm vào cuối file

export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export PATH=$PATH:$JAVA_HOME/bin

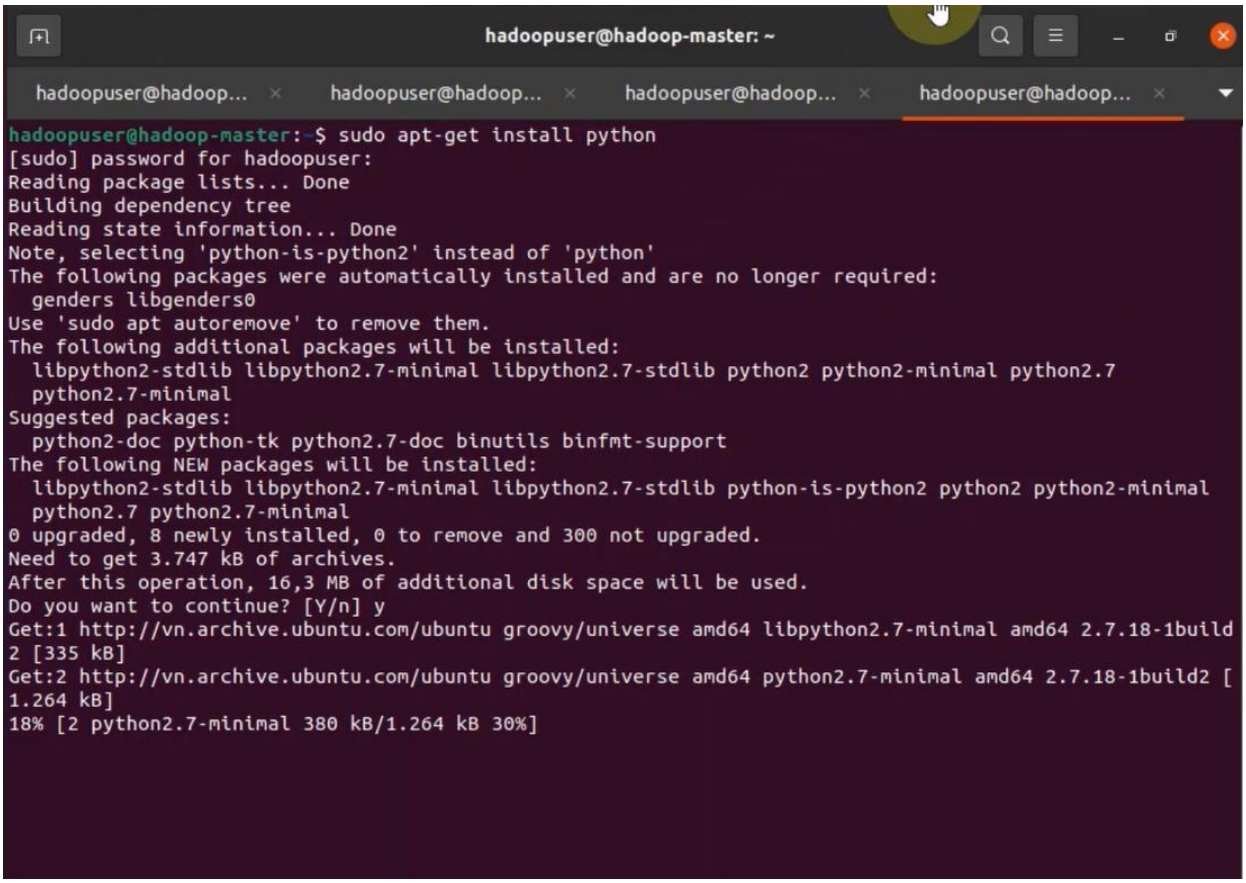


$ source ~/.bashrc



Bước 6: cài đặt python

$ sudo apt-get install python



kiểm tra phiên bản

$ python --version

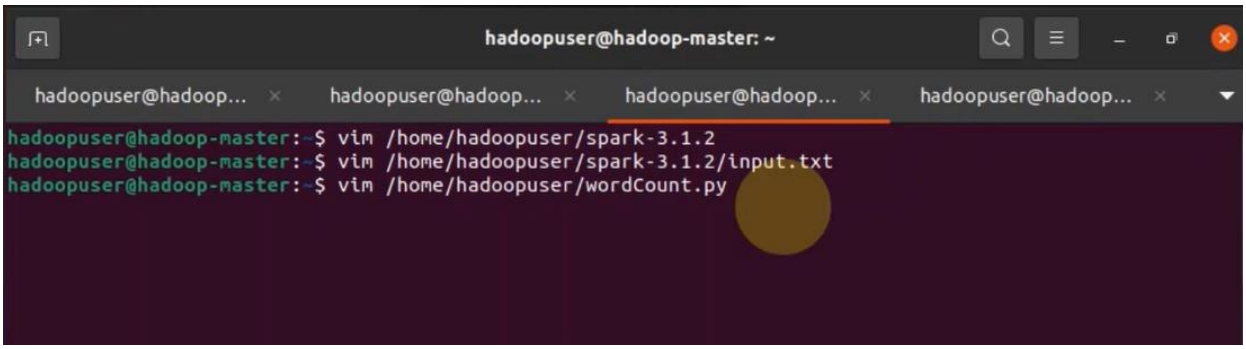


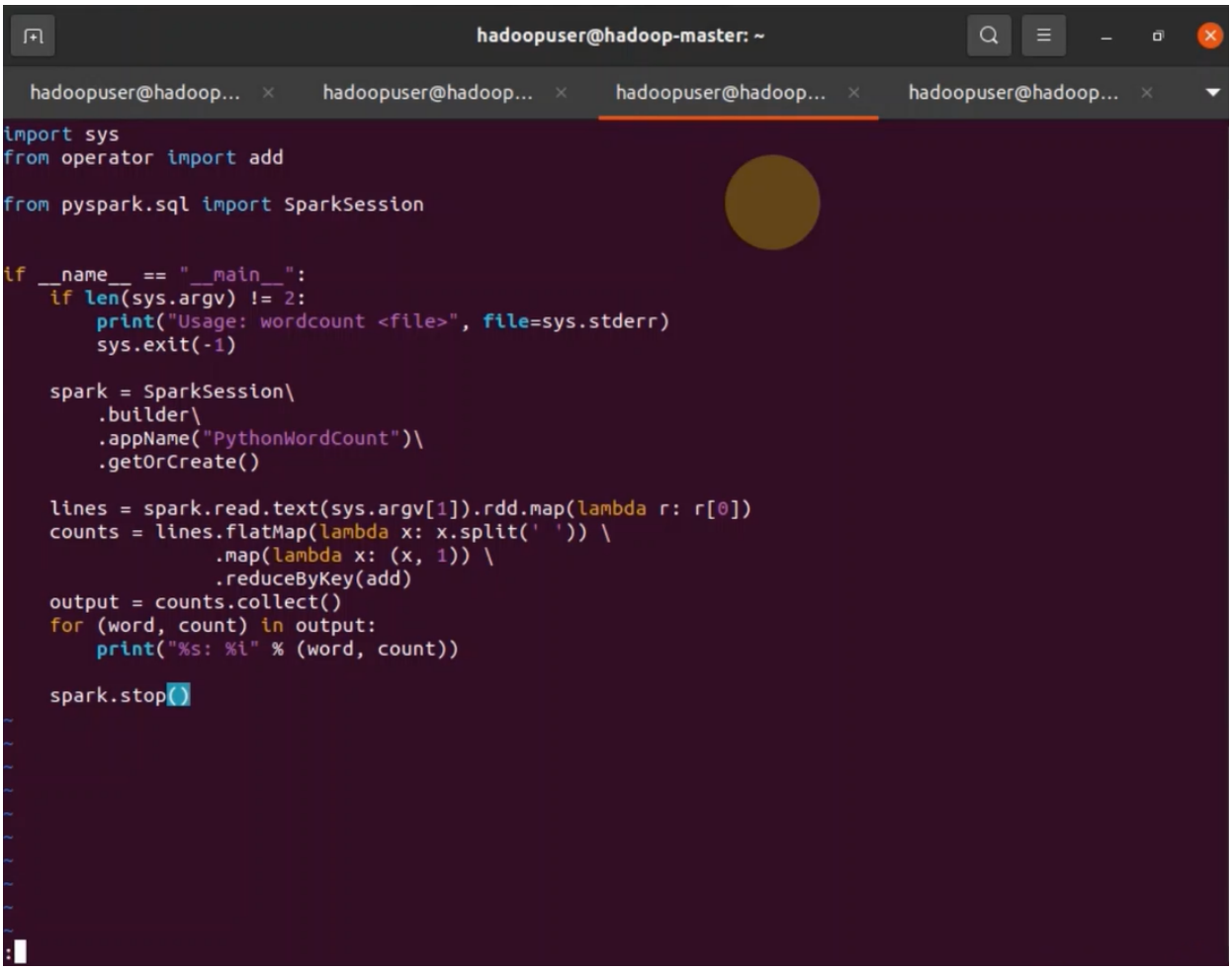
4. Demo bài toán wordcount

Bước 1: tạo file input.txt và file code wordCount.py

$ vim /home/hadoopuser/spark-3.1.2/input.txt

$ vim /home/hadoopuser/wordCount.py

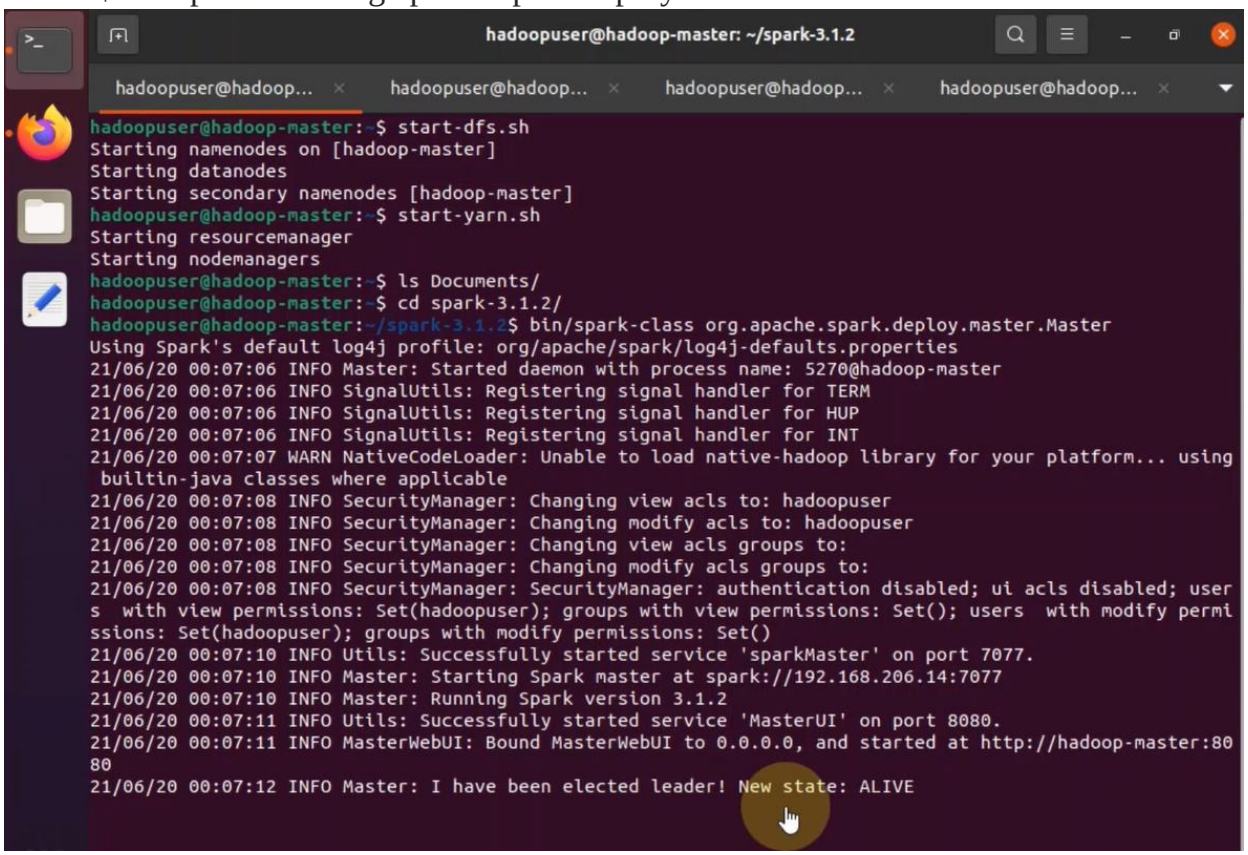




Bước 2: chạy spark trên máy master

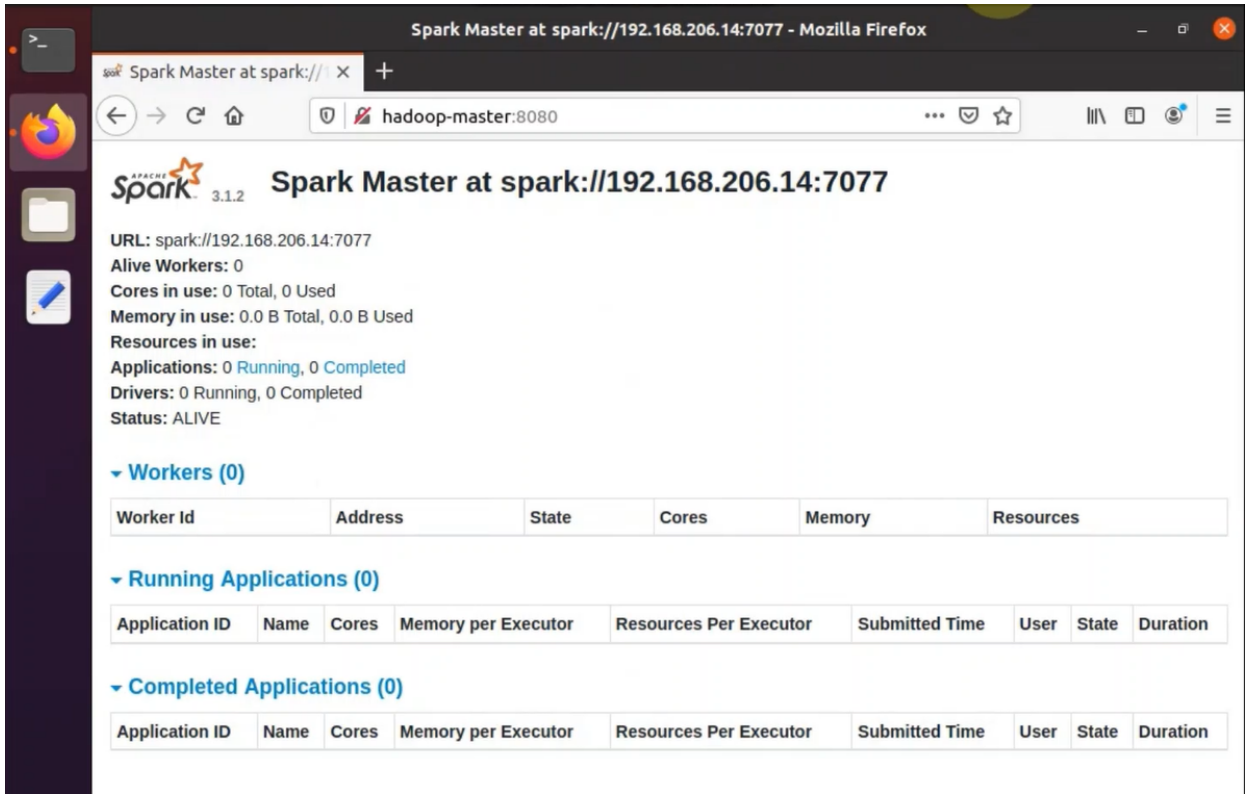
$ cd spark-3.1.2

$ bin/spark-class org.apache.spark.deploy.master.Master



Bước 3: truy cập web để kiểm tra

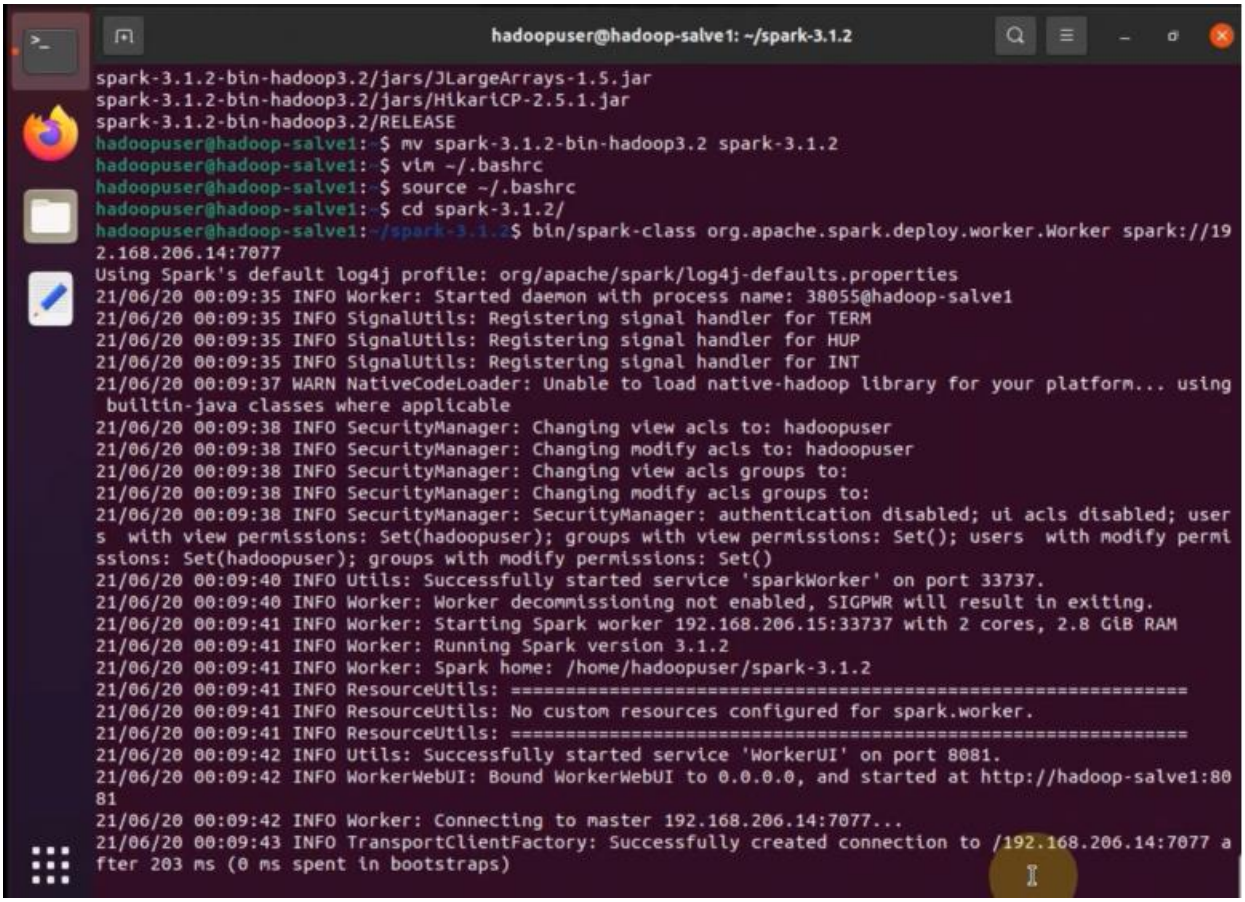
truy cập địa chỉ hadoop-master:8088


Spark Master at spark://192.168.206.14:7077 - Mozilla Firefox
Spark Master at spark://
hadoop-master:8080
Spark 3.1.2
Spark Master at spark://192.168.206.14:7077
URL: spark://192.168.206.14:7077
Alive Workers: 0
Cores in use: 0 Total, 0 Used
Memory in use: 0.0 B Total, 0.0 B Used
Resources in use:
Applications: 0 Running, 0 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE
Workers (0)
Worker Id
Address
State
Cores
Memory
Resources
Running Applications (0)
Application ID
Name
Cores
Memory per Executor
Resources Per Executor
Submitted Time
User
State
Duration
Completed Applications (0)
Application ID
Name
Cores
Memory per Executor
Resources Per Executor
Submitted Time
User
State
Duration

Bước 4: chạy spark trên máy salve1

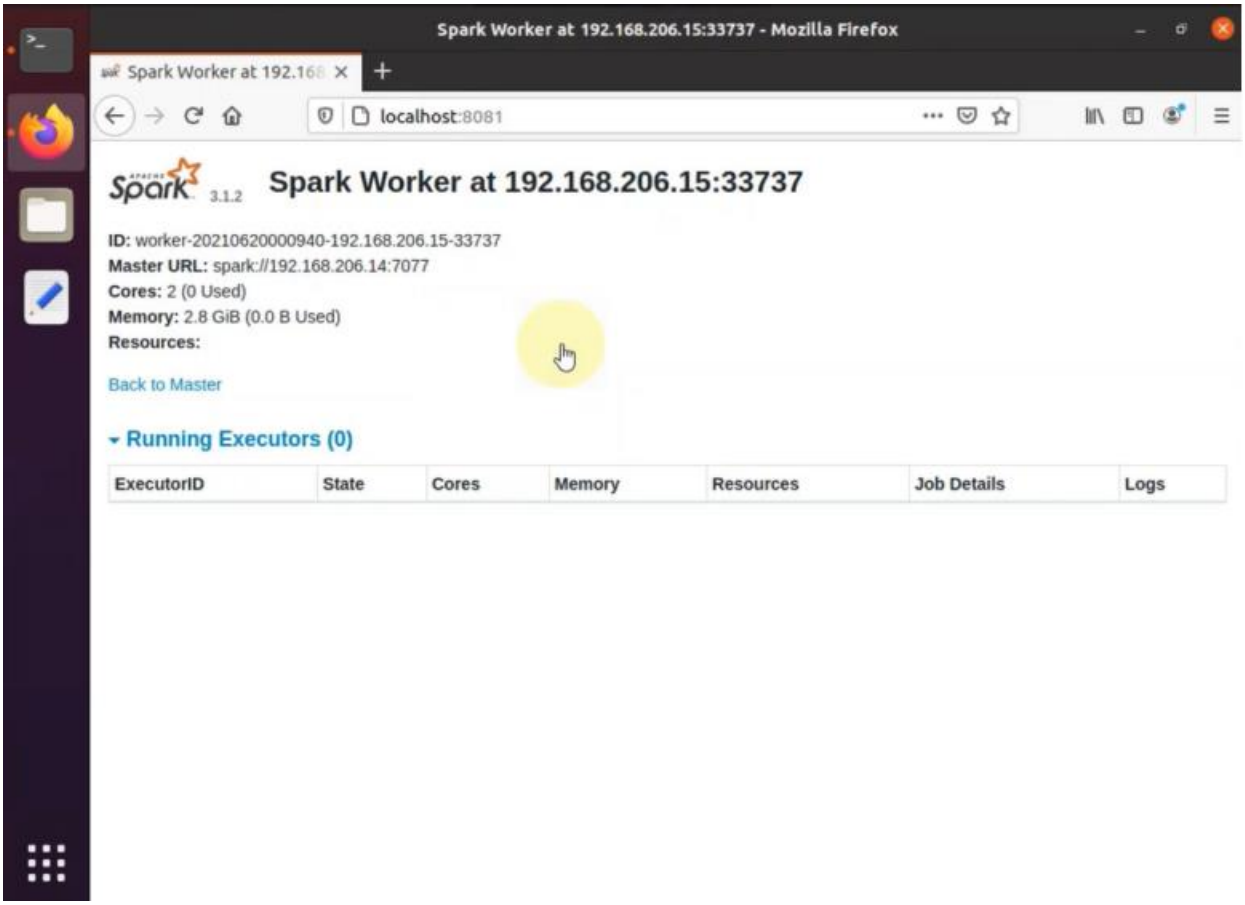
$ cd spark-3.1.2

$ bin/spark-class org.apache.spark.deploy.worker.Worker spark://192.168.206.14:7077 (truy cập địa chỉ hadoop-master:8088 trên máy master để có địa chỉ ip và cổng)

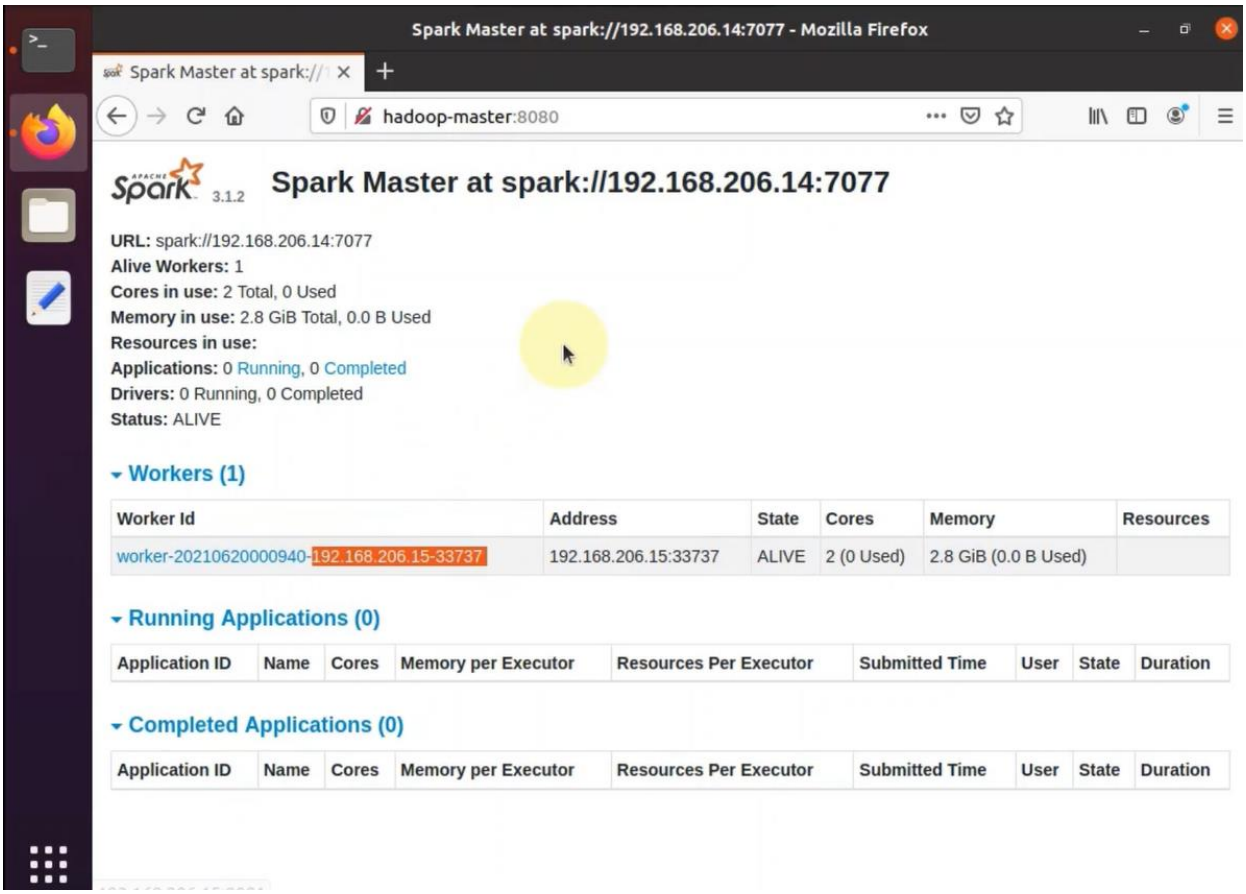


Bước 5: kiểm tra trên máy salve1

truy cập địa chỉ localhost:8081


Spark Worker at 192.168.206.15:33737 - Mozilla Firefox
Spark Worker at 192.168.206.15:33737
localhost:8081
Spark 3.1.2
Spark Worker at 192.168.206.15:33737
ID: worker-20210620000940-192.168.206.15-33737
Master URL: spark://192.168.206.14:7077
Cores: 2 (0 Used)
Memory: 2.8 GiB (0.0 B Used)
Resources:
Back to Master
Running Executors (0)
ExecutorID
State
Cores
Memory
Resources
Job Details
Logs

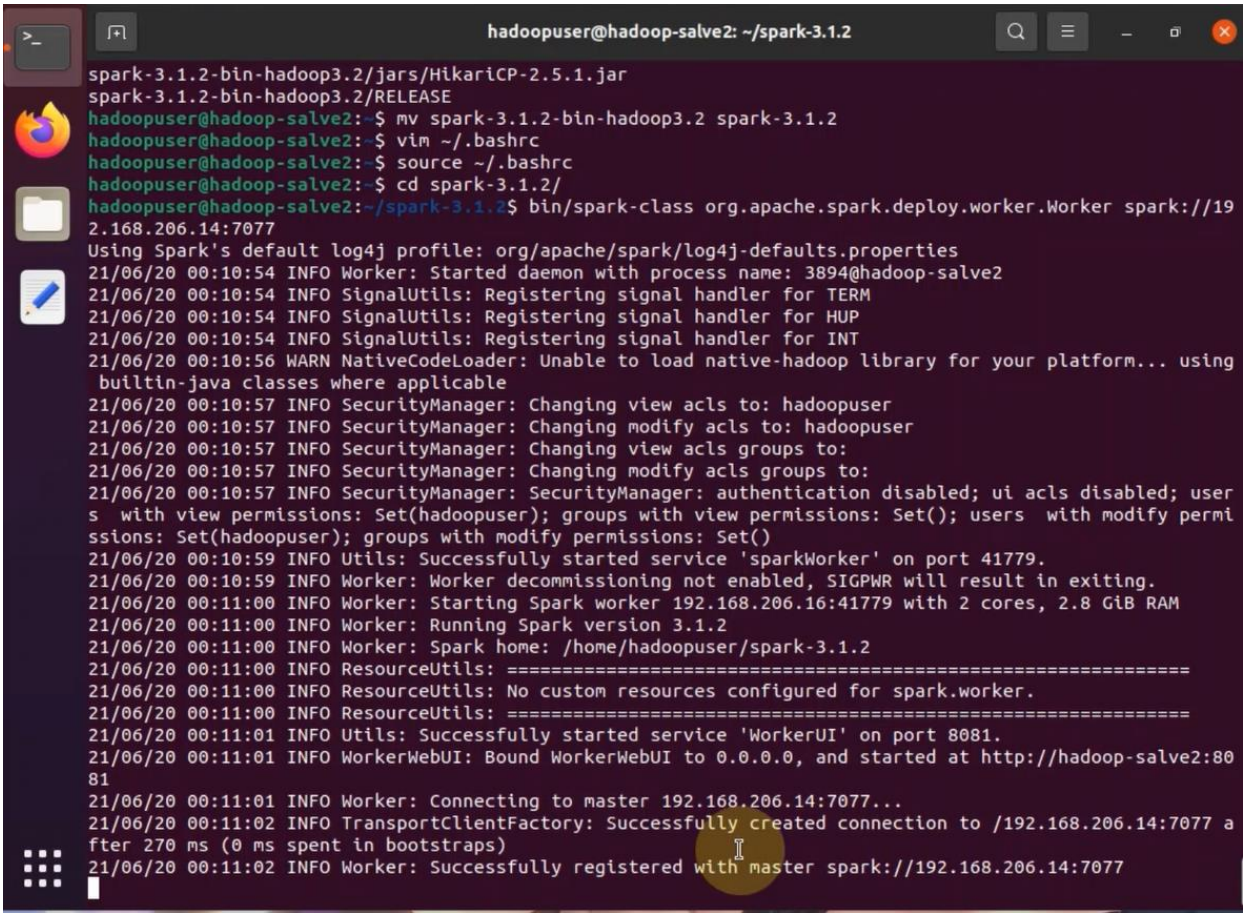
Bước 6: kiểm tra máy master



Bước 7: Chạy spark trên máy salve2

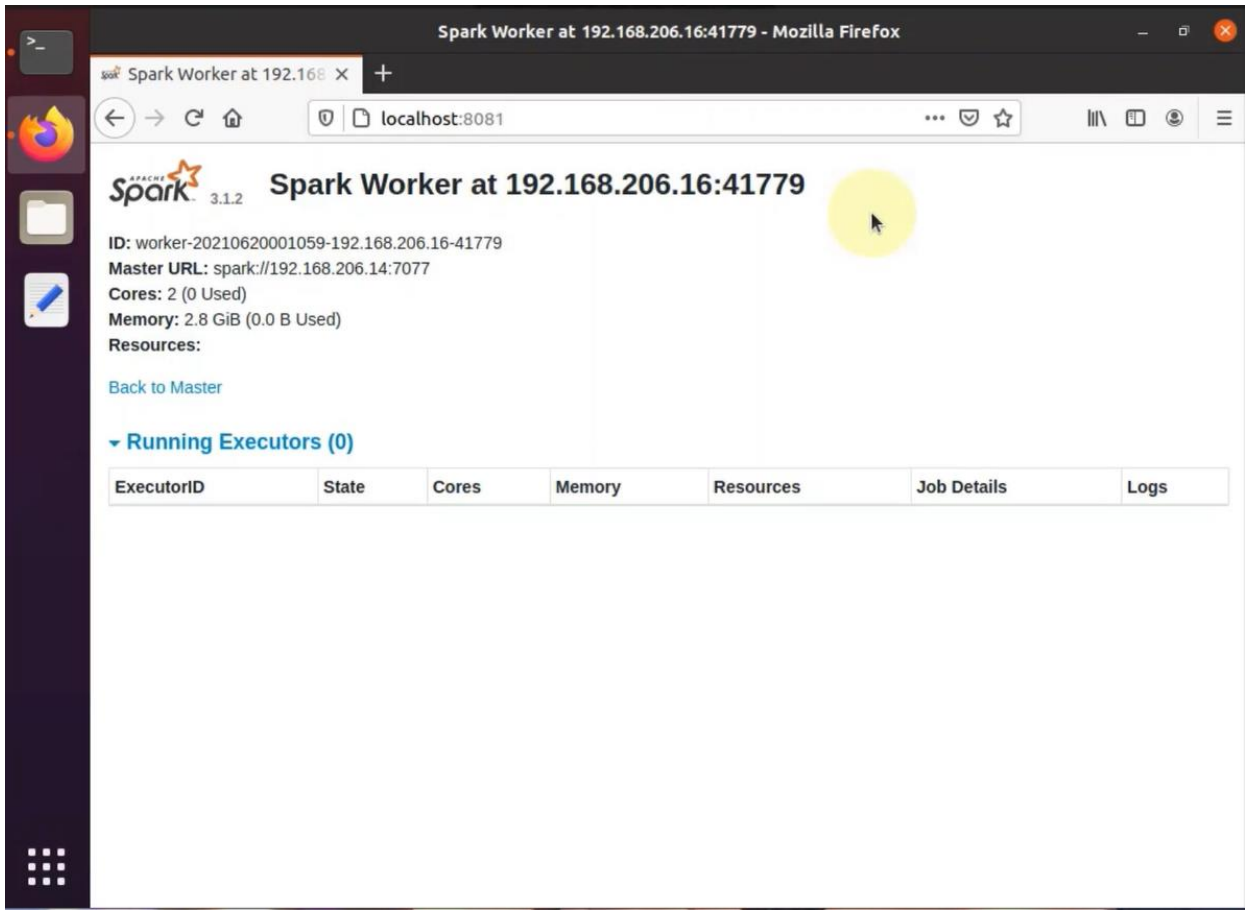
$ cd spark-3.1.2

$ bin/spark-class org.apache.spark.deploy.worker.Worker spark://192.168.206.14:7077

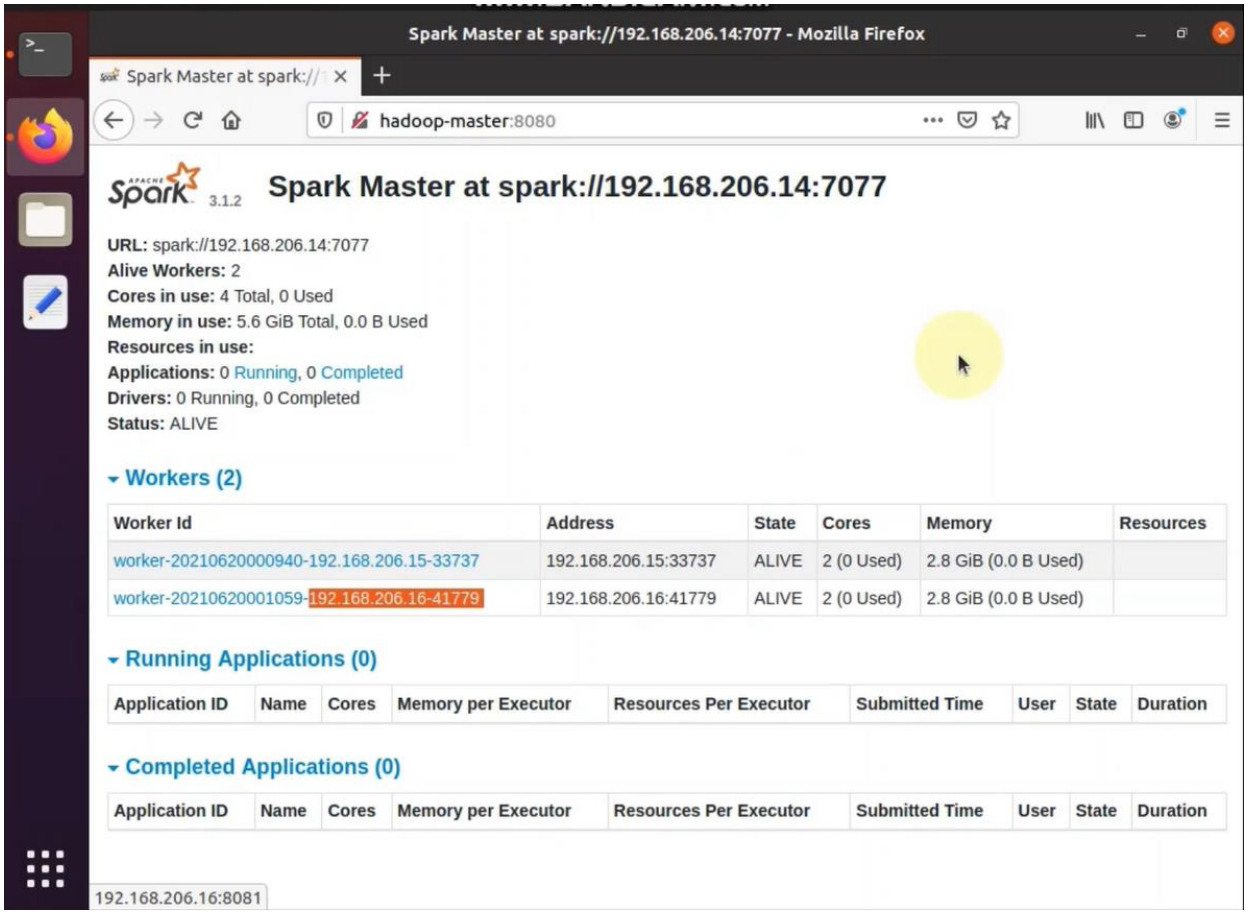


Bước 8: kiểm tra trên máy salve2

truy cập địa chỉ localhost:8081



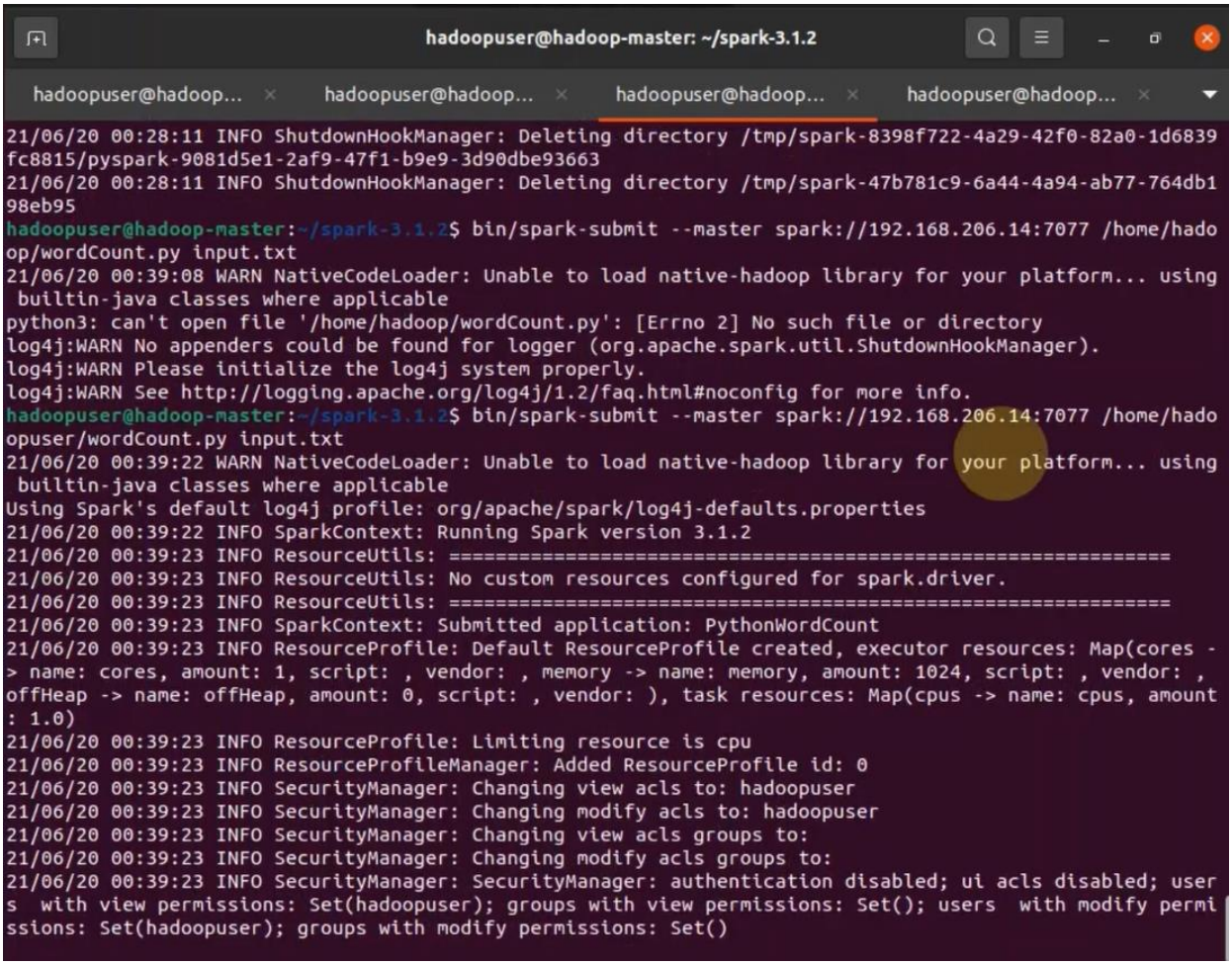
Bước 9: Kiểm tra máy master



Bước 10: chạy code wordCount.py trên spark

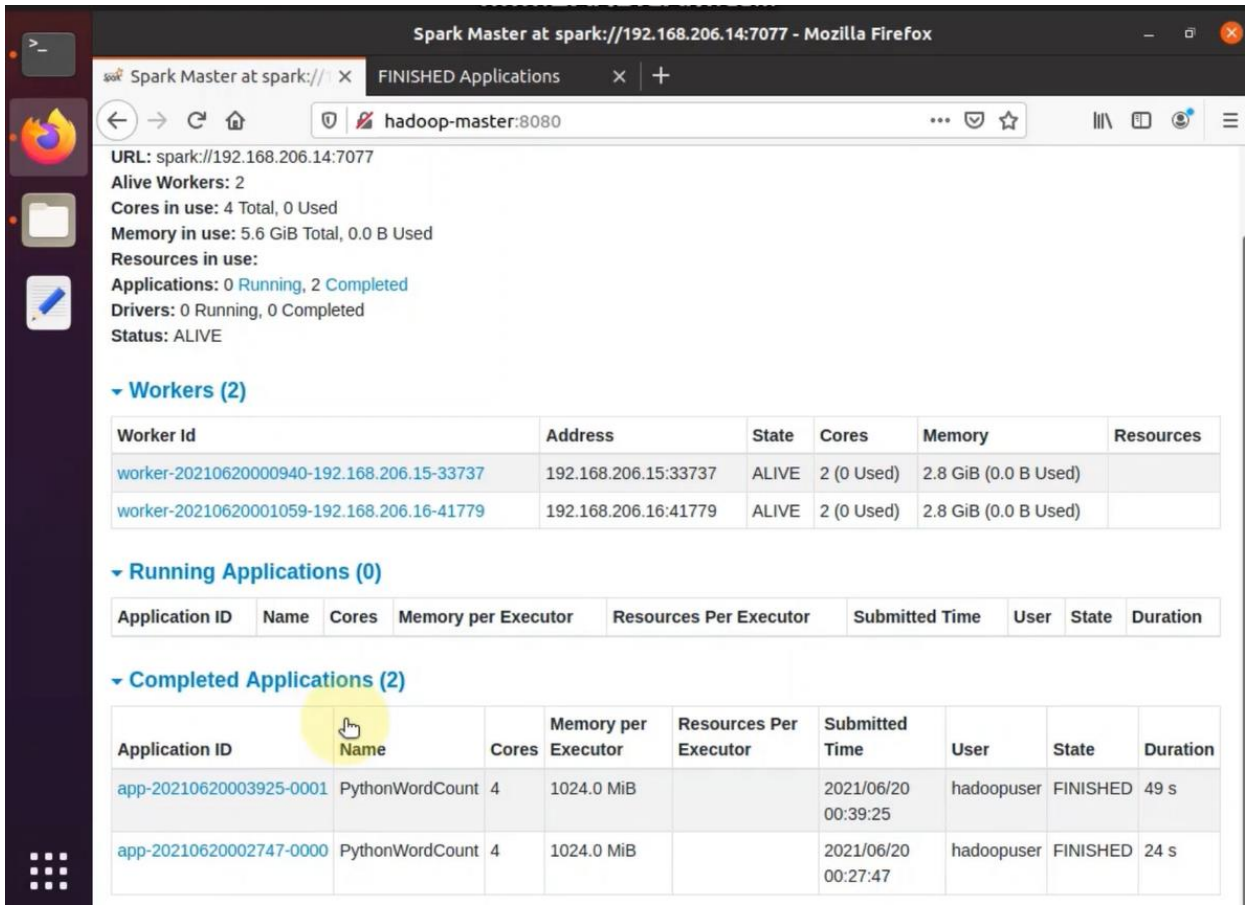
$ cd spark-3.1.2

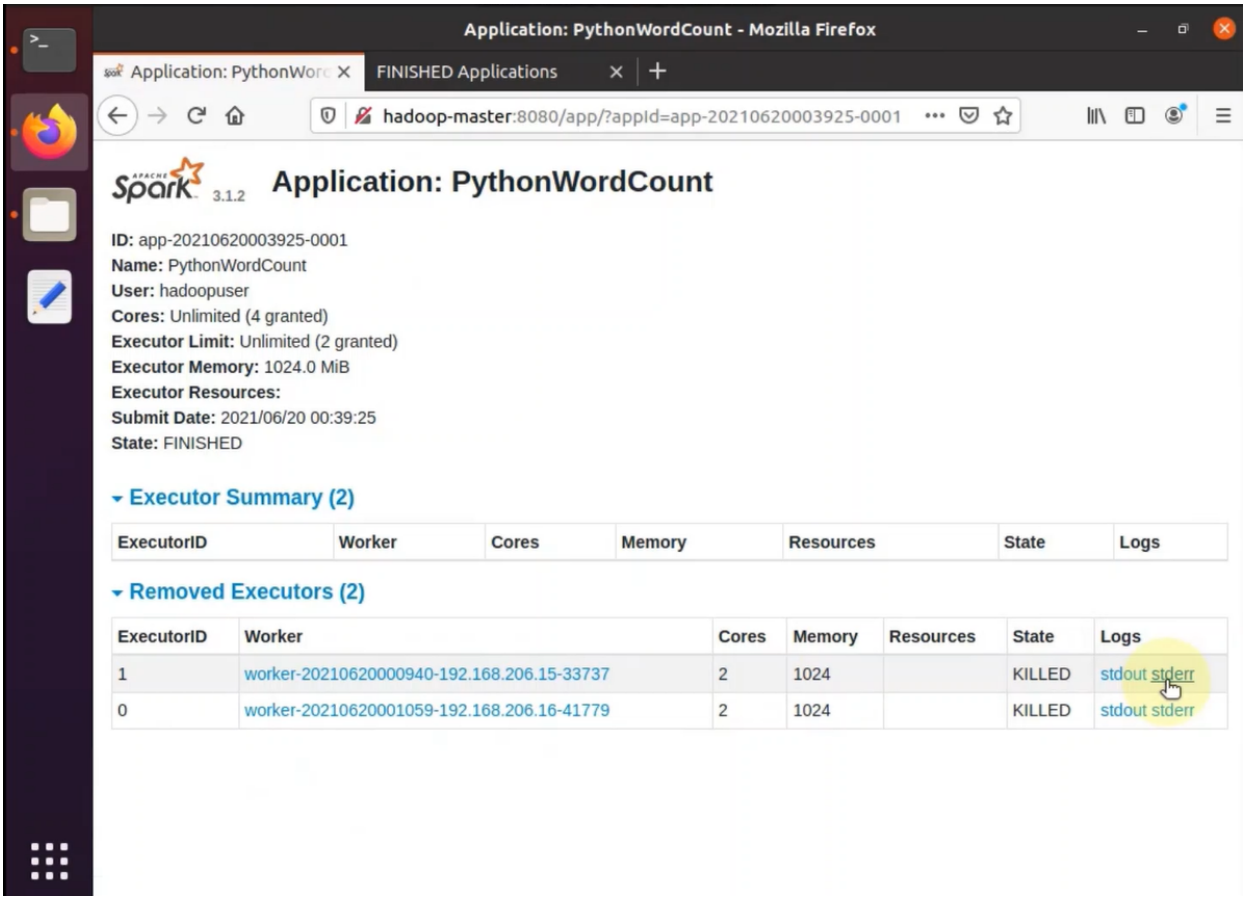
$ bin/spark-submit --master spark://192.168.206.14:7077 /home/hadoopuser/wordCount.py /home/hadoopuser/spark-3.1.2/input.txt



Kiểm tra kết quả

truy cập địa chỉ hadoop-master:8088




Application: PythonWordCount - Mozilla Firefox
Application: PythonWor
FINISHED Applications
hadoop-master:8080/app/?appId=app-20210620003925-0001
Spark 3.1.2
Application: PythonWordCount
ID: app-20210620003925-0001
Name: PythonWordCount
User: hadoopuser
Cores: Unlimited (4 granted)
Executor Limit: Unlimited (2 granted)
Executor Memory: 1024.0 MiB
Executor Resources:
Submit Date: 2021/06/20 00:39:25
State: FINISHED
Executor Summary (2)
ExecutorID
Worker
Cores
Memory
Resources
State
Logs
Removed Executors (2)
ExecutorID
Worker
Cores
Memory
Resources
State
Logs
1
worker-20210620000940-192.168.206.15-33737
2
1024
KILLED
stdout stderr
0
worker-20210620001059-192.168.206.16-41779
2
1024
KILLED
stdout stderr